

EHRM INFRASTRUCTURE UPGRADES

1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

PROJECT #: 581-22-700

BUILDINGS WITHIN THE SCOPE OF THIS PROJECT

BUILDING TYPE

BUILDING 01S - LEVEL 00B - ADMIN	BUILDING 07 - ATTIC - HEALTHCARE
BUILDING 01S - LEVEL 00G - HEALTHCARE	BUILDING 08 - LEVEL 00 - ADMIN
BUILDING 01S - LEVEL 01 - HEALTHCARE	BUILDING 08 - LEVEL 01 - ADMIN
BUILDING 01S - LEVEL 02 - HEALTHCARE	BUILDING 08 - LEVEL 02 - ADMIN
BUILDING 01S - LEVEL 03 - HEALTHCARE	BUILDING 08 - ATTIC - ADMIN
BUILDING 01S - LEVEL 04 - HEALTHCARE	BUILDING 09 - LEVEL 00 - HEALTHCARE
BUILDING 01S - LEVEL 05 - HEALTHCARE	BUILDING 09 - LEVEL 01 - HEALTHCARE
BUILDING 01S - LEVEL 06 - HEALTHCARE	BUILDING 09 - LEVEL 02 - HEALTHCARE
BUILDING 01W - LEVEL 00B - ADMIN	BUILDING 12 - LEVEL 00G - ADMIN
BUILDING 01W - LEVEL 00G - ADMIN	BUILDING 12 - LEVEL 01 - ADMIN
BUILDING 01W - LEVEL 01 - HEALTHCARE	BUILDING 12 - LEVEL 02 - ADMIN
BUILDING 01W - LEVEL 02 - HEALTHCARE	BUILDING 20 - LEVEL 01 - ADMIN
BUILDING 01W - LEVEL 03 - HEALTHCARE	BUILDING 20 - LEVEL 02 - ADMIN
BUILDING 01W - LEVEL 04 - ADMIN	BUILDING 23R - LEVEL 00B - HEALTHCARE
BUILDING 02 - LEVEL 00B - ADMIN	BUILDING 23R - LEVEL 01 - HEALTHCARE
BUILDING 02 - LEVEL 01 - ADMIN	BUILDING 23R - LEVEL 02 - HEALTHCARE
BUILDING 02 - LEVEL 02 - ADMIN	BUILDING 25 - LEVEL 01 - ADMIN
BUILDING 03 - LEVEL 01 - ADMIN	BUILDING 25 - LEVEL 02 - ADMIN
BUILDING 03 - LEVEL 02 - ADMIN	BUILDING 25 - LEVEL 03 - ADMIN
BUILDING 04 - LEVEL 00 - ADMIN	BUILDING 25 - LEVEL 04 - ADMIN
BUILDING 04 - LEVEL 01 - ADMIN	BUILDING 25 - LEVEL 05 - ADMIN
BUILDING 04 - LEVEL 02 - ADMIN	BUILDING 28 - LEVEL 01 - ADMIN
BUILDING 05 - LEVEL 00 - HEALTHCARE	BUILDING 34 - LEVEL 00 - ADMIN
BUILDING 05 - LEVEL 01 - HEALTHCARE	BUILDING 34 - LEVEL 01 - ADMIN
BUILDING 05 - LEVEL 02 - HEALTHCARE	BUILDING TR1 - LEVEL 01 - ADMIN
BUILDING 06 - LEVEL 00 - HEALTHCARE	BUILDING TR2 - LEVEL 01 - ADMIN
BUILDING 06 - LEVEL 01 - HEALTHCARE	BUILDING TR3 - LEVEL 01 - ADMIN
BUILDING 06 - LEVEL 02 - HEALTHCARE	BUILDING TR4 - LEVEL 01 - ADMIN
BUILDING 07 - LEVEL 00 - HEALTHCARE	BUILDING TR5 - LEVEL 01 - ADMIN
BUILDING 07 - LEVEL 01 - HEALTHCARE	
BUILDING 07 - LEVEL 02 - HEALTHCARE	

PROJECT INFORMATION

CODE ANALYSIS	APPLICABLE BUILDING
STATE: WEST VIRGINIA	BUILDING 1, 2, 3, 4, 5, 6, 7, 8, 9, 12, 20, 23, 23R,
COUNTY: CABELL	25, 28, 34, TRAILER 1, TRAILER 2, TRAILER 3,
JURISDICTION: HUNTINGTON	TRAILER 4, TRAILER 5, DATA CENTER
OWNER: DEPARTMENT OF VETERANS AFFAIRS	
ICRA CLASS REQUIREMENTS	PROJECT DESCRIPTION
REQUIREMENTS DEPENDENT ON EACH INDIVIDUAL ROOM LOCATION, CLASS II, III AND IV.	THE PROPOSED DESIGN SHALL IMPROVE, RECONFIGURE, EXPAND AND RENOVATE EXISTING SPACES, CONSTRUCT NEW EXTERIOR ROOMS FOR NEW TELECOMMUNICATION ROOMS, AND TO CONSTRUCT NEW DATA CENTER IN REPLACEMENT OF THE EXISTING IN THE INTERIOR COURTYARD.

DESIGN TEAM

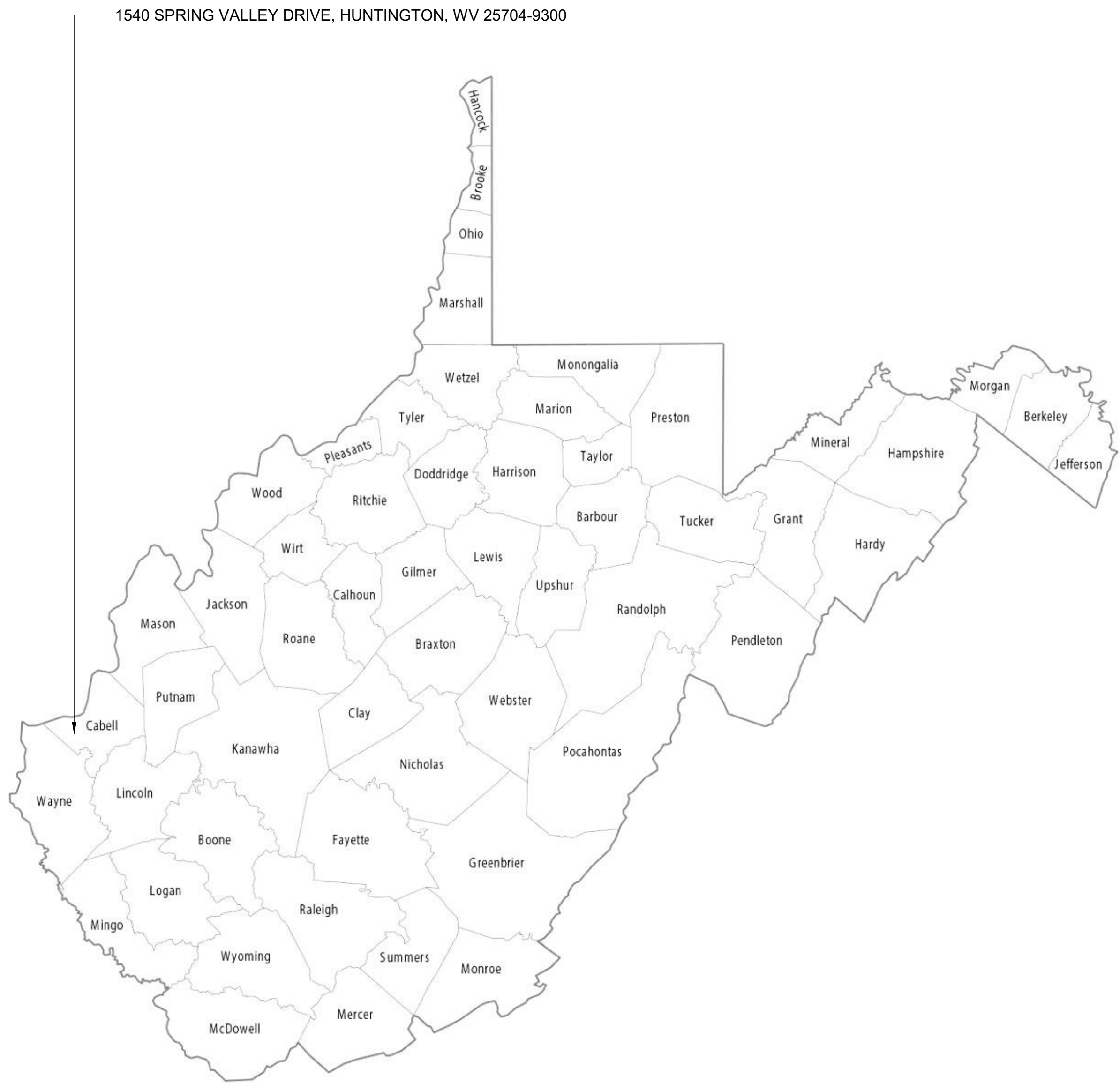
CIVIL	MECHANICAL
WSP USA, INC.	SPEES DESIGN BUILD
33301 9TH AVENUE SOUTH	23830 PACIFIC HIGHWAY S. SUITE 300
FEDERAL WAY, WA 98003-2600	KENT, WA 98032
PHONE: (206) 431-2300	PHONE: (206) 590-2118
ARCHITECTURAL	ELECTRICAL
SPEES DESIGN BUILD	SPEES DESIGN BUILD
23830 PACIFIC HIGHWAY S. SUITE 300	23830 PACIFIC HIGHWAY S. SUITE 300
KENT, WA 98032	KENT, WA 98032
PHONE: (206) 590-2118	PHONE: (206) 590-2118

STRUCTURAL

WSP USA, INC.
33301 9TH AVENUE SOUTH
FEDERAL WAY, WA 98003-2600
PHONE: (206) 431-2300

GENERAL NOTES

- COORDINATE WORK WITH ALL CONTRACT DOCUMENTS, INCLUDING BUT NOT LIMITED TO DRAWINGS & SPECIFICATIONS.
- SPECIFICATIONS SHALL SUPERSEDE DRAWINGS IN THE CASE OF ANY CONFLICTING INFORMATION. BRING TO THE AWARENESS OF AND VERIFY WITH VA COR PRIOR TO PROCEEDING.
- THE CONTRACTOR AND ANY SUB CONTRACTORS SHALL FAMILIARIZE THEMSELVES WITH EXISTING CONDITIONS AND PROJECT REQUIREMENTS PRIOR TO SUBMITTING A BID OR BEGINNING WORK. NO ALLOWANCES WILL BE MADE ON BEHALF OF THE CONTRACTOR OR SUB-CONTRACTORS FOR FAILURE TO VISIT THE SITE.
- THE CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS BEFORE BEGINNING WORK. ANY CONFLICTS BETWEEN EXISTING CONDITIONS AND THE CONTRACT DOCUMENTS SHALL BE BROUGHT IMMEDIATELY TO THE ATTENTION OF THE COR.
- ALL WORK SHALL BE PERFORMED WITH A HIGH-LEVEL OF WORKMANSHIP.
- THE CONTRACTOR SHALL VERIFY ALL PRODUCT TYPES AND QUANTITIES DURING SITE WALK THROUGH, PRIOR TO PURCHASING ANY MATERIALS, EQUIPMENT, PRODUCTS OR LABOR.
- THE CONTRACTOR SHALL VERIFY AND COORDINATE WITH ALL INSTALLERS TO ENSURE COMPLIANCE WITH THE CONTRACT DOCUMENTS.



1 LOCATION MAP

NOT TO SCALE

2 FACILITY SITE PLAN

1" = 20'-0"



CONSULTANT

ENGINEERING DISCIPLINE:

ARCHITECT/ENGINEER OF RECORD

A/E:
SPEES DESIGN BUILD
23830 PACIFIC HIGHWAY S. SUITE 300
KENT, WA 98032



STAMP



Office of
Construction
and Facilities
Management



U.S. Department
of Veterans Affairs

Drawing Title

COVER SHEET

Approved:

Phase

100% CONSTRUCTION
DOCUMENTS

Project Title

EHRM INFRASTRUCTURE
UPGRADES - HUNTINGTON

Project Number

581-22-700

Building Number

-

Drawing Number

G-001

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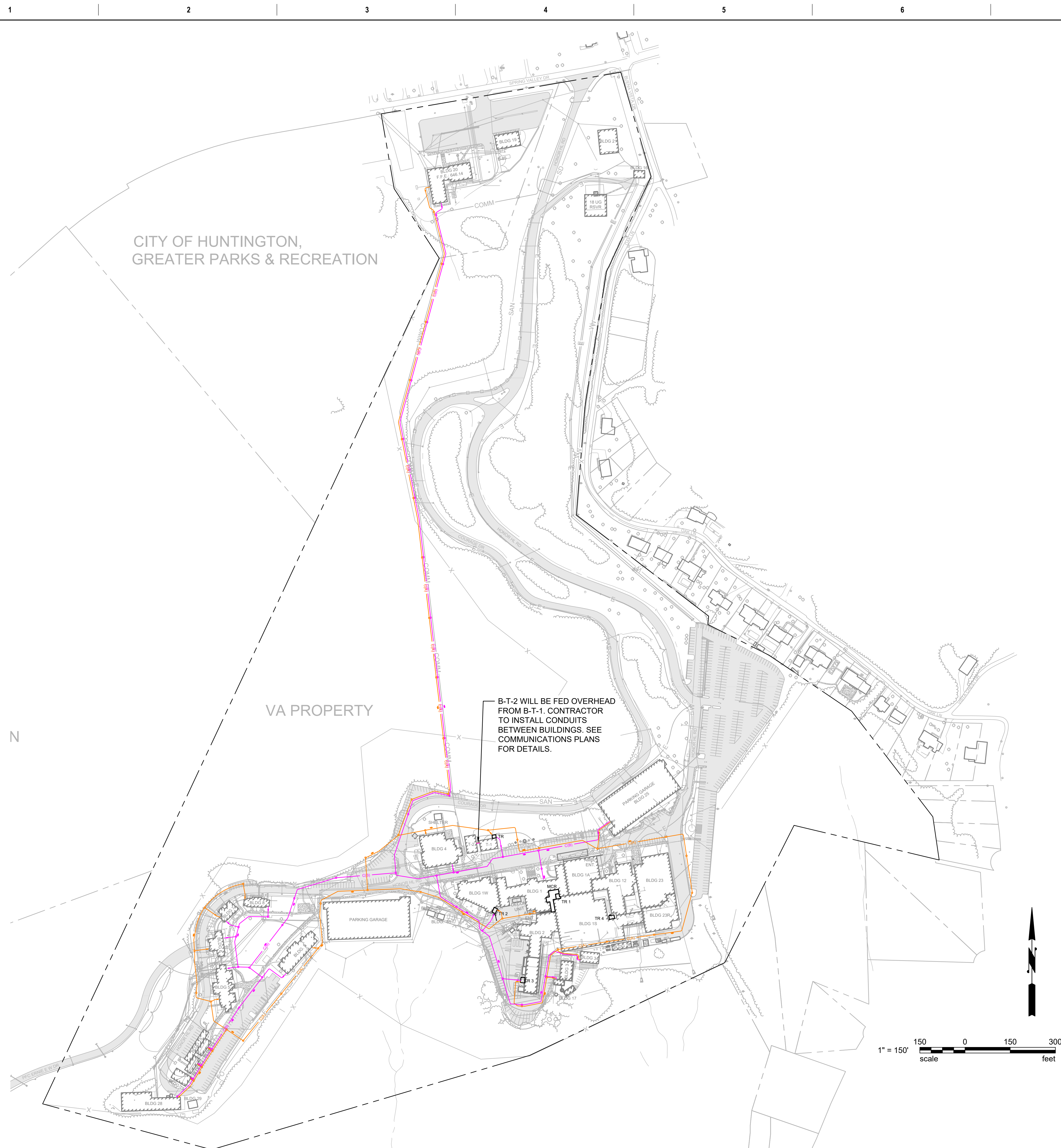
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- C**

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- ## E

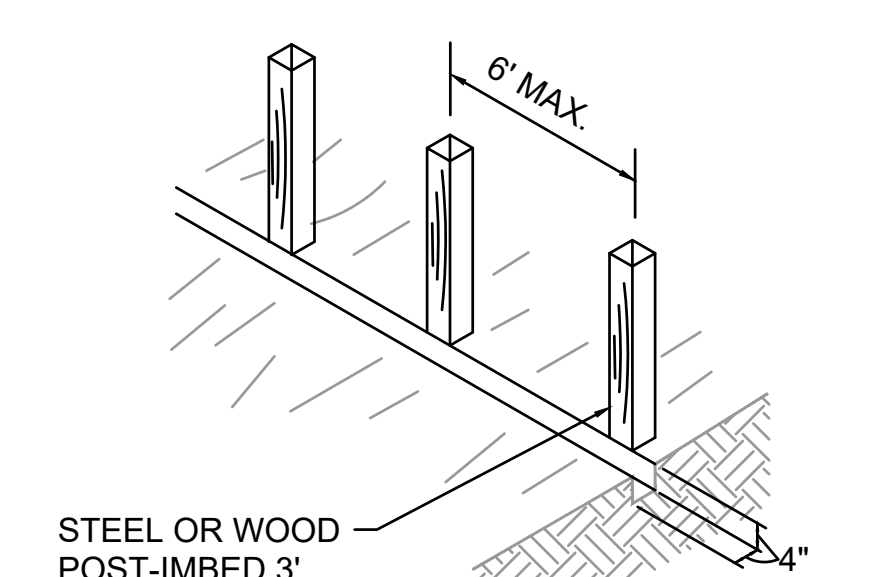
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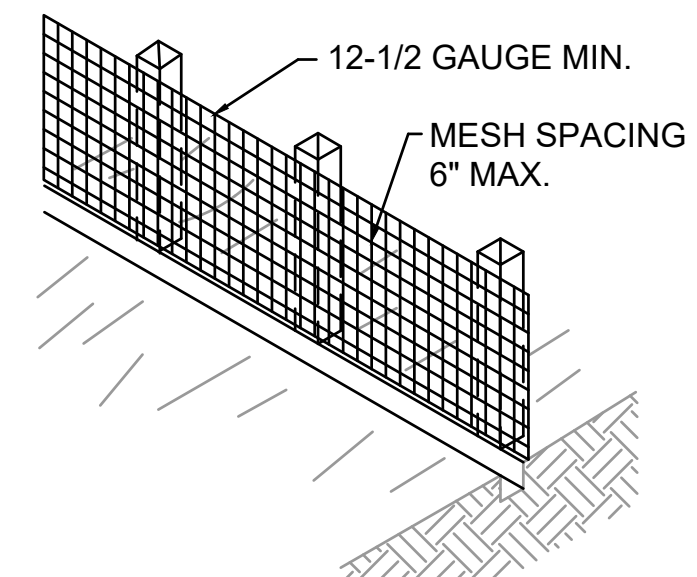
TEMPORARY EROSION AND SEDIMENT CONTROL NOTES:

1. THE CONTRACTOR MUST PROVIDE AND MAINTAIN TEMPORARY EROSION AND SEDIMENT CONTROL (TESC) FACILITIES. REFER TO THE APPLICABLE DRAWINGS AND THE SPECIFICATIONS FOR PLANS AND DETAILS THAT DESCRIBE THE MINIMUM TESC FEATURES REQUIRED. AS CONSTRUCTION PROGRESSES AND SEASONAL CONDITIONS DICTATE, THE CONTRACTOR MUST REVISE TESC FACILITIES AND CONFIGURATIONS AS NECESSARY TO ENSURE COMPLETE SILTATION CONTROL AND THAT NO SEDIMENT LADEN WATER ENTERS THE DOWNSTREAM NATURAL OR PIPED DRAINAGE SYSTEM.
2. DURING THE COURSE OF CONSTRUCTION, IT MUST BE THE OBLIGATION AND RESPONSIBILITY OF THE CONTRACTOR TO ADDRESS ANY NEW CONDITIONS THAT MAY BE CREATED BY ITS ACTIVITIES AND TO PROVIDE ADDITIONAL TESC FACILITIES THAT MAY BE NEEDED TO PROTECT THE SITE, ADJACENT LAND AND EXISTING DRAINAGE FEATURES.
3. TEMPORARY EROSION/WATER POLLUTION PREVENTION MEASURES MUST BE REQUIRED IN ACCORDANCE WITH THE CURRENT ENVIRONMENTAL PROTECTION AGENCY'S (EPA) CONSTRUCTION STORMWATER GENERAL PERMIT AND THE APPLICABLE STATE STANDARDS OR MANUAL. DURING THE COURSE OF CONSTRUCTION, IT MUST BE THE OBLIGATION AND RESPONSIBILITY OF THE CONTRACTOR TO ADDRESS ANY NEW CONDITIONS THAT MAY BE CREATED BY ITS ACTIVITIES AND TO PROVIDE ADDITIONAL TESC FACILITIES THAT MAY BE NEEDED TO PROTECT THE SITE, ADJACENT LAND, AND EXISTING DRAINAGE FEATURES.
4. THE CONTRACTOR MUST ENSURE THE TEMPORARY EROSION AND SEDIMENTATION CONTROL MEASURES AS SHOWN ON THESE DRAWINGS PROVIDE ADEQUATE CONTROL SO THAT NO SEDIMENT LADEN WATER LEAVES THE SITE. THE CONTRACTOR MUST INSTALL ADDITIONAL FACILITIES AS NECESSARY TO PROTECT ADJACENT PROPERTIES, SENSITIVE AREAS, NATURAL WATER COURSES, AND/OR STORM DRAINAGE SYSTEMS TO ENSURE COMPLETE SILTATION CONTROL.
5. PRIOR TO STARTING ANY WORK ON THE SITE, THE CONTRACTOR MUST NOTIFY THE OWNER AND MUST INSTALL ALL EROSION AND SEDIMENT CONTROL MEASURES AS SHOWN ON THE APPROVED PLANS. ALL TEMPORARY EROSION CONTROL MEASURES MUST BE MAINTAINED FOR THE DURATION OF PROJECT CONSTRUCTION.
6. THE BMP FACILITIES SHOWN ON THIS PLAN MUST BE CONSTRUCTED IN CONJUNCTION WITH ALL CLEARING AND GRADING ACTIVITIES, AND IN SUCH A MANNER AS TO ENSURE THAT SEDIMENT AND SEDIMENT LADEN WATER DO NOT ENTER THE DRAINAGE SYSTEM, ROADWAYS, OR VIOLATE APPLICABLE WATER STANDARDS.
7. PROVIDE STRAW WATTLE AND COMPOST FILTER SOCK SEDIMENT CONTROL MEASURES OR OTHER APPROVED PERIMETER PROTECTION SEDIMENT CONTROL MEASURES FOR EXISTING DRAINAGE CHANNELS NEARBY OR ADJACENT TO THE PROJECT SITE.
8. PROVIDE INLET PROTECTION AT ALL AREA STORM DRAIN STRUCTURES IN THE AREA OF CONSTRUCTION.
9. THE BMP FACILITIES MUST BE INSPECTED DAILY AND AFTER STORM EVENTS BY THE APPLICANT/CONTRACTOR AND MAINTAINED AS NECESSARY TO ENSURE THEIR CONTINUED FUNCTIONING. WRITTEN RECORDS MUST BE KEPT OF MONTHLY REVIEWS.
10. THE CONTRACTOR MUST BE RESPONSIBLE FOR CLEANING DIRT, MUD, AND OTHER CONSTRUCTION DEBRIS, WHICH MAY ACCUMULATE ON PAVED STREETS ADJACENT TO THE SITE AS A RESULT OF THE CONSTRUCTION ACTIVITY. AS A MINIMUM, DIRT, MUD AND OTHER DEBRIS MUST BE REMOVED FROM PAVEMENT AT THE CLOSE OF EACH CONSTRUCTION DAY.
11. EXPOSED SLOPES AND SURFACES MUST BE STABILIZED FROM SEDIMENT TRANSPORT USING APPROVED BMPS UNTIL THE SITE HAS BEEN STABILIZED AND THE CONTRACTING OFFICER APPROVES THE REMOVAL OF SLOPE STABILIZATION MEASURES.
12. THE CONTRACTOR MUST BE RESPONSIBLE TO ENSURE THAT NO SEDIMENT LADEN RUNOFF LEAVES THE CONSTRUCTION SITE, WORK AREA, OR HAUL ROUTES. THE CONTRACTOR MUST SUBMIT A SEDIMENT AND WATER POLLUTION PREVENTION PLAN TO THE CONTRACTING OFFICER FOR APPROVAL PRIOR TO BEGINNING ANY WORK.
13. THE CONTRACTOR MUST BE RESPONSIBLE FOR DEWATERING OF ANY EXCAVATION DURING INCLEMENT WEATHER. DEWATERING WATER MUST BE TESTED AND DISPOSED OF IN ACCORDANCE WITH STATE ENVIRONMENTAL REQUIREMENTS ACCORDING TO THE APPROVED SWPPP PLAN.
14. REMOVE ALL TEMPORARY EROSION AND SEDIMENT CONTROL MEASURES ONCE THE SITE IS STABILIZED AFTER CONSTRUCTION COMPLETION.

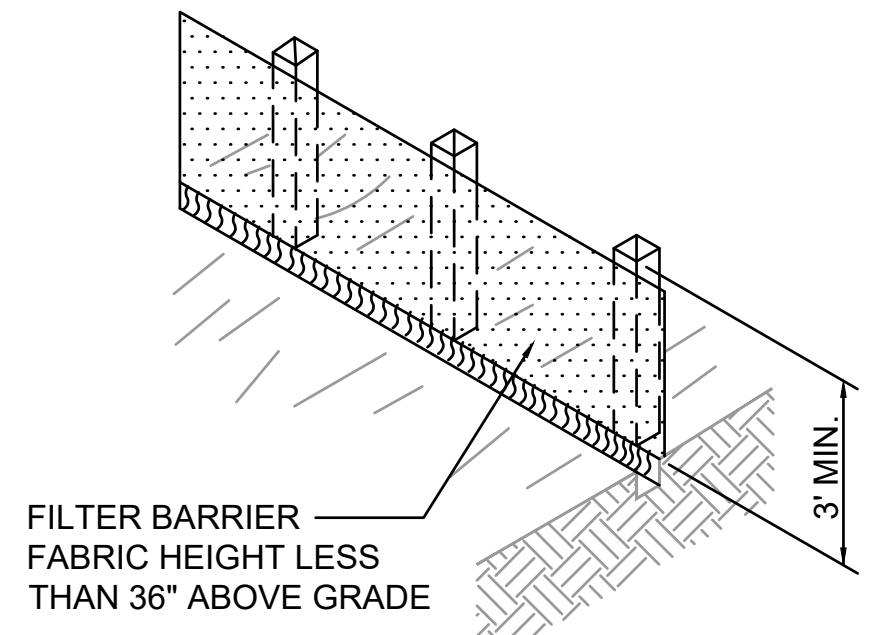
1. SET POSTS AND EXCAVATE A 4" x 4" TRENCH UPSLOPE ALONG THE LINE OF POSTS.



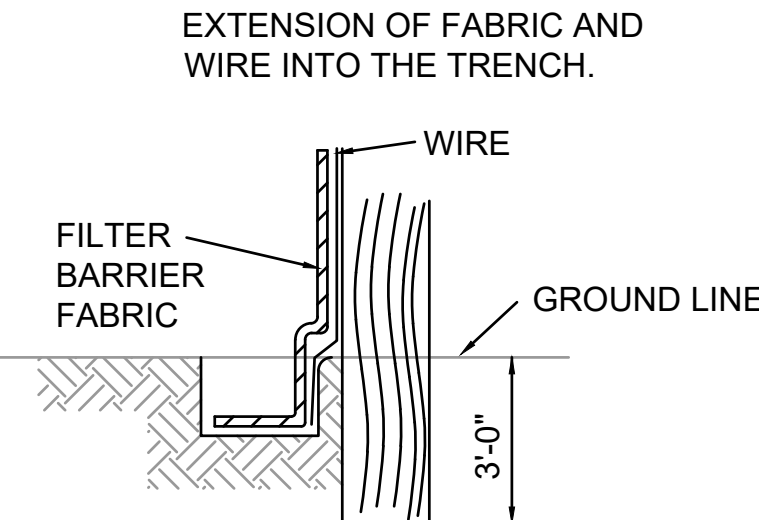
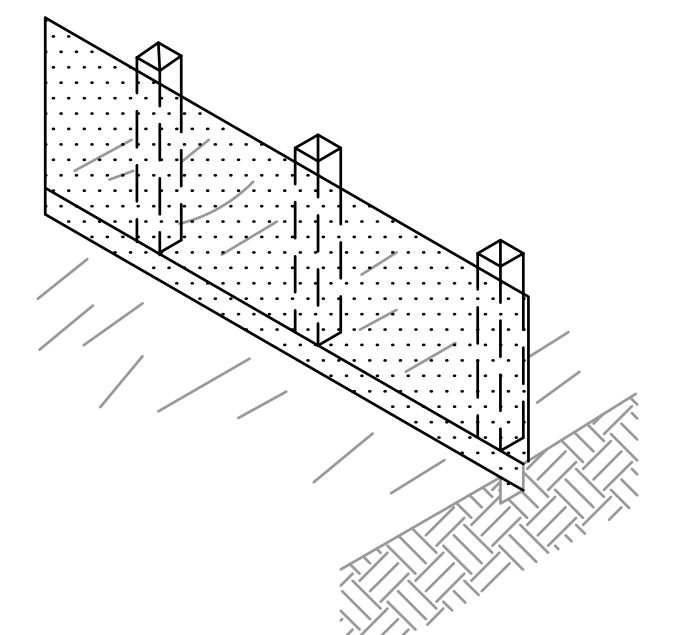
2. STAPLE WIRE FENCING TO THE POSTS.



3. ATTACH THE FILTER BARRIER FABRIC TO THE WIRE FENCE AND EXTEND IT INTO THE TRENCH.



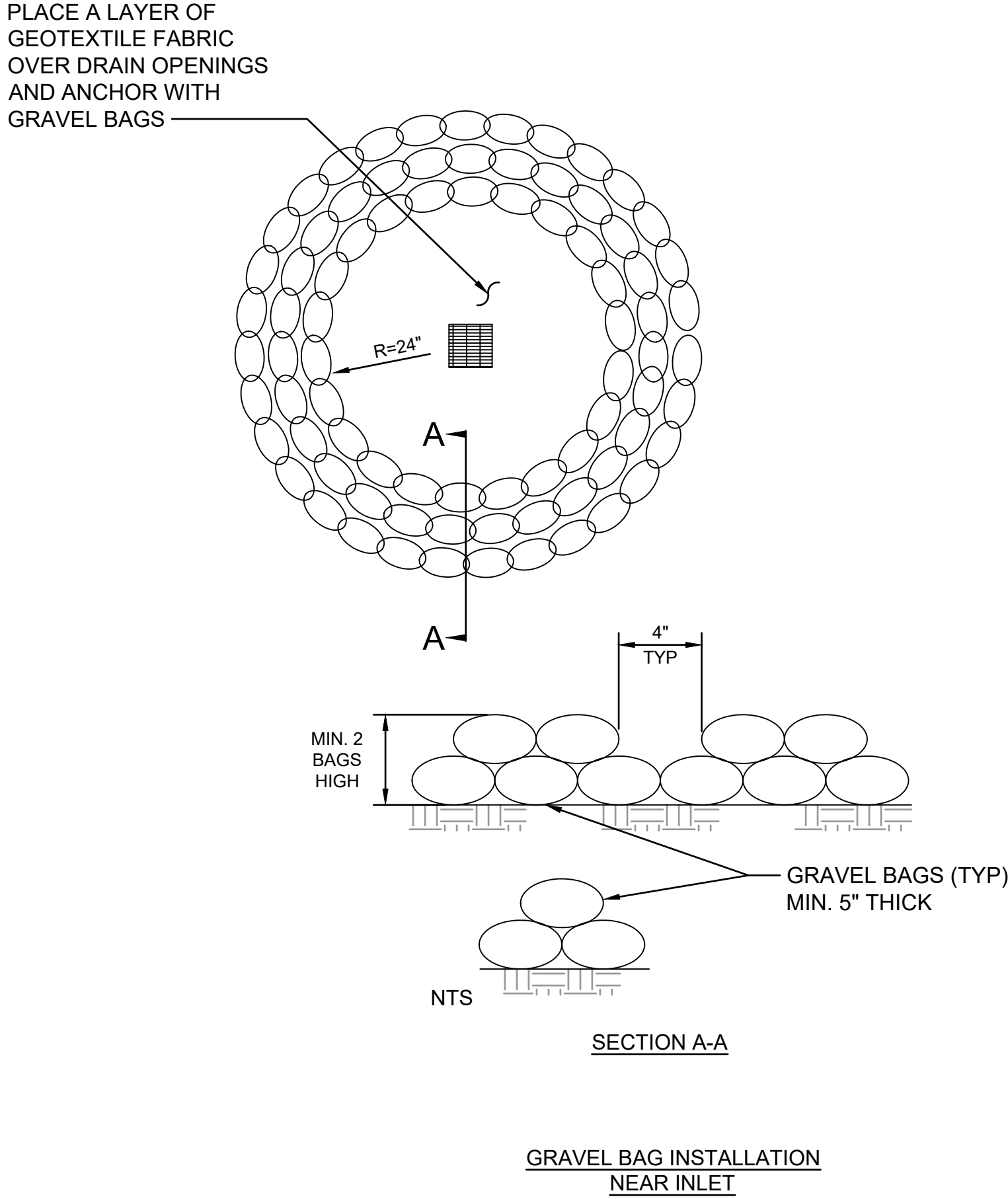
4. BACKFILL AND COMPACT THE EXCAVATED SOIL.



C5
CG101

TEMPORARY SEDIMENT FENCE

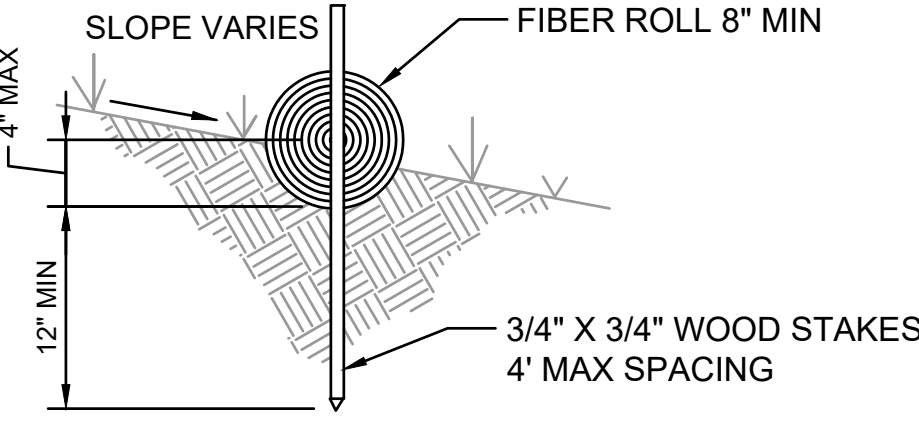
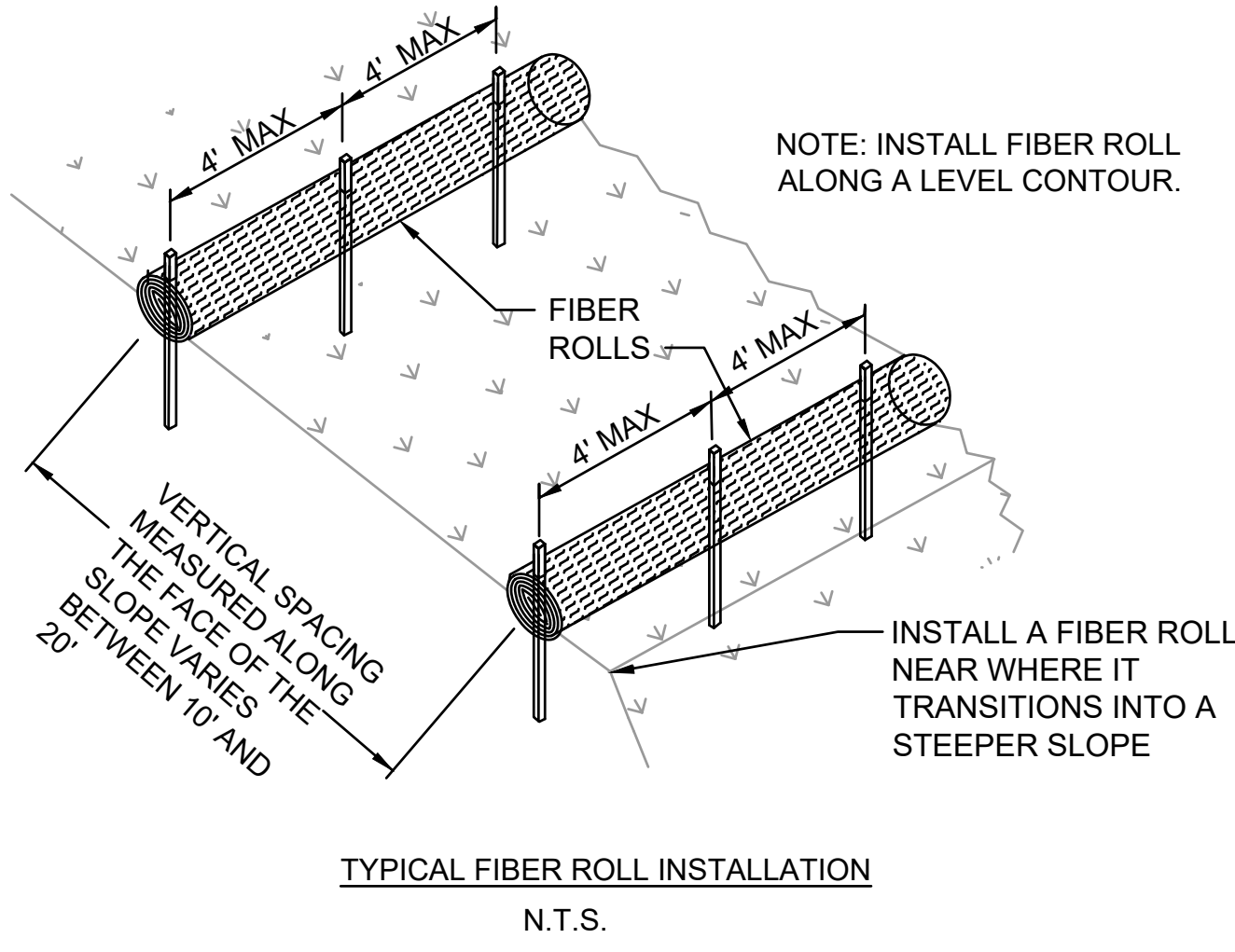
SCALE: NTS



F5
CG101

TEMPORARY GRAVEL BAG INSTALLATION

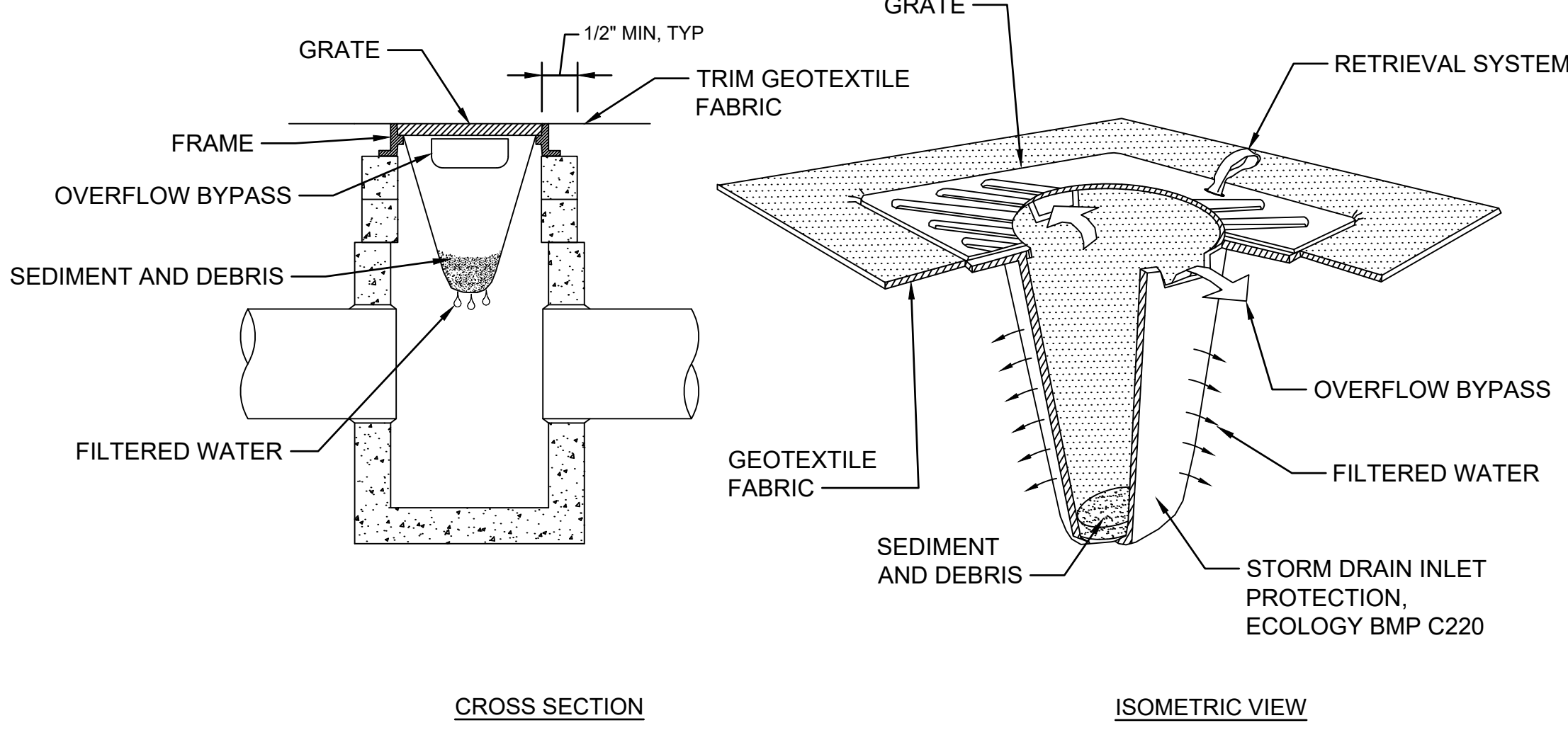
SCALE: NTS



C9
CG101

TEMPORARY FIBER ROLL

SCALE: NTS



F8
CG101

TEMPORARY INLET PROTECTION

SCALE: NTS

		CONSULTANT CIVIL ENGINEER: WSP WSP USA Solutions Inc. 1201 Pacific Avenue Suite 550 Tacoma, WA 98402 TEL: (206) 431-2300 FAX: (206) 431-2250	DESIGNER OF RECORD A/E: SPEES DESIGN BUILD 23830 PACIFIC HIGHWAY S, STE 300 KENT, WA 98032 (206) 209-2246 SDB SPEESDESIGNBUILD	STAMP SEAL OF THE DISTRICT OF COLUMBIA OFFICE OF THE ARCHITECT OF THE CAPITAL REGISTERED PROFESSIONAL ARCHITECT	Office of Construction and Facilities Management VAU.S. Department of Veterans Affairs	Drawing Title EROSION CONTROL GENERAL NOTES AND DETAILS	Phase 100% CONSTRUCTION DOCUMENTS	Project Title EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON	Project Number 581-22-700
						Building Number			
						Drawing Number CG101			
						Issue Date 09/03/2025	Checked PB	Drawn RDM	008 OF 693
Revisions:						Date:			





LEGEND

PROPOSED SIDE A FIBER PATHWAY

PROPOSED SIDE A UNDERGROUND HANDHOLE

PROPOSED SIDE B FIBER PATHWAY

PROPOSED SIDE B UNDERGROUND HANDHOLE

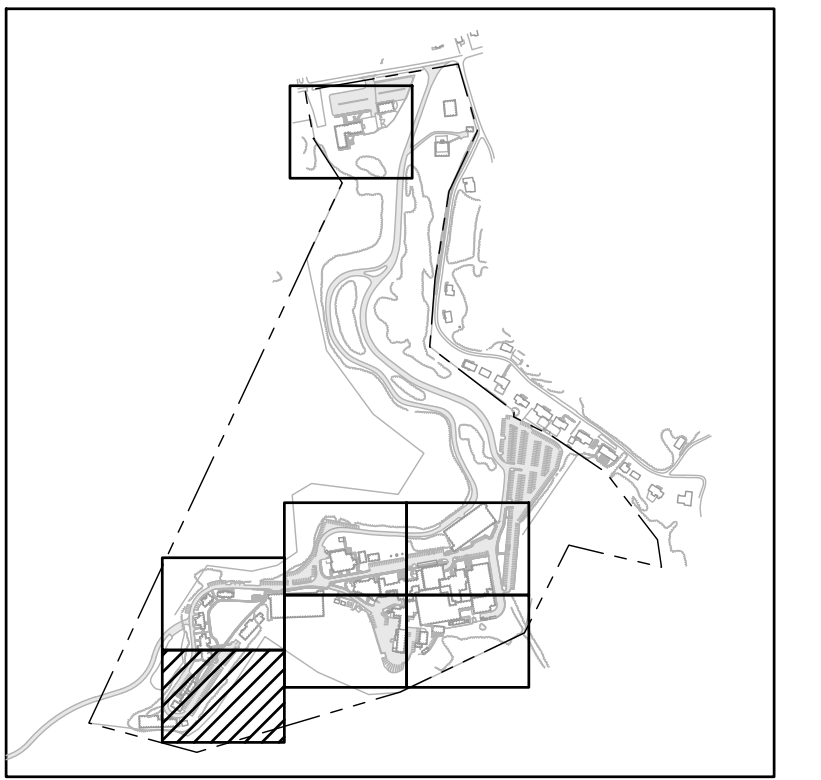
#	SITE KEYNOTES
①	TYPE 1 DUCT BANK PER DETAIL F3 ON SHEET CU500
②	TYPE 2 DUCT BANK PER DETAIL F6 ON SHEET CU500

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Revisions: <table><thead><tr><th>Rev</th><th>Description</th><th>Date</th></tr></thead><tbody><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr></tbody></table>		Rev	Description	Date																															CONSULTANT CIVIL ENGINEER: WSP USA Solutions Inc. 1201 Pacific Avenue Suite 550 Tacoma, WA 98402 TEL: (206) 431-2300 FAX: (206) 431-2250		DESIGNER OF RECORD A/E: SPEES DESIGN BUILD 23830 PACIFIC HIGHWAY S, STE 300 KENT, WA 98032 (206) 209-2246 SPEESDESIGNBUILD		STAMP PETER BAILEY STATE OF WASHINGTON REGISTERED PROFESSIONAL ENGINEER 2252		Office of Construction and Facilities Management VA U.S. Department of Veterans Affairs		Drawing Title FIBER OPTIC OUTSIDE PLANT UTILITY PLAN Approved:		Phase 100% CONSTRUCTION DOCUMENTS		Project Title EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON Location VA HUNTINGTON 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300 Issue Date 09/03/2025 Checked PB Drawn RDM		Project Number 581-22-700 Building Number Drawing Number CU301 011 OF 693	
Rev	Description	Date																																																

	CGA1	PROPOSED SIDE A FIBER PATHWAY
		PROPOSED SIDE A UNDERGROUND HANDHOLE
	CGB1	PROPOSED SIDE B FIBER PATHWAY
		PROPOSED SIDE B UNDERGROUND HANDHOLE

MATCHLINE SEE SHEET CU305

[illegible]

WSP)



A circular professional engineer seal for Peter Bailey, State of Washington, License No. 42527. The seal features a portrait of a man in the center, surrounded by the text "PETER BAILEY", "STATE OF WASHINGTON", "42527", and "REGISTERED PROFESSIONAL ENGINEER". The seal is stamped in blue ink on a white background.

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VA FORM 08-6231

Project Number 581-22-700
Building Number
Drawing Number CU304
014 OF 693

	PROPOSED SIDE A FIBER PATHWAY
	PROPOSED SIDE A UNDERGROUND HANDHOLE
	PROPOSED SIDE B FIBER PATHWAY
	PROPOSED SIDE B UNDERGROUND HANDHOLE

#	SITE KEYNOTES
①	TYPE 1 DUCT BANK PER DETAIL F3 ON SHEET CU500
②	TYPE 2 DUCT BANK PER DETAIL F6 ON SHEET CU500



KEYMAP
NOT TO SCALE



1" = 20'



scale feet

[illegible]



MATCHLINE SEE SHEET CU307

LEGEND

- PROPOSED SIDE A FIBER PATHWAY
 PROPOSED SIDE A UNDERGROUND HANDHOLE
 PROPOSED SIDE B FIBER PATHWAY
 PROPOSED SIDE B UNDERGROUND HANDHOLE

#	SITE KEYNOTES
1	TYPE 1 DUCT BANK PER DETAIL F3 ON SHEET CU500
2	TYPE 2 DUCT BANK PER DETAIL F6 ON SHEET CU500



KEYMAP
NOT TO SCALE



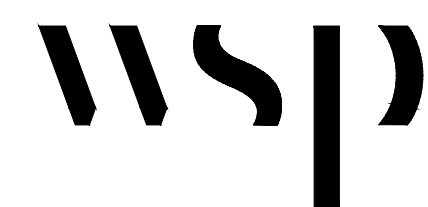
1" = 20'

scale feet

[illegible]

CONSULTANT

CIVIL ENGINEER:



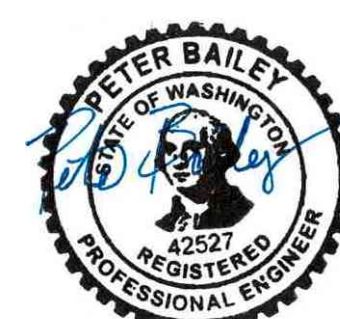
WSP USA Solutions Inc.
1201 Pacific Avenue
Suite 550
Tacoma, WA 98402
TEL: (206) 431-2300
FAX: (206) 431-2250

DESIGNER OF RECORD

A/E:
SPEES DESIGN BUILD
23830 PACIFIC HIGHWAY S, STE 300
KENT, WA 98032
(206) 209-2246



STAMP



Office of
Construction
and Facilities
Management

VA U.S. Department
of Veterans Affairs

Drawing Title	FIBER OPTIC OUTSIDE PLANT UTILITY PLAN
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Approved:

	Phase
1	Phase 1
2	Phase 2
3	Phase 3
4	Phase 4
5	Phase 5
6	Phase 6
7	Phase 7
8	Phase 8
9	Phase 9
10	Phase 10
11	Phase 11
12	Phase 12
13	Phase 13
14	Phase 14
15	Phase 15
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91	Phase 91
92	Phase 92
93	Phase 93
94	Phase 94
95	Phase 95
96	Phase 96
97	Phase 97
98	Phase 98
99	Phase 99
100	Phase 100

100% CONSTRUCTION
DOCUMENTS

Project Title

EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

Location VA HUNTINGTON 1540 SPRING VALLEY	Issue Date
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Checked

Checked _____

☐ Drawn

Drawn

Project Number

581-22-700

Drawing Number

CU306

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VA FORM 08-6231

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A

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A

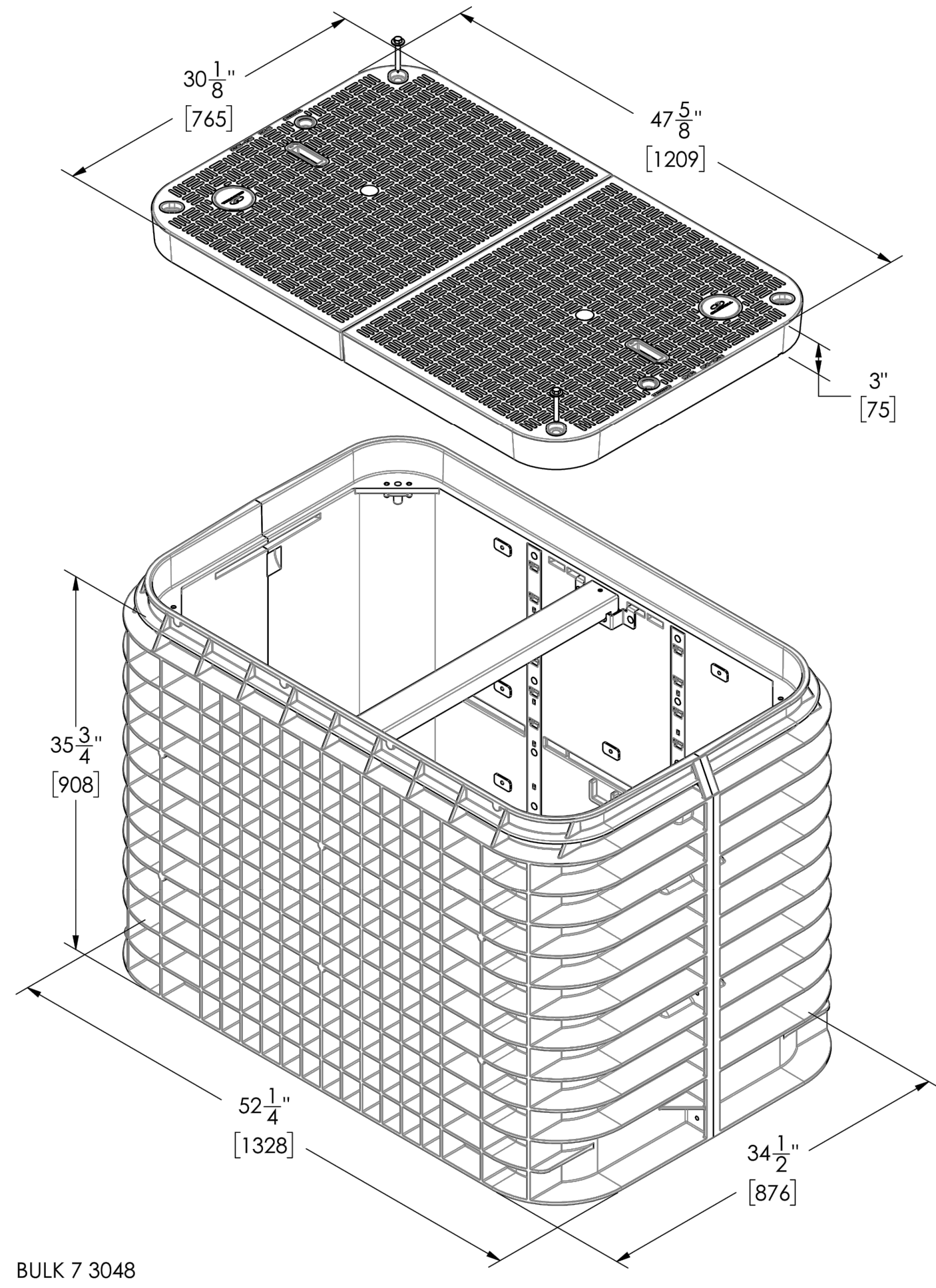
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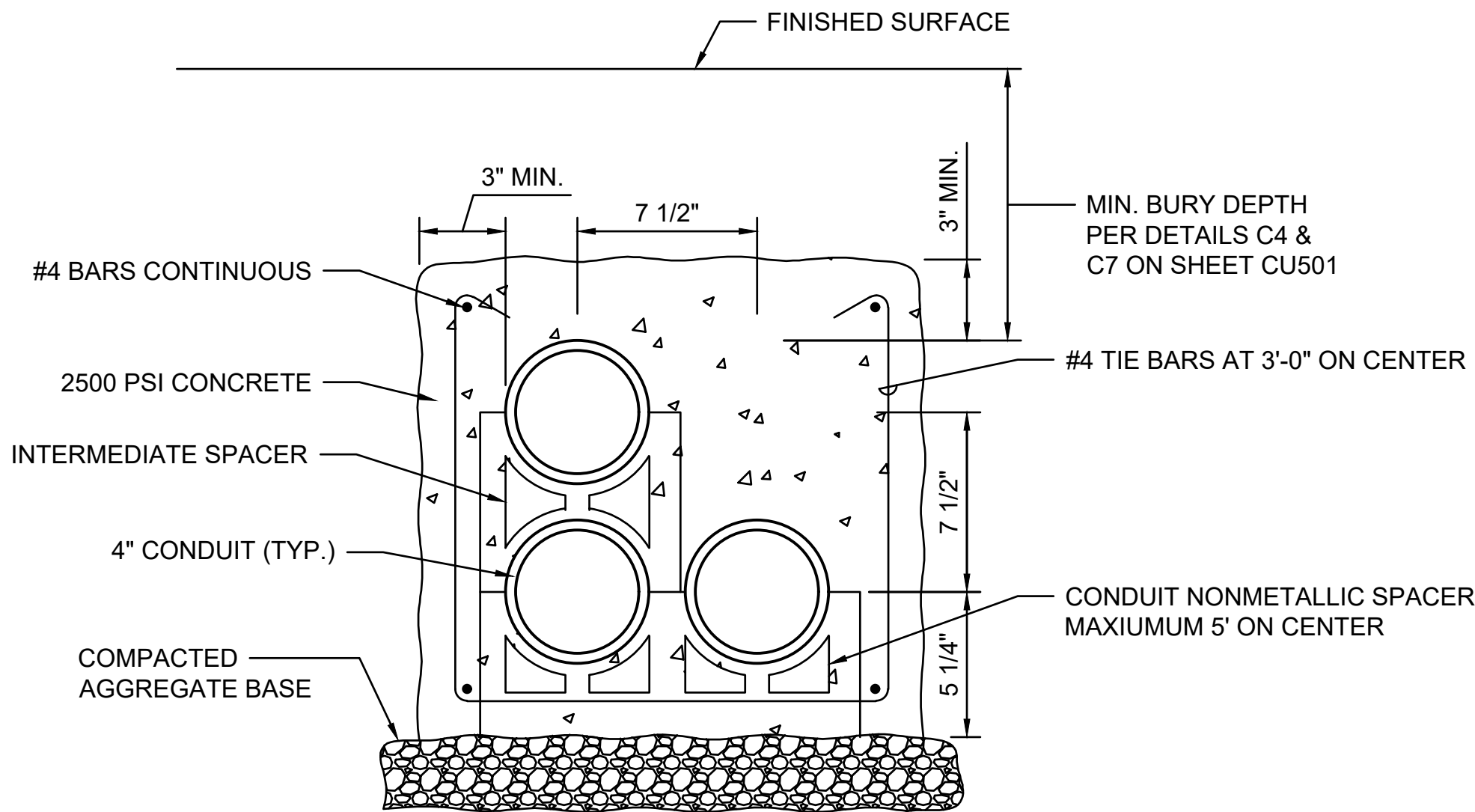
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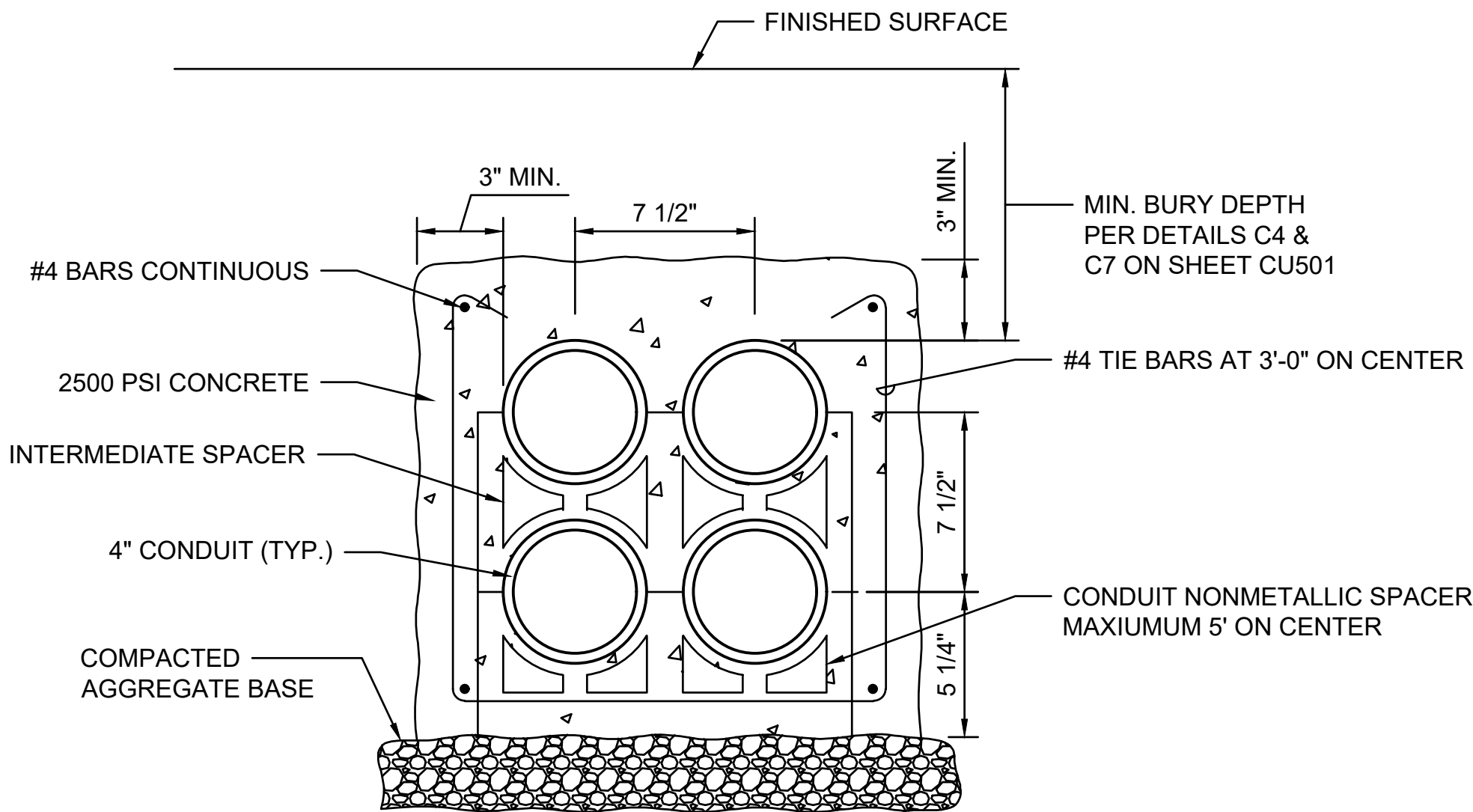


NOTE:
ALL EXTERIOR ENCLOSURE LIDS TO BE MARKED AS "COMMUNICATIONS."
VERIFY AND COORDINATE EXACT PLACEMENT BEFORE INSTALLATION.

D3 MEDIUM HANDHOLE DETAIL (TYP)
CU500 SCALE: NTS



F3 TYPE 1 DUCTBANK (TYP)
CU500 SCALE: NTS



F6 TYPE 2 DUCTBANK (TYP)
CU500 SCALE: NTS

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Revisions:	Date:

CONSULTANT

CIVIL ENGINEER:



WSP USA Solutions Inc.
1201 Pacific Avenue
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FAX: (206) 431-2250

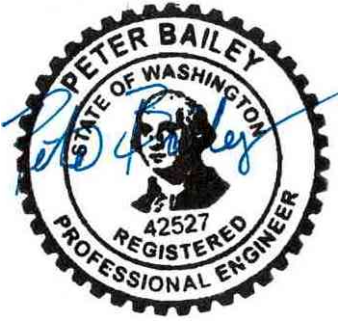
DESIGNER OF RECORD

A/E:



SPEED DESIGN BUILD
23830 PACIFIC HIGHWAY S, STE 300
KENT, WA 98032
(206) 209-2246

STAMP



Office of
Construction
and Facilities
Management

 U.S. Department
of Veterans Affairs

Drawing Title

COMMUNICATION DUCTBANK
DETAILS

Approved:

Phase

100% CONSTRUCTION
DOCUMENTS

Project Title

EHRM INFRASTRUCTURE
UPGRADES - HUNTINGTON

Location

VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date

09/03/2025

Checked

PB

Drawn

RDM

Project Number

581-22-700

Building Number

Drawing Number

CU500

018 OF 693

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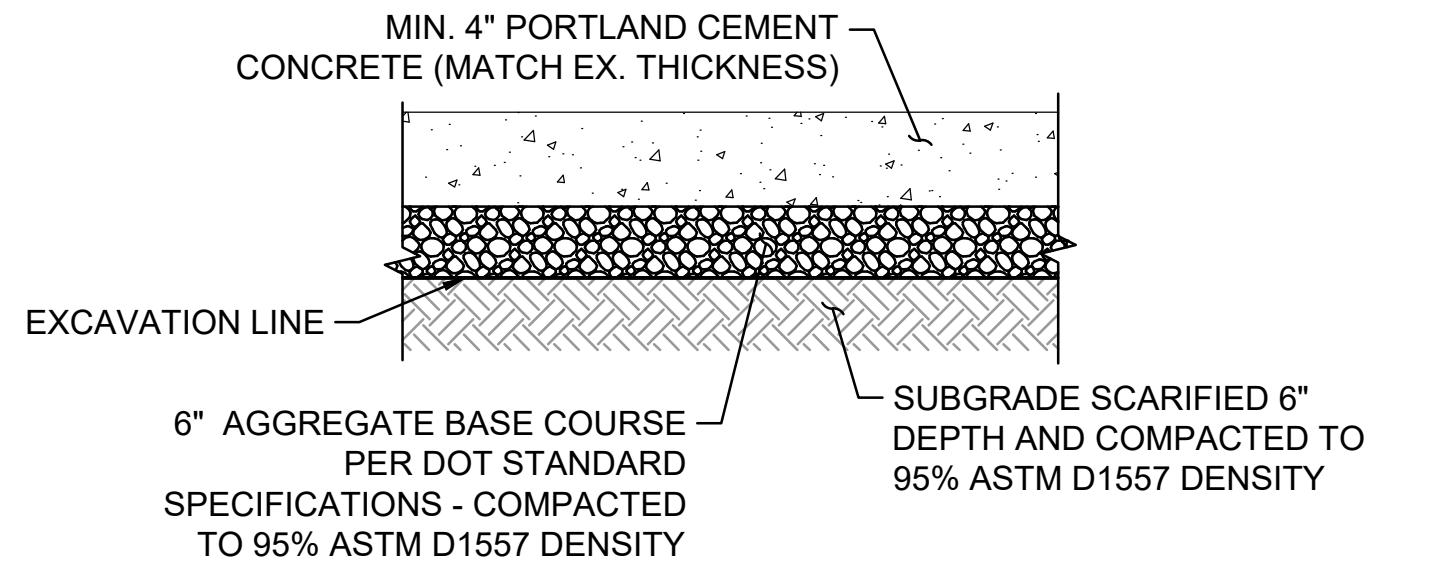
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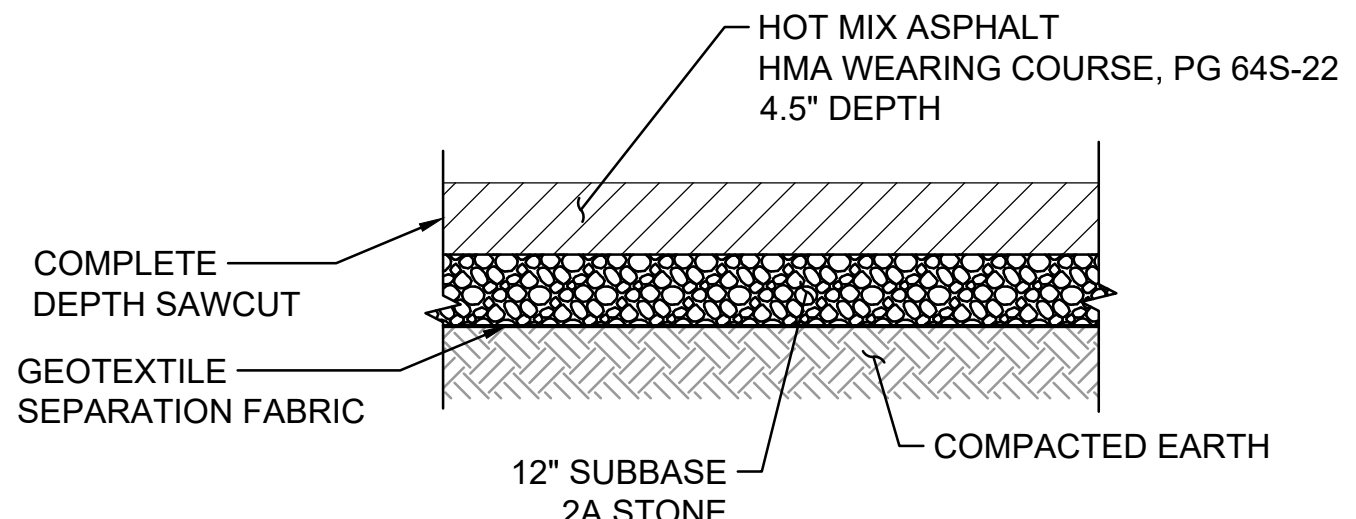
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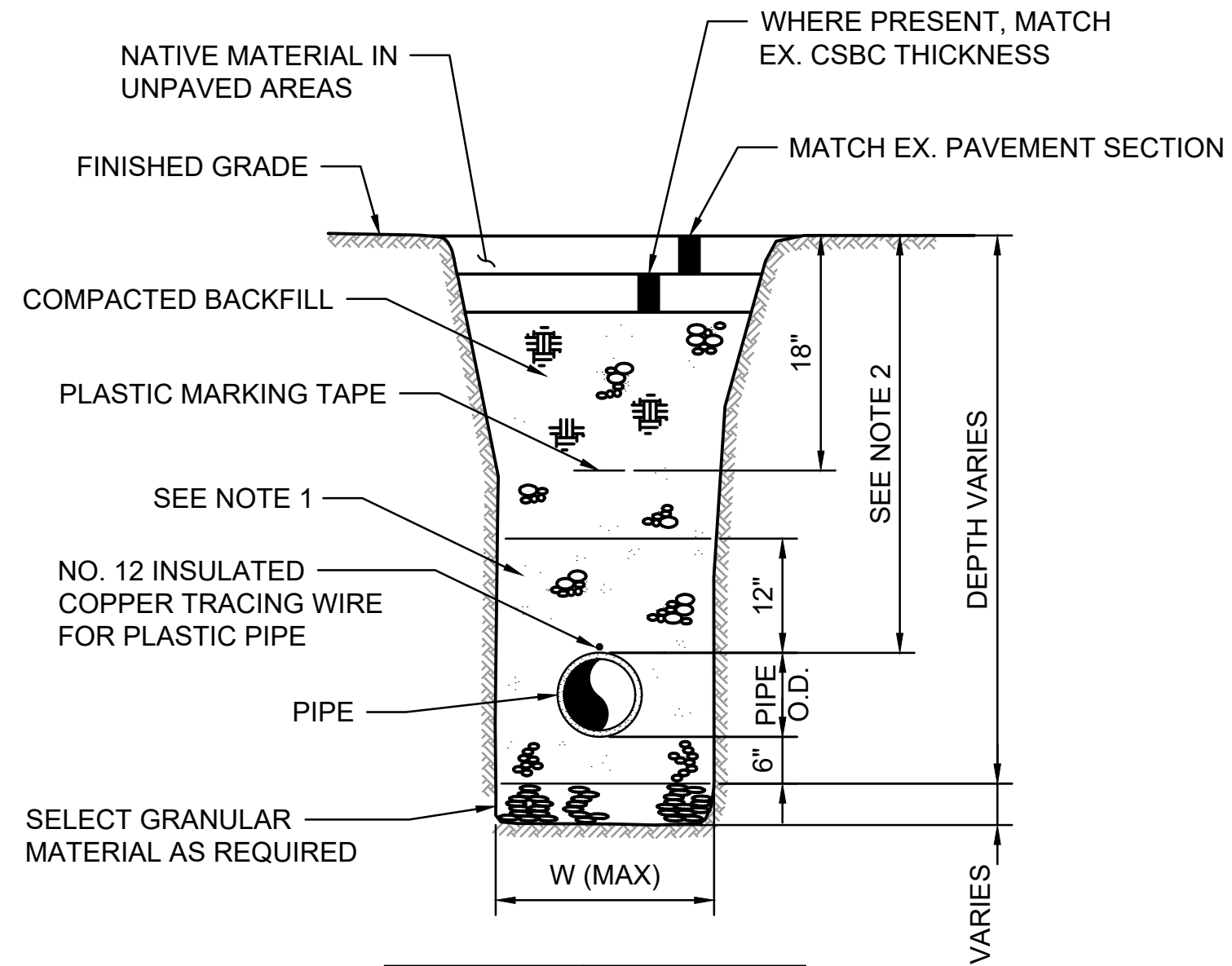
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B1 CONCRETE PAVEMENT SECTION
SCALE: NTS

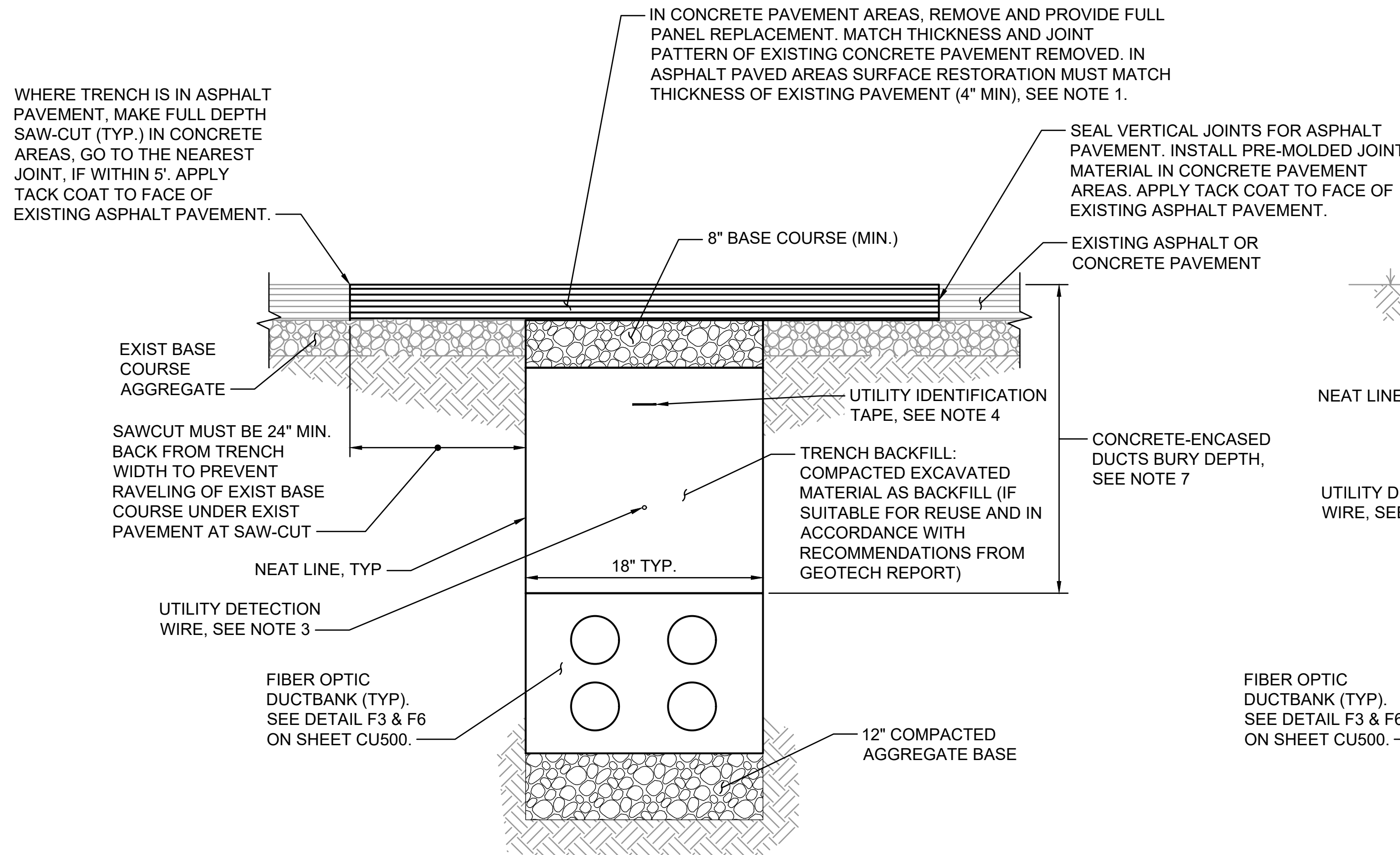


C1 ASPHALT PAVEMENT SECTION
SCALE: NTS

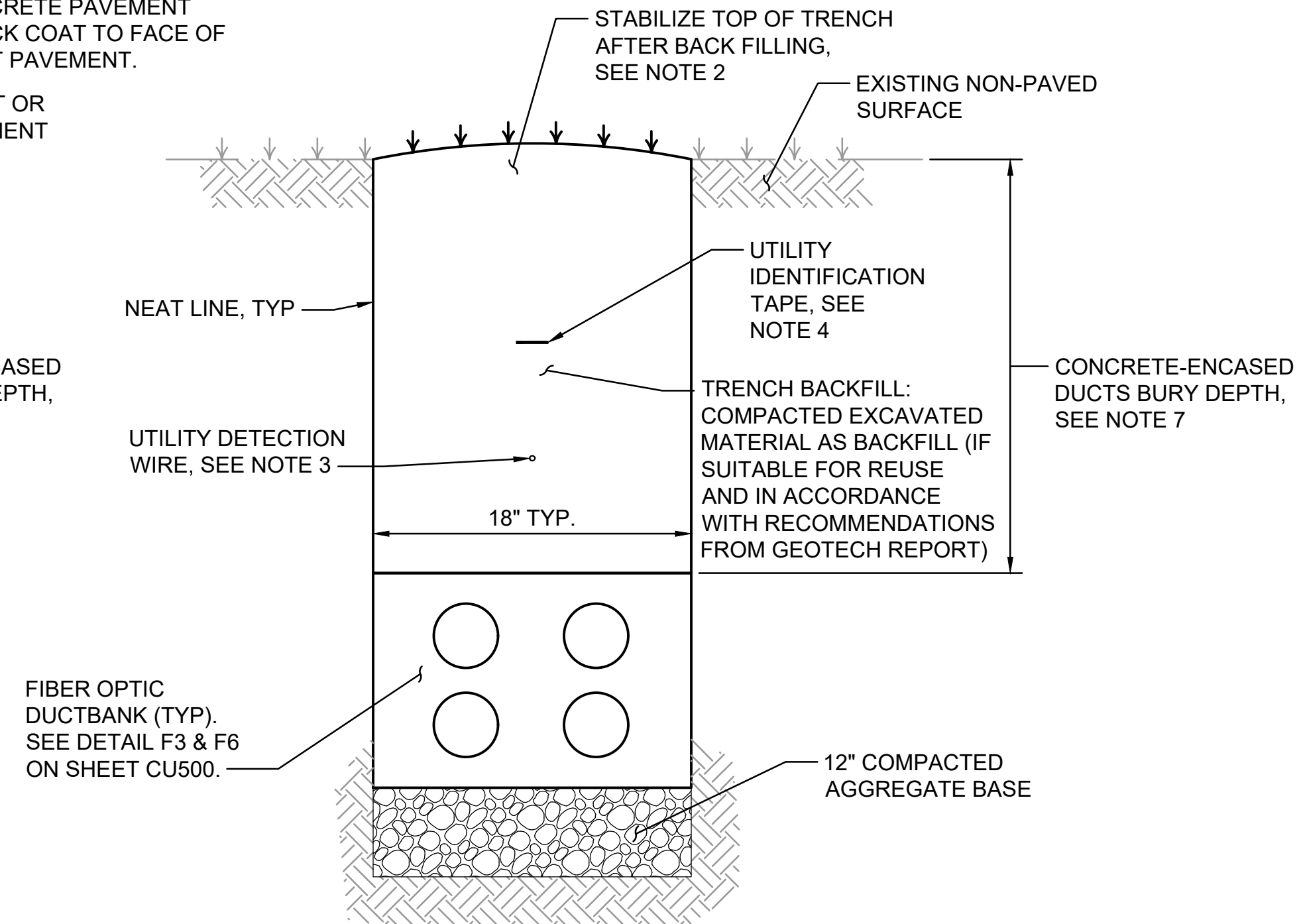


- NOTES:**
- MATERIAL MUST BE AS FOLLOWS:
 - RIGID PIPE: HAND-PLACED, COMPACTED INITIAL BACKFILL MATERIAL.
 - FLEXIBLE PIPE: HAND-PLACED, COMPACTED BEDDING MATERIAL. 4 INCHES FOR PIPE 18 INCH DIA. AND LESS.
 - CONTRACTOR TO MAINTAIN 48 INCHES MIN DEPTH OF COVER OVER WATER LINES PER NFPA 24 FIGURE A.10.4.1, AND 30 INCHES MIN COVER OVER OTHER UTILITIES UNLESS OTHERWISE NOTED ON PLANS.

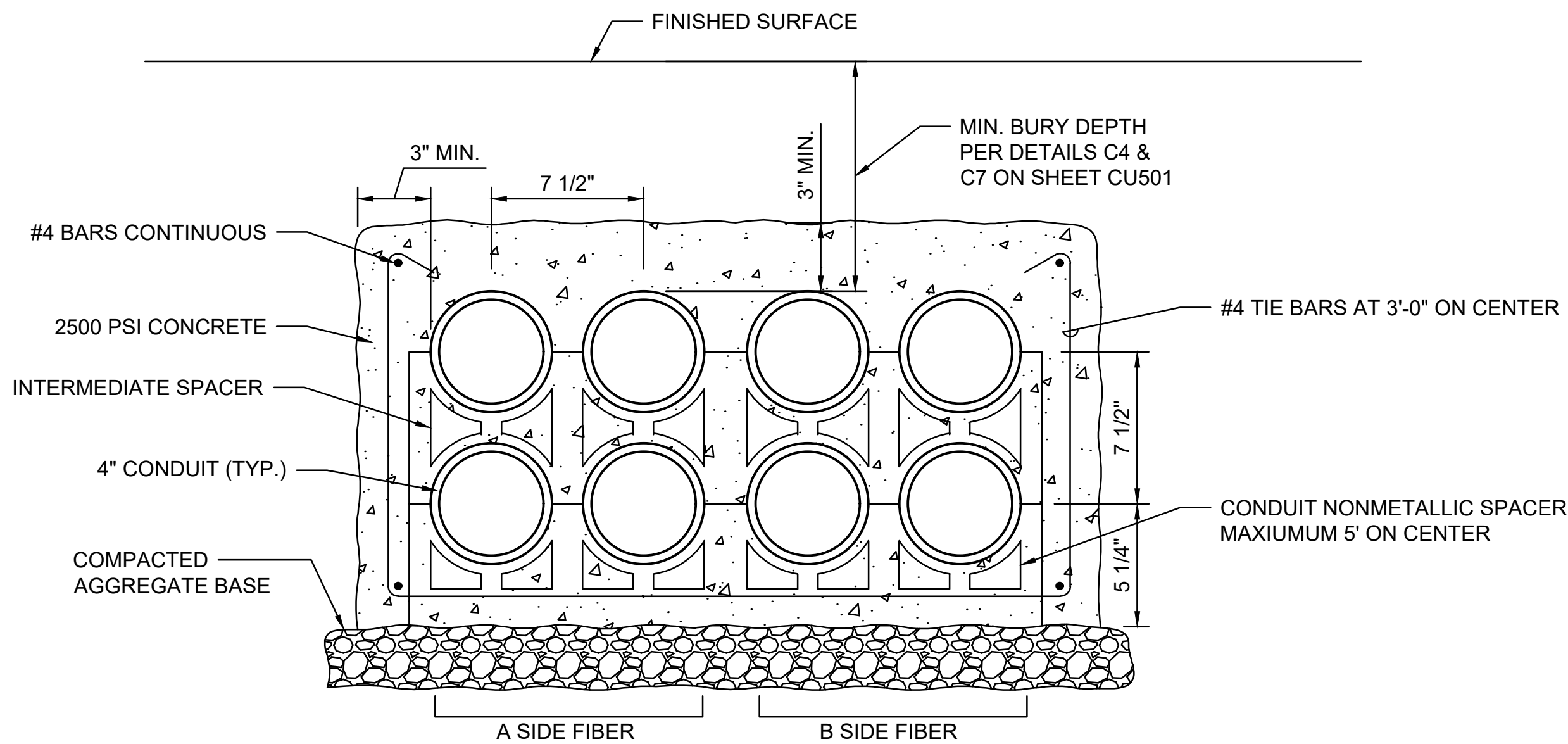
F1 UTILITY TRENCH SECTION
SCALE: NTS



C4 TRENCH IN PAVED AREAS (TYP.)
SCALE: NTS



C7 TRENCH IN NON-PAVED AREAS (TYP.)
SCALE: NTS

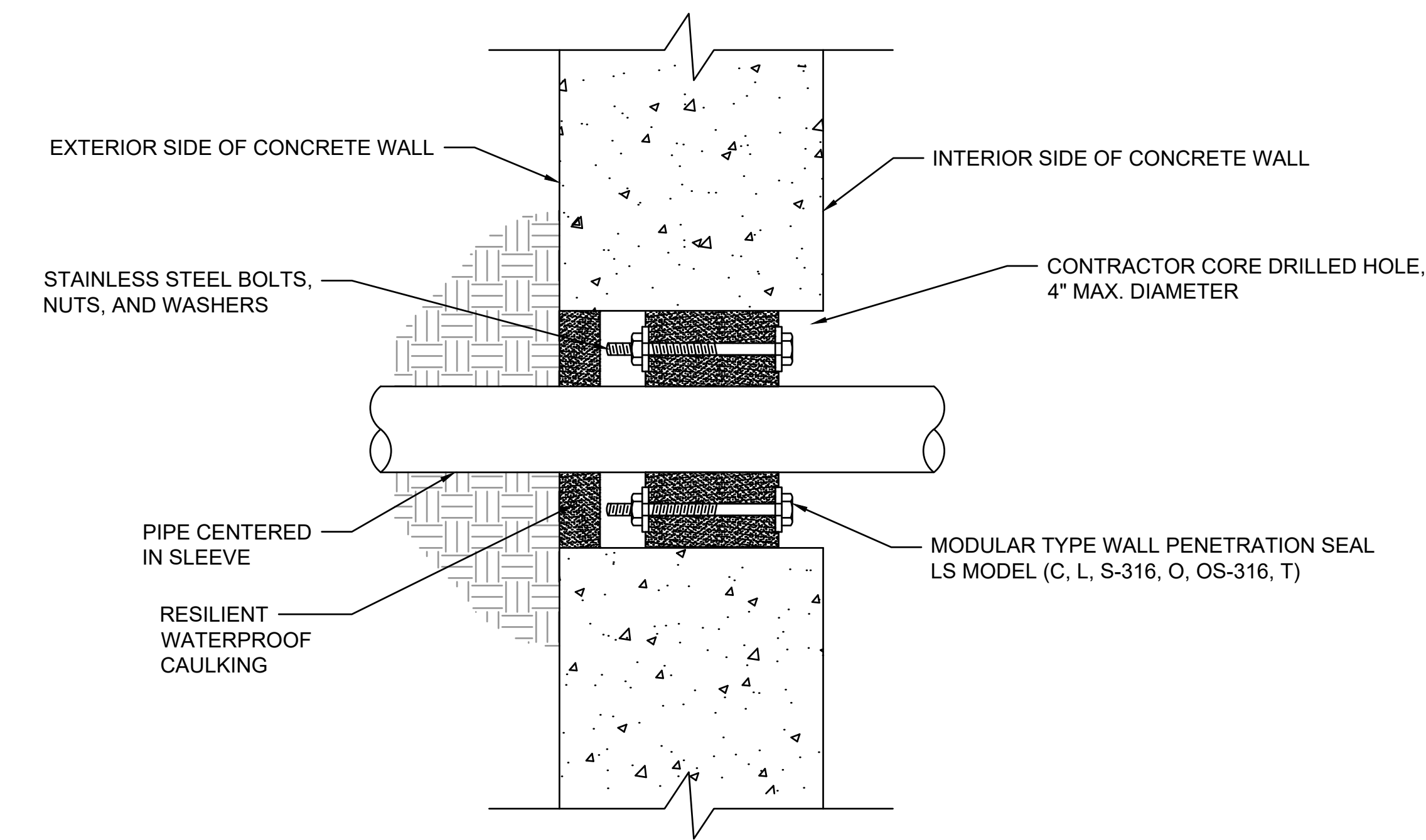


F4 JOINT COMMUNICATIONS DUCT BANK (TYP.)
SCALE: NTS

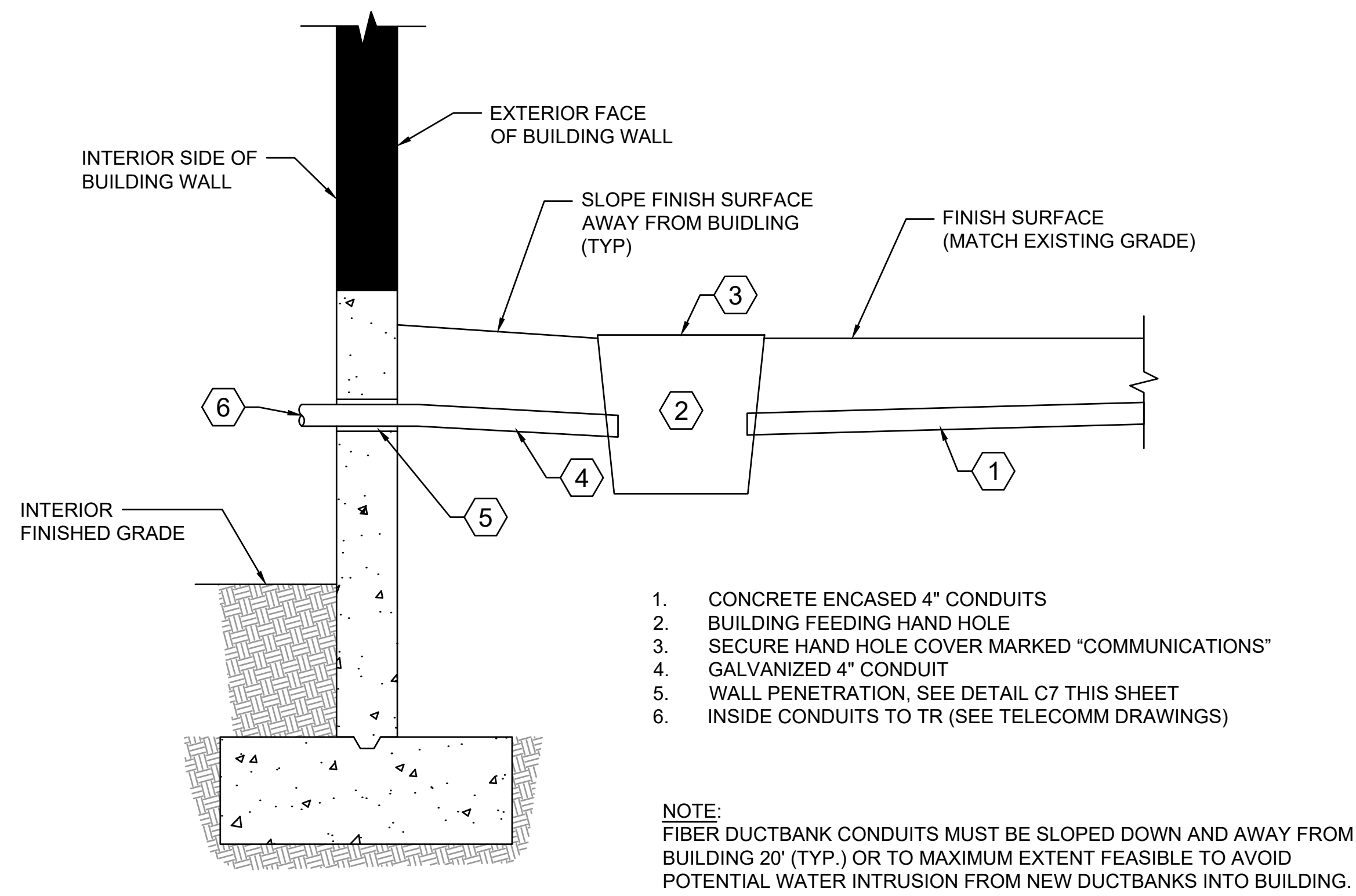
FIBER OPTIC DUCTBANK SURFACE RESTORATION NOTES:

- ALL DUCTBANK TRENCHING TO FOLLOW VA STANDARD SPECIFICATIONS.
- SURFACE RESTORATION IN EXISTING AREAS MUST MATCH EXISTING SURFACE. IN ASPHALT PAVING AREAS, HMA TO MATCH THICKNESS OF EXISTING PAVEMENT, BUT MUST NOT BE LESS THAN 4", PLACED OVER 8" OF BASE COURSE.
- FOR TRENCH IN NON-PAVED AREAS, PLACE 4" OF TOP SOIL SUCH THAT TOP IS FLUSH WITH EXISTING GRADE ON EACH SIDE AND SLIGHTLY MOUNDED BETWEEN. STABILIZE TOP OF TRENCH WITH GRASS SEED MIX AND MULCH, GRAVEL, BARK, OR WHATEVER MATERIAL MATCHES EXIST SURROUNDING CONDITIONS.
- UTILITY IDENTIFICATION TAPE MUST BE BURIED 12" BELOW FINISHED GRADE OR 6" BELOW TOP OF SUBGRADE IF UNDERNEATH PAVEMENT. DETECTION WIRE MUST BE BURIED DIRECTLY ABOVE NON-METALLIC PIPING AND 12" MAX. ABOVE TOP OF PIPE.
- TRENCH SAFETY SYSTEMS (SHORING) INCLUDING TRENCH BOXES MUST BE USED AS APPLICABLE AND AS DESCRIBED BY OSHA TO MAINTAIN VERTICAL SIDES FOR TRENCH AND MINIMIZE AMOUNT OF EXCAVATION, BACKFILL, AND SURFACE REPAIR. LAYING BACK TRENCH EXCAVATION (FLATTER SIDE SLOPES) IN LIEU OF SHORING MUST ONLY BE DONE IF APPROVED BY CONTRACTING OFFICER.
- DEWATER TRENCH AS NECESSARY FOR DEEPER EXCAVATIONS. DEWATERING AND THAWING AT CONTRACTORS SOLE EXPENSE IN BOTH TIME AND DOLLARS.
- TOP OF CONCRETE-ENCASED DUCTS MUST BE: 24" MIN. BELOW FINISHED GRADE IN NON-PAVED AREAS OR 30" MIN. BELOW PAVEMENT IN PAVED AREAS, AS WELL AS 6" MIN. BELOW FROST DEPTH.

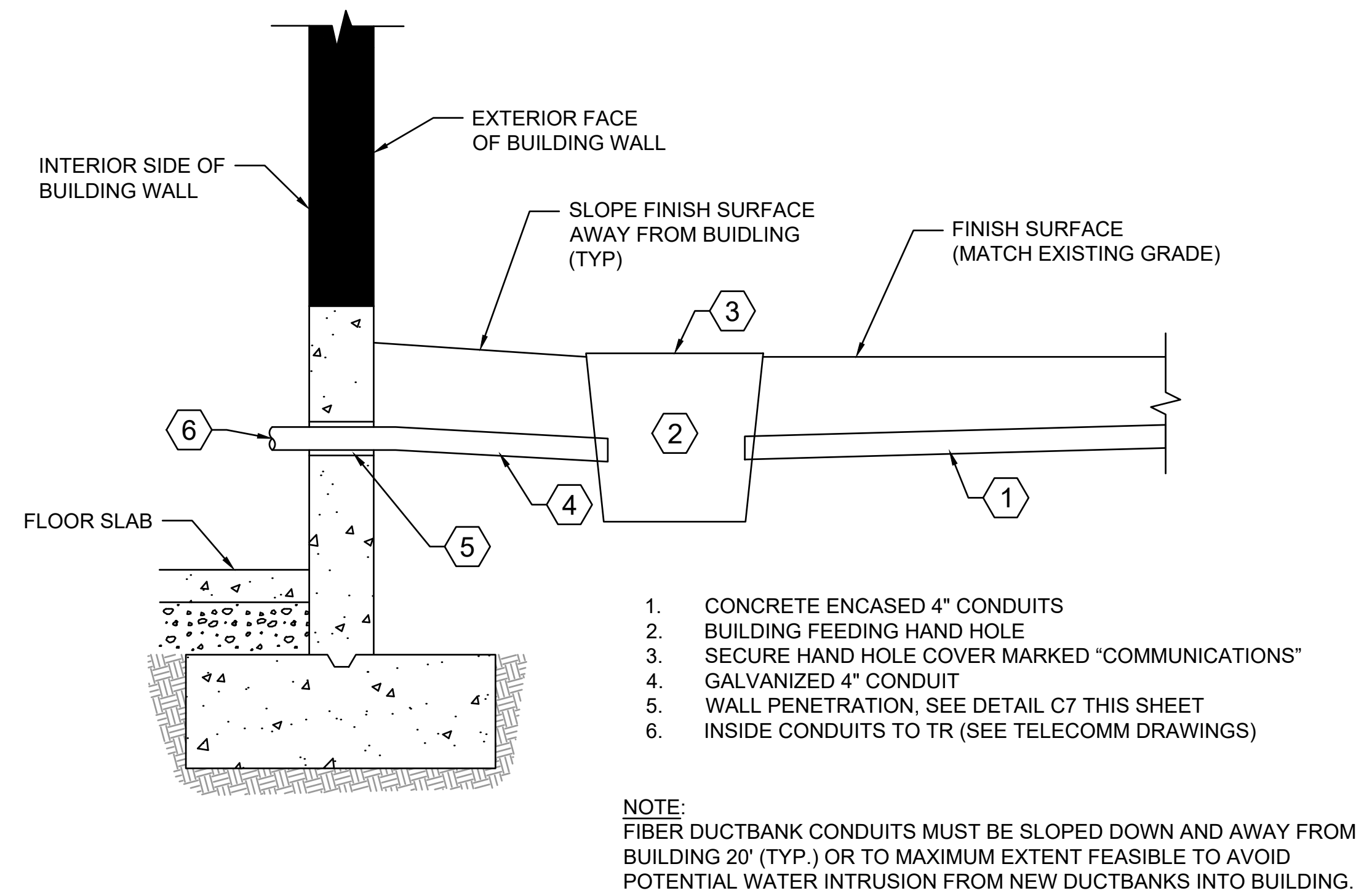
		CONSULTANT <u>CIVIL ENGINEER:</u>  WSP USA Solutions Inc. 1201 Pacific Avenue Suite 550 Tacoma, WA 98402 TEL: (206) 431-2300 FAX: (206) 431-2250		DESIGNER OF RECORD <u>A/E:</u> SPEES DESIGN BUILD 23830 PACIFIC HIGHWAY S, STE 300 KENT, WA 98032 (206) 209-2246  SPEESDESIGNBUILD		STAMP 	Office of Construction and Facilities Management  U.S. Department of Veterans Affairs		Drawing Title COMMUNICATION TRENCHING DETAILS Approved:		Phase 100% CONSTRUCTION DOCUMENTS		Project Title EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON Location VA HUNTINGTON 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300		Project Number 581-22-700 Building Number	
Revisions:		Date:										Issue Date 09/03/2025 Checked PB Drawn RDM		Drawing Number CU501 019 OF 693		



C7
CU502
PIPE PENETRATION THROUGH WALLS BELOW GRADE (TYP.)
SCALE: NTS



F2
CU502
UNDERGROUND CRAWL SPACE BUILDING ENTRY
SCALE: NTS



F7
CU502
UNDERGROUND BASEMENT ENTRY
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CODES AND STANDARDS

MATERIAL SPECIFIC STANDARDS ARE LISTED IN THE RESPECTIVE MATERIAL SECTION.

1. VA SEISMIC DESIGN REQUIREMENTS, H-16-8, JULY 1ST 2023

2. 2021 INTERNATIONAL BUILDING CODE (IBC) WITH STATE AND LOCAL AMENDMENTS

DESIGN LOADS

REFERENCES:

ASCE 7-16 MINIMUM DESIGN LOADS FOR BUILDINGS AND OTHER STRUCTURES
VA STRUCTURAL DESIGN MANUAL PG-18-10 OCTOBER 1ST 2022, REVISED JULY 1, 2023

DEAD LOADS

WEIGHT OF STRUCTURAL ELEMENTS AND PERMANENT FIXED EQUIPMENT.

SUPERIMPOSED DEAD LOADS

LOCATION	CATEGORY	UNIFORM LOAD (PSF)
ROOF	ROOFING MATERIALS	18
WALLS	EXTERIOR WALL SKIN	15
WALLS	PARTITIONS	20 (10 LATERAL)

LIVE AND ROOF LIVE LOADS

OCCUPANCY OR USE		UNIFORM LOAD (PSF)	CONCENTRATED LOAD (LBS)
CORRIDORS	FIRST FLOOR	100	
DATA CENTER		250	3500 (16 sq ft)
MECHANICAL ROOM		150	2000 (2.5 sq ft)
TOWER FLOORS		125	2000 (2.5 sq ft) 2300 (16 sq ft)
OFFICE	OFFICES FILE AND COMPUTER ROOMS	80 125	2000 (2.5 sq ft) 2000 (2.5 sq ft)
STAIRS & EXITS		100	300
ROOF		20	300
ROOF STAIRS AND PLATFORM		40	300

SNOW LOAD

DESIGN PARAMETER	VALUE
SNOW LOAD IMPORTANCE FACTOR, I _s	1.2
GROUND SNOW LOAD, P _g	20 PSF
SNOW LOAD ON FLAT ROOFS, P _f	24 PSF
EXPOSURE CATEGORY	C
SNOW LOAD EXPOSURE FACTOR, C _e	1.1
THERMAL FACTOR, C _t	1.0

WIND LOAD

REFER TO THE COMPONENTS AND CLADDING DIAGRAMS AT THE END OF THE GENERAL NOTES SECTION.

DESIGN PARAMETER	VALUE
BASIC WIND SPEED (3-SEC GUST), V	119 MPH
WIND EXPOSURE CATEGORY	C
KZT	1.62
ENCLOSURE CLASSIFICATION	ENCLOSED
INTERNAL PRESSURE COEFFICIENT	±0.18

DEFERRED SUBMITTALS

1. DEFERRED COMPONENT SUBMITTALS ARE INTENDED TO BE VENDOR DESIGNED AND HAVE NOT BEEN DESIGNED BY THE ARCHITECT/ENGINEER OF RECORD. THE COMPONENT DESIGNER IS RESPONSIBLE FOR CODE CONFORMANCE AND NECESSARY CONNECTIONS NOT CALLED OUT ON ARCHITECTURAL OR STRUCTURAL DRAWINGS.

2. DEFERRED SUBMITTALS INCLUDING CALCULATIONS MUST BE STAMPED AND SIGNED BY A REGISTERED PROFESSIONAL ENGINEER. DEFERRED SUBMITTALS MUST FIRST BE SUBMITTED TO THE ENGINEER OF RECORD FOR A CURSORY REVIEW FOR LOADS IMPOSED ON THE BASIC STRUCTURE.

3. A SUBMITTAL MAY THEN BE MADE BY THE CONTRACTOR TO THE CONTRACTING OFFICER'S REPRESENTATIVE TO SUBMIT TO THE AUTHORITY HAVING JURISDICTION FOR REVIEW AND APPROVAL.

4. THE LIST OF ITEMS FOR DEFERRED SUBMITTALS INCLUDES THE FOLLOWING ITEMS:

A. MICROPILES
B. MCR ROOF STAIRS AND PLATFORM
C. SEISMIC RESTRAINT FOR NON-STRUCTURAL COMPONENTS.
D. TEMPORARY ROOF OVER DATA CENTER.

5. SUBMITTAL MUST INCLUDE:

A. CALCULATIONS.
B. A DIAGRAM SHOWING LOAD MAGNITUDES AND LOCATIONS, SEPARATED INTO DEAD, LIVE, WIND AND/OR SEISMIC COMPONENTS - THAT ARE APPLIED TO THE PRIMARY STRUCTURE.

SEISMIC LOAD

DESIGN PARAMETER	VALUE
SEISMIC IMPORTANCE FACTOR, I _e	1.5
SITE LOCATION	LATITUDE: 38.3841745° N LONGITUDE: -82.5165789° E
SOIL SITE CLASS	C
MAPPED SPECTRAL ACCELERATION	SHORT PERIOD, S _s : 0.155 1-SECOND PERIOD, S ₁ : 0.069
DESIGN SPECTRAL ACCELERATION	SHORT PERIOD, S _{ds} : 0.134 1-SECOND PERIOD, S _{d1} : 0.069
SEISMIC DESIGN CATEGORY	C

TRANS AXIS

LONG AXIS

PRINCIPLE BUILDING AXES (NTS)

SEISMIC SYSTEM DESIGN	BOTH DIRECTION TOWER 3	BOTH DIRECTION MCR
SEISMIC FORCE RESISTING SYSTEM	STEEL ORDINARY MOMENT FRAME	LIGHT-FRAME (COLD-FORMED STEEL) WALLS SHEATHED WITH WOOD STRUCTURAL PANELS RATED FOR SHEAR RESISTANCE OR STEEL SHEETS
RESPONSE MODIFICATION FACTOR, R	3.5	6.5
OVERSTRENGTH FACTOR, Q ₀	3	2.5
DEFLECTION AMPLIFICATION FACTOR, C _d	3	4
SEISMIC WEIGHT, W	66 KIPS	36 KIPS
SEISMIC RESPONSE COEFF., C _s	0.082	0.038
DESIGN BASE SHEAR, V	4.1 KIPS	1.37 KIPS
ANALYSIS PROCEDURE	EQUIVALENT LATERAL FORCE ANALYSIS	EQUIVALENT LATERAL FORCE ANALYSIS

FOUNDATIONS

1. GROUND PREPARATION FOR THE SITE MUST BE IN ACCORDANCE WITH THE FINAL GEOTECHNICAL REPORT LISTED.

GEOTECHNICAL REPORT

REPORT TITLE	HUNTINGTON VA FIBER DUCTBANK
PREPARED BY	NGE
PROJECT NUMBER	W24044
DATED	APRIL 2024

2. THE CONTRACTOR MUST BE FAMILIAR WITH SUBSURFACE CONDITIONS AND RELEVANT ASPECTS OF THIS REPORT BEFORE COMMENCING EXCAVATION. THE CONTRACTOR MUST RETAIN A GEOTECHNICAL ENGINEER FOR INSPECTIONS, TO VERIFY CONDITIONS, OR TO PERFORM ADDITIONAL EXPLORATION, AS NECESSARY.

3. LOCATIONS OF SUBSURFACE SITE CONDITIONS, INCLUDING BURIED UTILITIES OR OBSTRUCTIONS AND OTHER FEATURES KNOWN TO THE DESIGN TEAM ARE IDENTIFIED IN DRAWINGS. VERIFY THE EXTENT AND LOCATIONS OF SITE UTILITIES AND OBSTRUCTIONS. REPORT ANY DISCREPANCY ENCOUNTERED AS SOON AS THEY ARE KNOWN TO THE CONTRACTING OFFICER.

4. FOOTING DESIGN IS BASED ON THE FOLLOWING ALLOWABLE SOIL PROPERTIES:

SOIL PROPERTY	LOAD CASE	VALUE
BEARING CAPACITY SHALLOW FOUNDATION: TOWER 3	SUSTAINED (INCL. W, E, OR BLAST)	5,000 PSF 6650 PSF
MCR	SUSTAINED (INCL. W, E, OR BLAST)	3000 PSF 3900 PSF
MODULUS OF SUBGRADE REACTION		100 PCF
COEFFICIENT OF FRICTION AT STRENGTH		0.40
FROST DEPTH		30"
MICROPILE GROUT IN BEDROCK SKIN FRICTION		5 KSF (ULTIMATE) DOWN 2.5 KSF (ULTIMATE) UPLIFT

5. EXTERIOR FOOTINGS MUST BEAR AT OR BELOW MINIMUM FROST DEPTH, UNLESS NOTED OTHERWISE.

6. MCR FOUNDATIONS TO BEAR ON BEDROCK OR OVER-EXCAVATE TO BEDROCK AND BACK FILL WITH 2000 PSI SLURRY BACKFILL. SEE GEOTECHNICAL REPORT FOR MORE INFORMATION.

7. SLAB ON GRADE TO BEAR ON EXISTING SOIL OR RECOMPENSED SOIL. SEE GEOTECHNICAL REPORT FOR MORE INFORMATION.

8. SPECIAL INSPECTION MUST BE PERFORMED DURING SITE EXCAVATION TO CONFIRM THAT EXISTING CONDITIONS COMPLY WITH CONDITIONS IN THE GEOTECHNICAL REPORT.

9. EXCAVATIONS MUST BE DEWATERED TO ALLOW FOR CASTING FOUNDATION CONCRETE IN DRY CONDITIONS.

10. USE FORMWORK TO CAST CONCRETE, SOIL FORMING CONCRETE WORK IS PROHIBITED.

11. USE A 15 MIL MIN VAPOR RETARDER AND A 4 INCH COMPACTED DRAINAGE LAYER BELOW SLABS-ON-GRADE, UNLESS NOTED OTHERWISE OR PER SPECIFIC REQUIREMENTS OF THE GEOTECHNICAL REPORT.

CAST IN PLACE CONCRETE

1. REFERENCES:

A. ACI 301-20 SPECIFICATIONS FOR CONCRETE CONSTRUCTION
B. ACI 318R-18 DETAILS AND DETAILING CONCRETE CONSTRUCTION
C. ACI 318-19 BUILDING CODE REQUIREMENTS FOR STRUCTURAL CONCRETE
D. ACI 347-14 GUIDE TO FORMWORK FOR CONCRETE

1. CONCRETE WORK MUST CONFORM TO ACI 301. FORMWORK MUST CONFORM TO ACI 347.

2. MINIMUM CONCRETE PROPERTIES AT 28-DAYS PER THE FOLLOWING TABLE.

LOCATION	MIN 28-DAY COMPRESSIVE STRENGTH (PSI)	ACI 318, CH. 19 EXPOSURE CATEGORIES						
		FREEZING & THAWING	SULFATE	WATER CONTACT	CORROSION PROTECTION	MAX WATER/CEMENT RATIO	CEMENT TYPE	NOMINAL MAX AGGREGATE SIZE
FOUNDATIONS	5,000	F2	S1	W1	C1	0.45	II	1"
CONC OVER METAL DECK (LT, WT, 115PCF MAX)	4,000	F0	S0	W0	C0	0.50	I OR II	3/4"
SLAB-ON-GROUND (INTERIOR)	4,000	F0	S0	W0	C0	0.50	I OR II	1 1/2"

3. CONCRETE EXPOSED TO THE ENVIRONMENT MUST HAVE 6% +/- 1% AIR ENTRAINMENT BY VOLUME PER ASTM C260.

4. PROVIDE A MUD SLAB (MINIMUM 2 INCHES OF CONCRETE, LEVELLED AT THE TOP SURFACE) UNDER SLABS AND FOOTINGS AT WET OR MUDDY GROUND SURFACES WHICH ARE UNSUITABLE FOR SUPPORTING REBAR CHAIRS, CONSTRUCTION WORKERS AND EQUIPMENT. PROVIDE WATERPROOF MEMBRANE TO TOP OF MUD SLAB AS REQUIRED.

5. HORIZONTAL CONSTRUCTION JOINTS IN SPREAD FOOTINGS, STRIP FOOTINGS, PILE CAPS, BEAMS AND SLABS ARE PROHIBITED UNLESS SHOWN ON THE DRAWINGS OR APPROVED BY THE SER.

6. THE EXTENT OF CONCRETE PLACEMENTS MUST BE LIMITED AS FOLLOWS, UNLESS NOTED OTHERWISE:

A. STRIP FOOTINGS AND WALLS: 60 FT (UNLESS INTERMEDIATE CONTROL JOINTS ARE PROVIDED).
B. SLABS-ON GROUND: 30 FT MAXIMUM LENGTH AND 900 SQ FT MAXIMUM AREA (UNLESS INTERMEDIATE CONTROL JOINTS ARE PROVIDED).

7. BEFORE CASTING THE SECOND SIDE OF A CONSTRUCTION JOINT, ROUGHEN SURFACE OF CONCRETE ELEMENTS AT CONSTRUCTION JOINTS TO A MINIMUM AMPLITUDE OF 1/4 INCH, UNLESS NOTED OTHERWISE. CLEAN AND APPLY BONDING AGENT TO THE HARDENED CONCRETE SURFACE RECEIVING FRESH CONCRETE.

8. SECTIONS AND DETAILS MAY NOT SHOW ALL REQUIRED DEFORMED BAR REINFORCEMENT FOR CLARITY. ADDITIONAL REINFORCEMENT MAY BE DESCRIBED IN PLANS, SCHEDULES, NOTES, OR OTHER DETAILS, AS APPLICABLE.

9. COORDINATE FLOOR FINISHES, SLAB SLOPES & DEPRESSIONS, FLOOR DRAINS, CURBS, CONCRETE PADS, AND OTHER CONCRETE FEATURES IDENTIFIED IN THE CONTRACT DOCUMENTS.

10. THE STRUCTURAL DRAWINGS DO NOT NECESSARILY SHOW ALL ITEMS EMBEDDED IN THE CONCRETE. COORDINATE AND PROVIDED EMBEDDED ITEMS AS SHOWN IN THE CONTRACT DOCUMENTS AND AS REQUIRED BY THE EQUIPMENT MANUFACTURERS, AS NEEDED.

11. NON-STRUCTURAL STEEL MEMBERS EMBEDDED IN CONCRETE MUST BE HOT-DIP GALVANIZED OR PAINTED. DAMAGED COATINGS OF EMBEDDED ITEMS MUST BE REPAIRED PRIOR TO CASTING CONCRETE.

12. PENETRATIONS THROUGH CONCRETE WALLS OR SLABS MUST CONFORM TO STANDARD DETAIL DRAWINGS.

13. PROVIDE 3/4" CHAMFER ON EXPOSED CONCRETE EDGES UNLESS NOTED OTHERWISE.

GENERAL CONDITIONS

1. CONFLICTS BETWEEN THE DRAWINGS AND SPECIFICATIONS MUST BE BROUGHT TO THE ATTENTION OF THE CONTRACTING OFFICER AND STRUCTURAL ENGINEER OF RECORD (SER) IN WRITTEN FORM AND RESOLVED BEFORE PROCEEDING WITH AFFECTED WORK.

2. STRUCTURAL DRAWINGS REPRESENT THE COMPLETED STRUCTURE. THE CONTRACTOR IS RESPONSIBLE FOR THE MEANS AND METHODS NECESSARY TO COMPLETE THE WORK, FOR JOB SITE SAFETY AND FOR COORDINATING WITH OTHER DISCIPLINES AS REQUIRED.

3. THE CONTRACTOR IS RESPONSIBLE FOR THE ERECTION PROCEDURE AND SEQUENCING TO ENSURE STABILITY OF THE STRUCTURE DURING CONSTRUCTION. PROVIDE SHORING, BRACING, AND FALSEWORK AS NEEDED TO PROTECT THE STRUCTURE THROUGHOUT CONSTRUCTION.

4. ERECTION AIDS FOR FACILITATING CONSTRUCTION NOT SHOWN IN THESE DRAWINGS MUST BE REMOVED AND ELEMENTS RESTORED TO ITS ORIGINAL DESIGN INTENT SHOWN PER THESE DRAWINGS.

5. STRUCTURAL DRAWINGS SHOW GENERAL AND TYPICAL DETAILS OF CONSTRUCTION, WHERE SPECIFIC CONDITIONS ARE NOT SHOWN, DETAILS OF SIMILAR TYPE AND CHARACTER MAY BE USED BUT REQUIRE APPROVAL FROM THE SER.

6. TYPICAL DETAILS MAY NOT BE REFERENCED FROM THE PLANS BUT APPLY FOR THE CONDITIONS AS DESCRIBED BY THE DETAIL.

7. CONTRACTOR-INITIATED CHANGES REQUIRE APPROVAL FROM THE SER.

8. FOR A CONTRACTOR REQUESTED DESIGN REVISION, THE PROPOSED ALTERNATIVE MUST BE SUBMITTED TO THE CONTRACTING OFFICER FOR REVIEW AND APPROVAL BY THE SER. THE SUBMITTAL MUST INCLUDE DRAWINGS, STRUCTURAL DETAILS AND RELATED CALCULATIONS, STAMPED AND SIGNED BY A PROFESSIONAL ENGINEER LICENSED TO PRACTICE IN THE JURISDICTION OF THE PROJECT LOCATION. CALCULATIONS MUST DEMONSTRATE COMPLIANCE WITH THE CODES AND DESIGN REQUIREMENTS LISTED PER THESE DRAWINGS, AND THAT THE ALTERNATIVE DESIGN DOES NOT ADVERSELY AFFECT THE STRUCTURE SHOWN IN THESE DRAWINGS.

9. CONTRACTOR MUST COORDINATE THESE DRAWINGS WITH THE DRAWINGS OF THE OTHER DISCIPLINES AND SPECIFICATIONS FOR ADDITIONAL INFORMATION NOT SHOWN ON THE STRUCTURAL DRAWINGS. EXAMPLES INCLUDE EMBEDDED ANCHORS, SLEEVES AND INSERTS, MISCELLANEOUS ATTACHMENT DETAILS, FLOOR FINISHES, DOOR THRESHOLDS, SLOPES TO DRAINS, OPENINGS IN STRUCTURAL ELEMENTS, ETC.

10. VERIFY DIMENSIONS AND DETAILS PRIOR TO FABRICATION AND CONSTRUCTION. THIS INCLUDES COORDINATING THE SIZE AND LOCATION OF OPENINGS THROUGH STRUCTURAL ELEMENTS WITH THE RESPECTIVE SUBCONTRACTOR, AS WELL AS COORDINATING INSERTS AND ATTACHMENTS FOR USE BY OTHER TRADES.

11. NO PIPES OR DUCTS MAY BE PLACED IN STRUCTURAL MEMBERS UNLESS DETAILED BY THE STRUCTURAL PLANS OR APPROVED BY THE SER.

12. DIMENSIONS MUST NOT BE SCALED FROM THE DRAWINGS.

13. PROVIDE SUBMITTALS AND (FABRICATION, AND ERECTION, ETC.) SHOP DRAWINGS FOR THE WORK SHOWN IN THESE DRAWINGS TO THE CONTRACTING OFFICE FOR APPROVAL FROM THE SER. WORK MUST NOT PROCEED WITHOUT REVIEW AND APPROVAL BY THE SER.

14. SUBMIT A DETAILED PLAN OF THE CONSTRUCTION INCLUDING A DESCRIPTION OF DEMOLITION, CONSTRUCTION METHODS, ERECTION SEQUENCING, METHODS OF STABILIZING THE STRUCTURE, SHORING LAYOUT, ELEMENT SIZES, CONNECTIONS AND ACCESSORIES WITH SUPPORTING CALCULATIONS STAMPED AND SIGNED BY A REGISTERED PROFESSIONAL ENGINEER, TO THE CONTRACTING OFFICER FOR REVIEW AND APPROVAL PRIOR TO BEGINNING WORK.

15. SHOP DRAWINGS MUST BE SUBMITTED TO THE CONTRACTING OFFICER FOR REVIEW AND APPROVAL PRIOR TO FABRICATION. UNLESS NOTED OTHERWISE, PROVIDE SHOP DRAWINGS FOR:

A. DEFORMED BAR REINFORCING LAYOUT FOR CONCRETE AND MASONRY.
B. STRUCTURAL AND MISCELLANEOUS STEEL.
C. COLD-FORMED STEEL FRAMING.
D. ROOF METAL.
E. MICROPILES.

16. BY SUBMITTING FOR ENGINEER'S REVIEW, THE CONTRACTOR CONFIRMS THE ACCURACY OF THE SUBMITTALS FOR DIMENSIONS, QUANTITIES, ELEMENT SIZES, CONSTRUCTION MATERIALS, FIELD MEASUREMENTS AND THAT THE CONTENTS HAVE BEEN REVIEWED FOR MEANS, METHODS, AND OPERATIONS OF CONSTRUCTION, AND FOR ITS ADHERENCE TO SAFETY PROTOCOLS AT THE SITE.

17. SHOP DRAWING REVIEW DOES NOT RELIEVE THE CONTRACTOR FROM HAVING SOLE RESPONSIBILITY FOR THE ACCURACY OF ELEMENT SIZES, DIMENSIONS AND DETAILS. SUCH REVIEW DOES NOT CONSTITUTE ACCEPTANCE BY THE ENGINEER OF THE CORRECTNESS OR ADEQUACY OF SUCH SUBMITTALS, NOR IS A WARRANTY THAT THE SUBMITTALS SATISFY THE REQUIREMENTS OF THE CONTRACT.

REINFORCING STEEL

1. REFERENCES:

A. ACI 301-20 SPECIFICATIONS FOR CONCRETE CONSTRUCTION.
B. ACI SP-66-04 ACI DETAILING MANUAL.
C. ACI 318R-18 DETAILS AND DETAILING CONCRETE CONSTRUCTION.
D. AWS D14.2:018 STRUCTURAL WELDING CODE -- REINFORCING STEEL.

2. STEEL REINFORCEMENT MUST CONFORM TO:

CONDITION	STANDARD AND GRADE	MIN YIELD (KSI)
TYPICAL DEFORMED BAR	ASTM A615 GR 60 OR ASTM A706	60

3. PROVIDE BAR SUPPORTS AND SPACERS PER ACI 315 UNLESS NOTED OTHERWISE. DEFORMED BAR REINFORCEMENT AND INSERTS MUST BE SECURED IN-PLACE PRIOR TO PLACING CONCRETE. DO NOT PLACE REBAR MATS DIRECTLY AGAINST SOIL. REBAR MATS ADJACENT TO SOIL OR OVER MUD SLABS MUST BE SUPPORTED BY CHAIRS, DO NOT LIFT INTO PLACE DURING POURING.

4. UNLESS INDICATED OTHERWISE, REINFORCEMENT BARS ARE CONTINUOUS AND MUST BE LAP SPLICED OR COUPLED WITH MECHANICAL CONNECTORS. LAP LENGTH NOTES:

A. SEE DEVELOPMENT AND LAP SPICE LENGTH SCHEDULES FOR REINFORCEMENT DETAILS.
B. LAP SPICE LENGTH MUST BE USED FOR THE LARGER BAR SIZE WHEN BAR SIZE TRANSITIONS AT LAP.
C. OVERLAP WELDED WIRE LAYER REINFORCEMENT A LENGTH OF 1.5X WIRE DEVELOPMENT LENGTH OR ONE CROSSWIRE SPACING PLUS 2 INCHES, WHICHEVER IS LARGER.
D. MECHANICAL CONNECTORS MUST HAVE A MINIMUM CAPACITY OF 1.25 X YIELD STRENGTH OF THE BAR.
E. MECHANICAL CONNECTORS IN MEMBERS DESIGNATED AS PART OF THE SEISMIC-FORCE-RESISTING SYSTEM MUST HAVE A MINIMUM CAPACITY EQUAL TO THE TENSILE STRENGTH OF THE BAR.

5. TYPE 1 MECHANICAL CONNECTORS MUST DEVELOP 125% OF THE YIELD STRENGTH (F_y) OF THE STRONGER OF THE TWO BAR BEING SPLICED. TYPE 2 MECHANICAL CONNECTORS MUST DEVELOP THE FULL TENSILE STRENGTH OF THE STRONGER OF THE TWO BARS BEING SPLICED. WHERE MECHANICAL CONNECTORS ARE USED IN LIEU OF LAP SPLICES, PROVIDE TYPE 2 MECHANICAL CONNECTORS EXCEPT WHERE TYPE 1 ARE PERMITTED AS INDICATED.

6. HEADED DEFORMED BARS (TERMINATORS) MUST BE INTEGRALLY FORMED OR T-THREADED DEVICES MEETING THE REQUIREMENTS OF ACI 318, SECTION 25.4.4.1.

7. UNLESS NOTED OTHERWISE ON THE DRAWINGS:

A. DEFORMED BAR COVER MUST CONFORM TO SPECIFIED CONCRETE COVER REQUIREMENTS OF ACI 318.
B. REINFORCEMENT MUST BE CONTINUOUS THROUGH CONSTRUCTION JOINTS.
C. APPROVAL FROM SER REQUIRED BEFORE WELDING NON-ASTM A706 Gr. 60 REINFORCEMENT.
D. APPROVAL FROM SER REQUIRED BEFORE FIELD BENDING REINFORCEMENT PARTIALLY EMBEDDED CONCRETE.
E. MATCH DOWELS TO THE LARGER OF MULTIPLE BAR SIZES.

8. DOWELS MUST BE SECURED WITH A RIGID TEMPLATE TO ENSURE SPLICE IS LOCATED INSIDE THE CONFINED REGION (REGION ENCLOSED BY SHEAR OR HORIZONTAL REINFORCEMENT) OF CROSS-SECTION.

Revisions:

Date:

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Office of
Construction
and Facilities
Management

VA U.S. Department
of Veterans Affairs

Drawing Title

STRUCTURAL GENERAL NOTES

Approved:

Phase

100% CONSTRUCTION
DOCUMENTS

Project Title

EHRM INFRASTRUCTURE
UPGRADES - HUNTINGTON

Location
VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date
09/03/2025

Checked
CH

Drawn
JKM

Project Number
581-22-700

Building Number

Drawing Number
S-001

021 OF 693

VA FORM 08 - 6231

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STRUCTURAL STEEL

1. REFERENCES:

A. AISC 303-22 CODE OF STANDARD PRACTICE FOR STEEL BUILDINGS AND BRIDGES.

B. AISC 360-16 SPECIFICATION FOR STRUCTURAL STEEL BUILDINGS.

C. AISC 341-16 SEISMIC PROVISIONS FOR STEEL BUILDINGS.

D. AISC RESEARCH COUNCIL ON STRUCTURAL CONNECTIONS (RCSC) SPECIFICATION FOR STRUCTURAL JOINTS USING HIGH-STRENGTH BOLTS 2014.

2. STRUCTURAL STEEL WORK MUST BE PERFORMED BY AISC CERTIFIED FABRICATORS & ERECTORS.

3. STEEL CONSTRUCTION AND DESIGN AS PART A DEFERRED DESIGN SUBMITTAL MUST CONFORM TO AISC 360 AND AISC 303.

4. USE STRUCTURAL STEEL SHAPES (AS APPLICABLE PER THE DRAWINGS), WITH CROSS-SECTIONS PER ASTM A6, PER THE FOLLOWING TABLE.

SHAPE	STANDARD & GRADE (UNO)	MIN YIELD (KSI)
WIDE FLANGES (W) AND TEES (WT)	ASTM A992	50
CHANNELS (C & MC) AND ANGLES (L)	ASTM A36	36
PLATE AND FLAT BAR STOCK (PL)	ASTM A36 GR 36	36
HSS RECTANGULAR OR SQUARE ROUND	ASTM A500 GR C ASTM A500 GR C	50 46
PIPE	ASTM A53 GR B	35
OTHER STEEL	ASTM A572	50

5. PROVIDE PROTECTIVE COATING AT STEEL EXPOSED TO THE ENVIRONMENT AND WEATHER PER ARCHITECTURAL REQUIREMENTS.

6. MACHINE COLUMN ENDS AT SPLICES TO PROVIDE A FLUSH BEARING CONTACT SURFACE, UNLESS NOTED OTHERWISE.

7. EQUALLY SPACE STEEL BEAMS, NOT SPECIFICALLY DIMENSIONED ON PLAN, BETWEEN COLUMN GRID LINES OR THE INDICATED DIMENSIONED POINTS.

8. CAMBER BEAMS AS INDICATED. WHERE CAMBER IS NOT SHOWN, FABRICATE BEAMS WITH INCIDENTAL CAMBER SO THE HIGH SIDE IS PLACED UPWARD, OR OUTWARD AT WIND GIRTS.

9. CAMBER MUST BE COLD FORMED PER AISC SPECIFICATIONS AND TOLERANCES.

10. USE HIGH STRENGTH (HS) BOLTS PER ASTM F3125, GRADES AS INDICATED IN THE FOLLOWING TABLE, UNLESS NOTED OTHERWISE.

TYPICAL HIGH STRENGTH BOLTS	STANDARD, GRADE, AND NOTES
BOLT DIAMETER BOLT ASTM F3125 GRADE BOLT FINISH/COATING HEAVY HEX NUT HARDENED WASHER (CIRCULAR, BEVELED OR CLIPPED, AS REQ'D) DIRECT TENSION INDICATING WASHERS BOLT DIAM. 1-1/2 IN AND SMALLER BOLT DIAM. UP TO 2-1/2 INCHES TENSION CONTROLLED BOLTS (USE MANUFACTURER SUPPLIED NUTS AND WASHERS)	7/8 INCH DIAMETER, UNO A325, TYPE 1 UNO ASTM F2329 HDG ASTM A563 TYPE DH ASTM F436 TYPE 1 ASTM F959 ASTM F2437 F1852, TYPE 1

11. SLIP-CRITICAL (SC) BOLTS MUST BE PER THE RCSC SPECIFICATION AND INSPECTED IN ACCORDANCE WITH THE IBC.

12. USE SNUG-TIGHT (ST) BOLTS PER THE RCSC WHEN SLIP-CRITICAL IS NOT IDENTIFIED IN THE DRAWINGS.

13. USE ASTM A449 BOLTS WITH ASTM F563 NUTS AND ASTM F436 HARDENED WASHERS FOR THROUGH BOLTING HSS SHAPES. BOLTS MUST BE SNUG TIGHT ONLY. PROVIDE JAMB NUT TIGHTENED AGAINST STANDARD NUT TO FASTEN CONNECTOR.

14. DIRECT TENSION INDICATING (DTI) WASHERS, IF USED FOR HIGH STRENGTH BOLTS, MUST NOT BE TIGHTENED TO A FULLY CRUSHED CONDITION. DTI MUST HAVE THE MINIMUM ACCEPTABLE NUMBER OF TAPERED GAGE REJECTIONS BETWEEN LUGS. REJECTION OF GAGE AT EACH LUG LOCATION IS CAUSE FOR REJECTION OF BOLT INSTALLATION.

15. CONTRACTOR MUST SUBMIT A BOLTING PROCEDURE PLAN TO THE CONTRACTING OFFICER FOR REVIEW AND APPROVAL BY THE SER BEFORE BEGINNING WORK.

16. BOLT HOLES CUT IN THE FIELD MUST BE DRILLED, NOT BURNED. BOLT HOLES CUT IN THE SHOP MAY BE DRILLED, WATER-JET OR LASER CUT; PLASMA CUTTING HOLES NOT PERMITTED.

17. LOCATION OF EMBEDDED ELEMENTS MUST BE COORDINATED WITH CONTRACT DOCUMENTS.

18. STEEL COMPONENTS EMBEDDED IN CONCRETE OR MASONRY AT EXTERIOR LOCATIONS, EXPOSED TO EARTH, OR LOCATED IN UNCONDITIONED SPACE, MUST BE HOT-DIPPED GALVANIZED PER ASTM A123.

19. AT INTERIOR CONDITIONS (NOT EXPOSED TO WEATHER) AND COMPONENTS EMBEDDED IN CONCRETE OR MASONRY MUST BE LEFT UNPAINTED.

20. ERECTION AIDS, WHICH ARE THE SOLE RESPONSIBILITY OF THE CONTRACTOR, MUST BE REMOVED AND THE AFFECTED ELEMENTS RESTORED TO ORIGINAL DESIGN INTENT. ERECTION AIDS AND OTHER ATTACHMENTS ARE NOT PERMITTED IN PROTECTED SEISMIC ZONES OF ELEMENTS WITHOUT PRIOR APPROVAL FROM THE SER.

21. NOTIFY CONTRACTING OFFICER OF ANY FABRICATION OR ERECTION ERRORS, OR DEVIATIONS. WRITTEN APPROVAL IS REQUIRED BEFORE FIELD CORRECTIONS ARE MADE.

ANCHOR RODS

1. ANCHOR RODS MUST BE ASTM F1554 GRADE 55 WITH S-1 (WELDABLE) SUPPLEMENT, UNLESS NOTED OTHERWISE.

2. PROVIDE SMOOTH ANCHOR RODS THREADED ONLY AT EACH END OF THE ANCHOR ROD. BEFORE CASTING CONCRETE, GREASE THE UPPER 2/3 LENGTH OF EACH ANCHOR ROD. IN LIEU OF GREASE, PROVIDE TWO, INDEPENDENT LAYERS OF VISQUEEN OVER UPPER 2/3 THE LENGTH OF THE ANCHOR.

3. USE TWO HEAVY HEX NUTS ON EACH END OF ANCHOR ROD. AT TOP, USE DOUBLE NUTS FOR SECURING A WASHER AND WASHER PLATE TO THE TOP SURFACE

4. WHERE AISC OVERSIZED HOLES ARE USED AT BASE PLATES PROVIDE SINGLE-PLY MINIMUM 3/8 INCH ASTM A572 GR 50 PLATE WASHERS WELDED ALL AROUND TO THE BASE PLATE.

5. USE A RIGID STEEL TEMPLATE TO LOCATE AND SECURE ANCHOR RODS BEFORE PLACING CONCRETE. TEMPLATES MUST BE REMOVED AFTER THE CONCRETE HAS HARDENED WITHOUT DAMAGING THE STRUCTURE.

6. AFTER BASE PLATE INSTALLATION, FULLY TIGHTEN ANCHOR ROD NUTS UNLESS NOTED OTHERWISE.

7. HEATING OR BENDING ANCHOR RODS IS PROHIBITED AND GROUNDS FOR REJECTION.

COLD-FORMED STEEL

1. REFERENCES:

A. AISI S100-16 (2020) WITH S2-20 NORTH AMERICAN SPECIFICATION FOR THE DESIGN OF COLD-FORMED STEEL STRUCTURAL MEMBERS.

B. AISI S202-20 CODE OF STANDARD PRACTICE FOR COLD-FORMED STEEL STRUCTURAL FRAMING.

C. AISI S240-20 NORTH AMERICAN STANDARD FOR COLD-FORMED STEEL STRUCTURAL FRAMING.

D. AISI S400-20 NORTH AMERICAN STANDARD FOR SEISMIC DESIGN OF COLD-FORMED STEEL STRUCTURAL SYSTEM.

E. AWS D1.3-18 STRUCTURAL WELDING CODE – SHEET STEEL.

2. CFS STRUCTURAL MEMBERS MUST MEET THE FOLLOWING MATERIAL REQUIREMENTS. PROVIDE MINIMUM PROTECTION COATING EQUAL TO G60 GALVANIZED FINISHED. (THIS TABLE DOES NOT APPLY TO METAL DECK)

MEMBER THICKNESS	STANDARD AND GRADE	MIN YIELD (KSI)
54 MIL AND THICKER	ASTM A1003 GR 50	50
43 MIL AND THINNER	ASTM A1003 GR 33	33

3. CFS FRAMING MEMBERS MUST MEET THE TYPE, SIZE AND THICKNESS AS INDICATED ON THE STRUCTURAL PLANS AND SPECIFICATIONS, AND MUST BE MANUFACTURED BY A MEMBER OF THE STEEL STUD MANUFACTURER'S ASSOCIATION (SSMA), OR PRE-APPROVED EQUAL, IN ACCORDANCE WITH SSMA ICC EVALUATION REPORT ESR-3064P.

4. FASTENING OF COMPONENTS MUST BE WITH SCREWS OR WELDS AS NOTED ON THE DRAWINGS. WELDERS MUST BE CERTIFIED FOR WELDING OF SHEET STEEL PER AWS AND WELDING MUST CONFORM TO AWS D1.3.

5. POWDER ACTUATED FASTENERS (PAF) MUST HAVE A MINIMUM SHANK DIAMETER OF 0.145 INCHES WITH A MINIMUM WASHER BEARING AREA OF 0.321 SQUARE INCHES. THE PAF MUST COMPLY WITH THE IBC AND MUST BE PROVIDED IN ACCORDANCE WITH THE MANUFACTURER'S PUBLISHED INSTRUCTIONS AND APPROVED ICC-ES REPORT OR IAPMO-ES REPORT. INSPECTION AND TESTING MUST BE PROVIDED IN ACCORDANCE WITH THE PROJECT SPECIFICATIONS, GENERAL STRUCTURAL NOTES AND APPROVED ICC-ES REPORT OR IAPMO-ES REPORT.

6. FRAMING COMPONENTS MUST BE CUT TO FIT SQUARELY AGAINST ABUTTING MEMBERS. MEMBERS MUST BE HELD FIRMLY IN POSITION UNTIL PROPERLY FASTENED. SPLICES IN STUDS AND BRACES ARE NOT PERMITTED.

7. PROVIDE ACCESSORIES INCLUDING CLIPS, STRUTS, BRACES, FASTENING DEVICES, AND OTHER ACCESSORIES REQUIRED FOR A COMPLETE AND PROPER INSTALLATION. BRIDGING PER MANUFACTURER'S REQUIREMENTS AND AS SHOWN IN THE STRUCTURAL DRAWINGS MUST BE IN PLACE PRIOR TO PLACING OF CONSTRUCTION LOADS. RUNS MUST BE RIGIDLY ANCHORED TO END WALLS.

8. CONTRACTOR MUST SUBMIT SHOP DRAWINGS AND CALCULATIONS STAMPED AND SIGNED BY AN SSE AND ALL SUPPORTING DESIGN DOCUMENTATION TO THE CONTRACTING OFFICER FOR REVIEW AND APPROVAL BY THE SER BEFORE CONSTRUCTION BEGINS.

9. CALCULATIONS MUST DEMONSTRATE THE ADEQUACY OF THE PROPOSED MEMBERS AND CONNECTIONS FOR STRUCTURAL DESIGN CRITERIA SHOWN IN DOCUMENTS. DRAWINGS MUST SHOW CFS SIZE, THICKNESS, LAYOUT, CONNECTIONS, MATERIAL DESIGNATION, METHOD OF INSTALLATION AND ACCESSORIES. DESIGN MUST BE IN ACCORDANCE WITH AISI STANDARDS.

10. MANUFACTURED PRODUCTS DEMONSTRATING ADEQUATE CAPACITY MAY BE USED IN LIEU OF THE CONNECTORS AND MEMBERS SPECIFIED. A COPY OF THE ICC EVALUATION REPORTS OF THE PRODUCTS SELECTED MUST BE INCLUDED IN THE SUBMITTAL.

13. WHERE NOTE SPECIFICALLY CALLED OUT ON THE DRAWINGS, CONTRACTOR MAY USE STANDARD WALL CONNECTIONS TO THE ROOF MEMBERS.

STRUCTURAL WELDING

1. REFERENCES:

A. AWS D1.1-18 STRUCTURAL WELDING CODE – STEEL.

2. WELDING MUST BE DONE BY AWS CERTIFIED WELDERS AND PER AWS D1.1.

3. USE LOW HYDROGEN E70XX ELECTRODE WITH TENSILE STRENGTH PER AWS D1.1, UNLESS NOTED OTHERWISE.

4. WELDS THAT ARE PART OF THE SEISMIC FORCE-RESISTING SYSTEM MUST HAVE A MINIMUM CHAMPP V-NOTCH TOUGHNESS OF 20 FOOT-POUNDS AT 0°F.

5. FIELD WELDING SYMBOLS HAVE NOT NECESSARILY BEEN USED IN THE DRAWINGS. CONTRACTOR MUST COORDINATE THE USE OF SHOP AND FIELD WELDS AS NECESSARY.

6. FILLET WELD SIZES SHOWN ON DRAWINGS INDICATE THE MINIMUM LEG SIZE REQUIRED AT THE JOINT. INCREASE WELD SIZE BY ROOT GAP DIMENSION IF A GAP EXISTS AT THE JOINT. IF THE INCLUDED ANGLE OF THE TWO CONNECTED PLATES EXCEEDS 90°, THE WELD MUST BE CONTOURED TO PROVIDE AN EFFECTIVE THROAT DIMENSION NO LESS THAN 72% OF WELD SIZE INDICATED. IF NOT SHOWN THE MINIMUM FILLET WELD SIZE AT INTERIOR LOCATIONS MUST BE NO LESS THAN 3/16 INCHES, AND NO LESS THAN 1/4 INCHES AT EXTERIOR LOCATIONS.

7. PARTIAL JOINT PENETRATION (PJP) GROOVE WELDS SHOWN ON THE DRAWINGS REFER TO EFFECTIVE THROAT THICKNESS AND MUST BE SIZED ACCORDINGLY FOR THE WELD PROCESS USED.

8. COMPLETE JOINT PENETRATION (CJP) MUST HAVE A BACKUP BAR WITH RUNOFF TABS TO ACHIEVE FULL DEPTH OVER FULL LENGTH OF WELD. BACKGROUING THE ROOT IS REQUIRED. BACKUP BAR MUST BE REMOVED AND GROUND SMOOTH, UNLESS NOTED OTHERWISE.

9. ULTRASONICALLY TEST (UT) CJP WELDS UPON COMPLETION OF THE WELDED JOINT. PLATE THICKNESS OF 5/16-INCH OR LESS MAY USE MAGNETIC PARTICLE INSPECTION (MPI) OR EDDIE CURRENT ARRAY (ECA) TESTING.

10. QUALIFICATIONS AND WELDING PROCEDURE SPECIFICATION (WPS) PER AWS D1.1 AND D1.8 MUST BE SUBMITTED TO THE ENGINEER FOR REVIEW BEFORE STARTING FABRICATION OR ERECTION. COPIES OF THE WPS MUST BE ON SITE AND AVAILABLE TO WELDERS AND THE SPECIAL INSPECTOR.

STEEL FLOOR AND ROOF DECK

1. REFERENCES:

A. AISI S01 S0-22 STANDARD FOR STEEL DECK.

B. AISI S100-16 (2020) WITH S2-20 NORTH AMERICAN SPECIFICATION FOR THE DESIGN OF COLD-FORMED STEEL STRUCTURAL MEMBERS.

C. AWS D1.3-18 STRUCTURAL WELDING CODE – SHEET STEEL.

2. METAL FLOOR AND ROOF DECK MUST CONFORM TO THE STEEL DECK INSTITUTE (SDI) STANDARD FOR STEEL DECK.

3. METAL DECK AND ACCESSORIES MUST BE FORMED FROM STEEL SHEET CONFORMING TO ASTM A563, SS GR. 50, WITH A ZINC COATING FINISH CONFORMING TO ASTM A525, CLASS G-60 (OR G-90 WHERE SPECIFIED). METAL DECK SURFACE USED FOR COMPOSITE CONSTRUCTION IN CONTACT WITH CONCRETE MUST BE PHOSPHATIZED OR GALVANIZED.

4. METAL DECK EXPOSED TO THE WEATHER MUST BE HOT-DIPPED GALVANIZED AFTER FABRICATION.

5. ALL DECK MUST BE INSTALLED USING 3 OR MORE SPANS. WHERE ONLY 1 OR 2 SPAN CONDITIONS ARE POSSIBLE, PROVIDE A DECK WITH SUFFICIENT SECTION PROPERTIES TO EQUAL BOTH LOAD AND DEFLECTION CONDITIONS OF THE 3 SPAN CONDITION. DECK LAPS MUST OCCUR OVER SUPPORT.

6. FLOOR DECK MUST HAVE A 3-INCH MINIMUM BEARING AND ROOF DECKING MUST HAVE A 2-INCH MINIMUM BEARING.

7. FURNISH ALL ACCESSORIES, INCLUDING REINFORCING ANGLES, CLOSURE STRIPS, FOUR STOPS, AND OTHER ELEMENTS TO ADEQUATELY SUPPORT DECK GRAVITY LOADS AROUND OPENINGS TO COMPLETE INSTALLATION.

ANCHORED MASONRY VENEER

1. ANCHORED MASONRY VENEER AND ATTACHMENTS MUST CONFORM TO THE REQUIREMENTS OF IBC CHAPTER 14 AND THE PROJECT SPECIFICATIONS.

2. ANCHOR TIES AND JOINT REINFORCEMENT MUST BE HOT-DIPPED GALVANIZED PER ASTM A153, CLASS B-2 AND MUST BE MANUFACTURED BY OUR O-WALL OR APPROVED EQUAL.

3. PROVIDE #9 GAUGE HORIZONTAL JOINT REINFORCEMENT WIRE. JOINT REINFORCEMENT MUST BE CONTINUOUS WITH BUTT SPLICES BETWEEN TIES PERMITTED.

4. ANCHOR TIES MUST BE SPACED 16" O.C. EACH WAY MAXIMUM AND MUST HAVE A LIP OR HOOK ON THE EXTENDED LEG THAT WILL ENGAGE OR ENCLOSE HORIZONTAL JOINT REINFORCEMENT WIRE. ATTACH TO BACKING PER THE FOLLOWING TABLE.

BACKING CONDITION	DUR-O-WALL VENEER TIE	ATTACHMENT TO BACKING
METAL STUDS	DA 2135	DA 807 VENEER SCREW
STRUCTURAL STEEL	DA 700	WELD DA 709 TO STEEL
CONCRETE WALL	DA 904 WITH DA 931	1/4 IN DIAM EXP ANCHOR

POST INSTALLED ANCHOR IN CONCRETE AND MASONRY

1. EXCEPT WHERE INDICATED ON THE DRAWINGS, POST-INSTALLED ANCHORS CONSIST OF THE FOLLOWING ANCHOR TYPES OR APPROVED EQUAL.

CONDITION	TYPE	ACCEPTABLE ANCHORS
CONCRETE	SCREW	HILTI KWIK HUS-EZ SCREW ANCHORS PER ICC ESR-3027 SIMPSON TITEN HD SCREW ANCHOR PER ICC ESR-2713 DEWALT WEDGE-BOLT+ PER ICC ESR-2526
	EXPANSION	HILTI KWIK-BOLT T22 ANCHOR PER ICC ESR-4266 SIMPSON STRONG BOLT 2 ANCHOR PER ICC ESR-3037 DEWALT POWER-STUD+ SD1 ANCHOR PER ICC ESR-2818
	ADHESIVE	HILTI HIT-HY 200 ADHESIVE ANCHOR SYSTEM PER ICC ESR-3187 SIMPSON SET-XP ADHESIVE ANCHOR SYSTEM PER ICC ESR-2508 DEWALT PE1000+ ADHESIVE ANCHOR SYSTEM PER ICC ESR-2583
	UNDERCUT	HILTI HDA-P UNDERCUT ANCHOR PER ICC ESR-1546 SIMPSON TORO-CUT SELF UNDERCUTTING ANCHOR PER ICC ESR-2705 DEWALT CCU+ UNDERCUT ANCHOR CARBON STEEL PER ICC ESR-4810
SOLID-GROUTED MASONRY	SCREWS	HILTI KWIK HUS-EZ SCREW ANCHORS PER ICC ESR-3056 SIMPSON TITEN HD SCREW ANCHOR PER ICC ESR-2713 DEWALT SCREW-BOLT+ PER ICC ESR-4042
	EXPANSION	HILTI KWIK-BOLT 3 ANCHOR PER ICC ESR-1385 SIMPSON STRONG BOLT 2 ANCHOR PER UES ER-240 DEWALT POWER-STUD+ SD1 ANCHOR PER ICC ESR-2966
	ADHESIVE	HILTI HIT-HY 270 ADHESIVE SYSTEM PER ICC ESR-4143 SIMPSON SET-XP ADHESIVE SYSTEM PER IAPMO UES ER-265 DEWALT AC108+GOLD ADHESIVE SYSTEM PER ICC ESR-3200 (STEEL ANCHOR MUST BE ASTM F1554 GRADE 55 CONTINUOUSLY THREADED ROD PER DRAWINGS)

2. ANCHOR CAPACITY USED IN DESIGN MUST BE BASED ON THE TECHNICAL DATA PUBLISHED IN THE ICC APPROVAL REPORT. SUBSTITUTION REQUESTS MUST INCLUDE ICC ESR REPORT SHOWING COMPLIANCE WITH RELEVANT BUILDING CODE AND INSTALLATION CATEGORY AND BE APPROVED IN WRITING BY THE CONTRACTING OFFICER FOR REVIEW AND APPROVAL BY THE SER PRIOR TO USE.

3. INSTALL ANCHORS PER THE MANUFACTURER INSTRUCTIONS, AS INCLUDED IN THE ANCHOR PACKAGING.

4. INSTALL ANCHORS IN ACCORDANCE WITH SPACING AND EDGE CLEARANCES INDICATED ON THE DRAWINGS.

5. EXISTING DEFORMED BAR REINFORCEMENT IN THE STRUCTURE MAY CONFLICT WITH SPECIFIC ANCHOR LOCATIONS. UNLESS NOTED OTHERWISE, THE CONTRACTOR MUST LOCATE THE POSITION OF THE EXISTING REINFORCING BARS AND AVOID CUTTING.

6. DRILLING, CORING, SAW-CUTTING, ETC. INTO CONCRETE SHALL MEET THE LATEST OSHA REGULATIONS FOR SILICA DUST EXPOSURE.

SPECIAL INSPECTION AND TESTING

1. SPECIAL INSPECTION AND TESTING MUST CONFORM TO SECTION 1704 OF THE INTERNATIONAL BUILDING CODE (IBC) AS AMENDED BY H-18-8.

2. ONE OR MORE QUALIFIED SPECIAL INSPECTION & TESTING AGENCIES MUST BE EMPLOYED TO PERFORM THE NECESSARY INSPECTION OF IBC SECTION 1705, AS APPLICABLE. QUALIFICATIONS FOR SPECIAL INSPECTION AND TESTING AGENCIES AND INSPECTORS MUST BE SUBMITTED TO THE CONTRACTING OFFICER FOR REVIEW AND APPROVAL.

3. SPECIAL INSPECTOR QUALIFICATIONS MUST MEET THE REQUIREMENTS OF SPECIFICATION SECTION 01 45 36.

• FOR STATEMENT OF SPECIAL INSPECTION SEE PROJECT SPECIFICATIONS.

5. REPORTING ON SPECIAL INSPECTION AND TESTING MUST CONFORM TO IBC SECTION 1704.5 AND BE SUBMITTED TO THE CONTRACTING OFFICER WEEKLY, OR AS REQUIRED BY THE SPECIFICATIONS.

STRUCTURAL OBSERVATIONS

1. STRUCTURAL OBSERVATIONS MUST BE PERFORMED BY THE DESIGNATED REPRESENTATIVE BY THE GOVERNMENT, FOR THE FOLLOWING:

A. FOUNDATION:

B. CONCRETE WALLS:

C. STRUCTURAL STEEL:

D. ANCHORAGE

PLACEMENT OF DEFORMED BAR REINFORCEMENT, BEFORE CONCRETE.

FINAL PLACEMENT OF DEFORMED BARS REINFORCEMENT AND ANCHORS PRIOR TO POUR.

BEAMS AND METAL DECK.

ANCHORAGE OF TOWERS TO EXISTING BUILDINGS.

2. CONTRACTOR MUST NOTIFY CONTRACTING OFFICER 14 DAYS IN ADVANCE OF WHEN WORK IS EXPECTED TO COORDINATE THE DATE OF OBSERVATION.

3. STRUCTURAL OBSERVATION DOES NOT WAIVE THE RESPONSIBILITY OF SPECIAL INSPECTION REQUIREMENTS.

Revisions:

Date:

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VA U.S. Department of Veterans Affairs

Drawing Title

STRUCTURAL GENERAL NOTES

Approved:

Phase

100% CONSTRUCTION DOCUMENTS

Project Title

EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

Location
VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date
09/03/2025

Checked
CH

Drawn
JKM

Project Number

581-22-700

Building Number

Drawing Number

S-002

022 OF 693

VA FORM 08 - 6231

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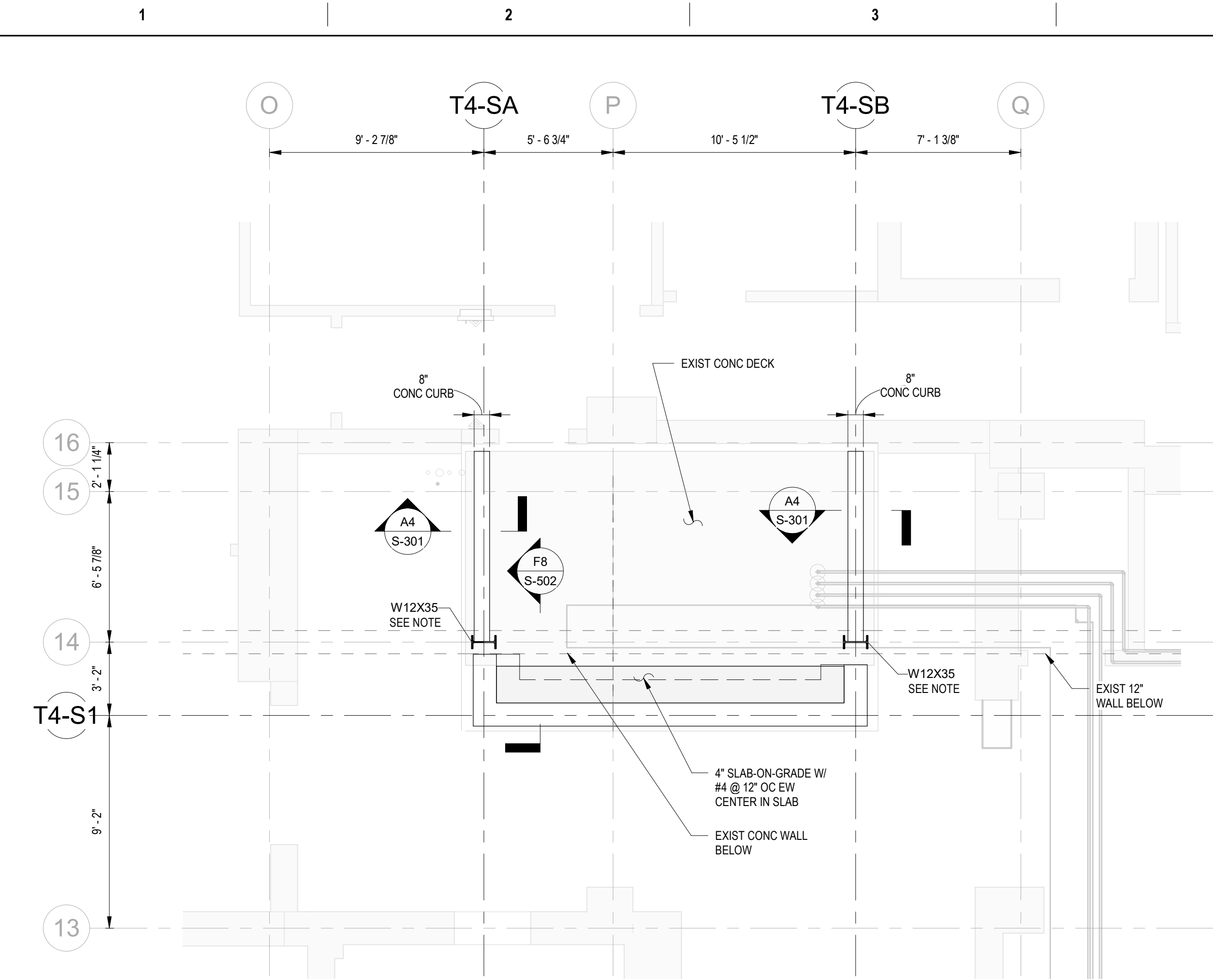
C1 FOUND
1/4" = 1'-0"

C5 FRAM
1/4" = 1'-0"

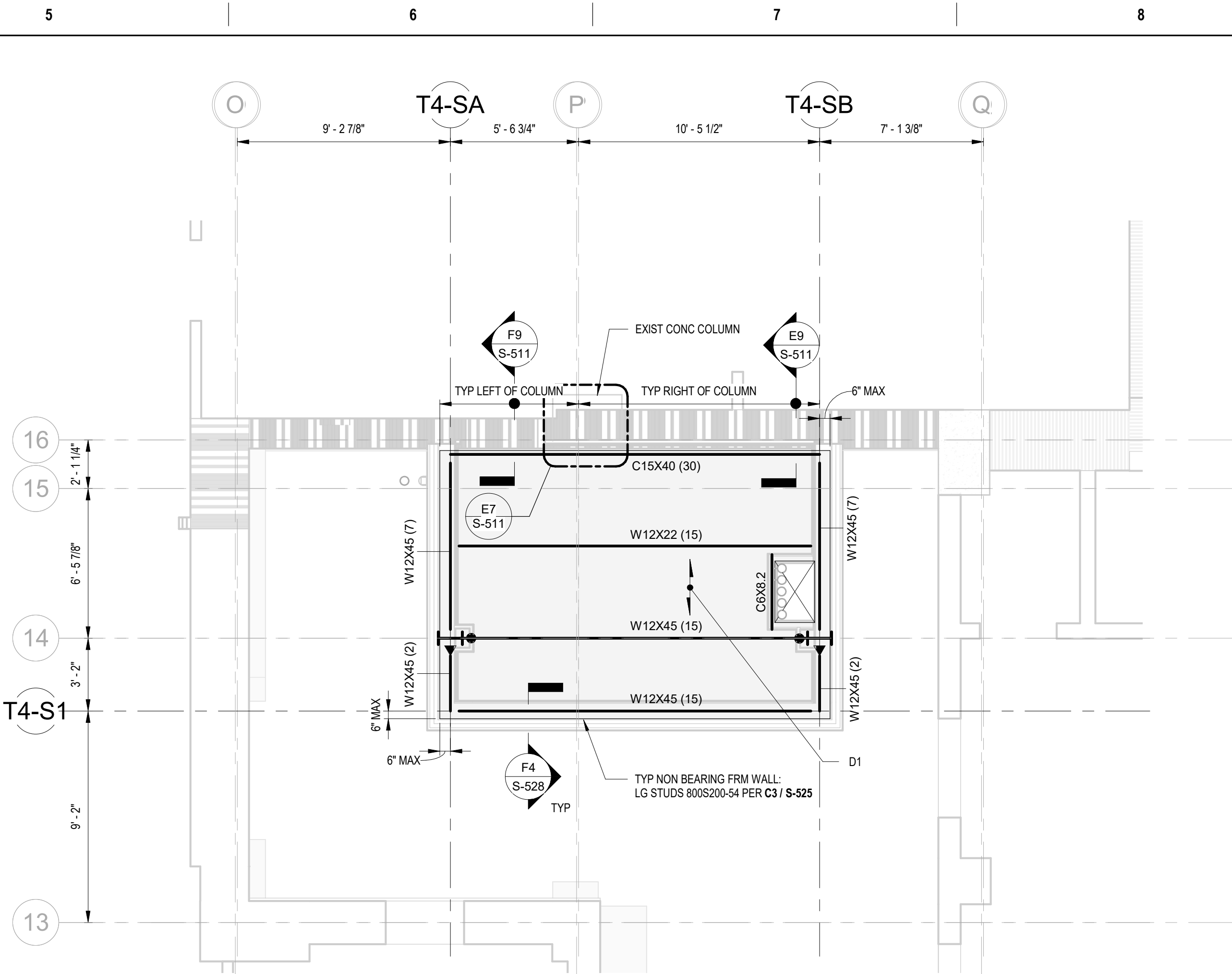
(F1) ROOF
1/4" = 1'-0"



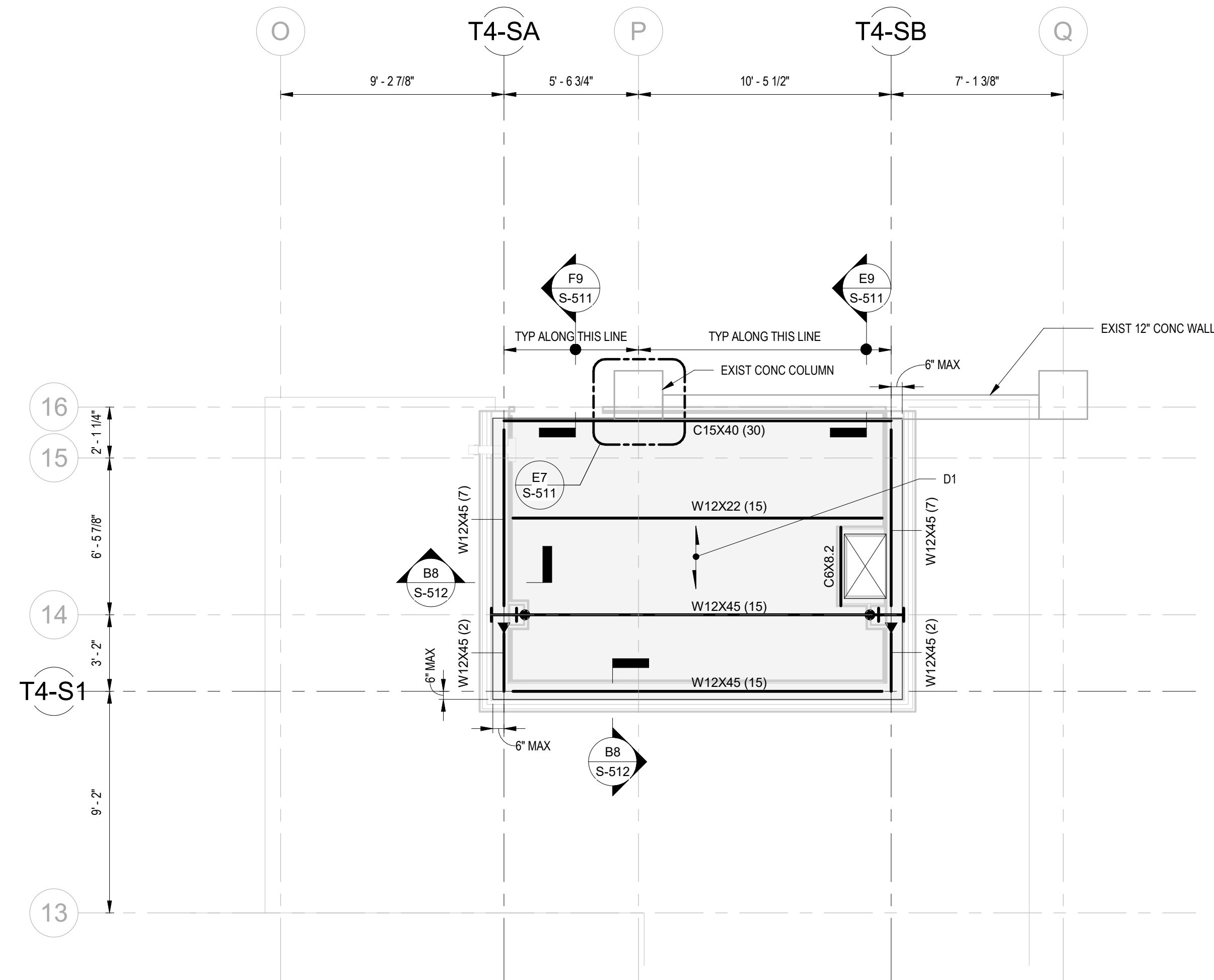




C1 GROUND LEVEL PLAN T4
1/4" = 1'-0"



C5 FRAMING PLAN - LEVELS 01 THROUGH 05 T4
1/4" = 1'-0"



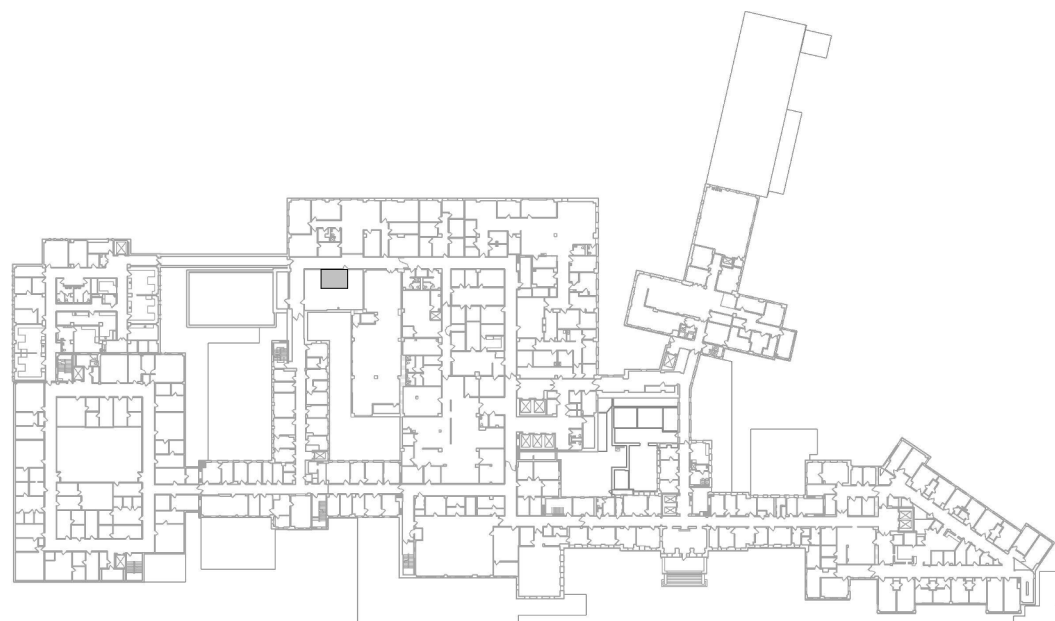
F1 ROOF FRAMING PLAN T4
1/4" = 1'-0"

- PLAN NOTES:**
- REFER TO SHEETS S-001 THRU S-003 FOR GENERAL NOTES. THESE NOTES & DETAILS ON SHEET S-511, S-512, AND S-515 SHALL BE USED WHERE APPLICABLE, WHETHER SPECIFICALLY REFERENCED OR NOT.
 - REFER TO FOUNDATION AND SLAB-ON-GRADE NOTES ON SHEET S-001 & SPECIFICATIONS FOR GEOTECHNICAL RECOMMENDATIONS AND PAD PREPARATION.
 - REFER TO ARCHITECTURAL AND MECHANICAL DRAWINGS FOR ANY DIMENSIONS, OPENINGS, AND ELEVATIONS NOT SHOWN.
 - REFER TO ARCHITECTURAL AND MECHANICAL DRAWINGS FOR THE FOLLOWING:
A. DIMENSIONS NOT SHOWN.
B. OPENINGS NOT SHOWN.
C. FLOOR ELEVATIONS.
 - FOR TYPICAL OPENINGS EXACT LOCATION AND SIZE, SEE ARCHITECTURAL AND MECHANICAL DRAWINGS.
 - DIMENSIONS AND ELEVATIONS SHALL BE VERIFIED PRIOR TO STEEL FABRICATION AND ERECTION.
 - FOR TYPICAL CONCRETE DETAILS, REFER TO SHEET S-501.
 - FOR FOUNDATION DETAILS, REFER TO SHEET S-502.
 - FOR TYPICAL STRUCTURAL STEEL DETAILS, REFER TO SHEET S-511, S-512, AND S-515.
 - BEAM-TO-BEAM CONNECTION PER B1/S-511 UNO.

LEGEND:

- W14x22 (1) INDICATES BEAM AND BEAM SIZE (#OF STUDS)
- GRAVITY MOMENT FRAME CONNECTION PER D7 / S-511
- LATERAL MOMENT FRAME CONNECTION PER D7 / S-511
- CFH INDICATES COLD FORMED HEADER PER C3 / S-525
- INDICATES DIRECTION OF DECK SPAN.
- SW1 INDICATES METAL STUD SHEAR WALL PER 3 / S-527
- X-XX" INDICATES APPROXIMATE LENGTH OF SHEAR WALL
- INDICATES LOCATION OF SHEAR WALL SHEATHING
- SHD/4 INDICATES SIMPSON HOLDOWN TYPE AND SIZE PER
- D1 COMPOSITE DECK PER B1 / S-512
- DECK OPENING
- MICROPILE
- TB TIE BEAM PER D1 / S-503

KEY PLAN



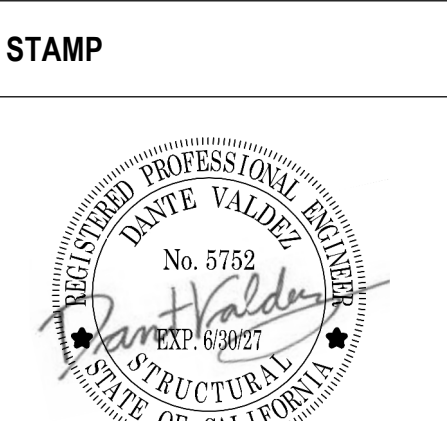
SHOWN AREA



Revisions:	Date:

CONSULTANT
ENGINEERING DISCIPLINE:
wsp
WSP USA Inc.
33301 9th Avenue South
Suite 300
Federal Way, WA 98003-2600
TEL: (206) 431-2300
FAX: (206) 431-2250

ARCHITECT/ENGINEER OF RECORD
A/E:
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RAY SPEES
SDE SPEESDESIGNBUILD



Office of
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and Facilities
Management
VA U.S. Department
of Veterans Affairs

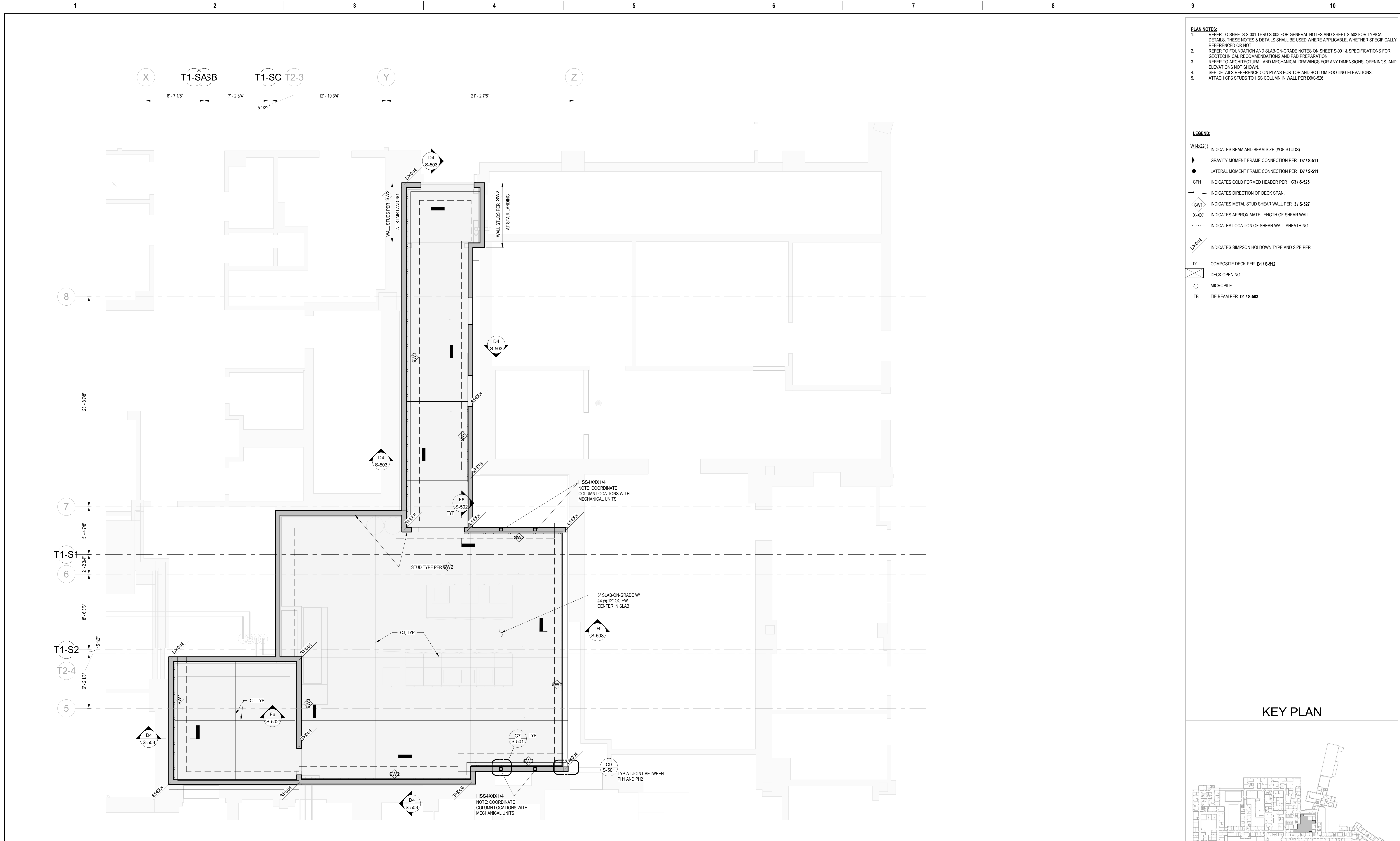
Drawing Title
STRUCTURAL PLANS - TOWER T4
Approved:

Phase
**100% CONSTRUCTION
DOCUMENTS**

Project Title
**EHRM INFRASTRUCTURE
UPGRADES - HUNTINGTON**

Location
VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300
Issue Date
09/03/2025
Checked
CH
Drawn
JKM

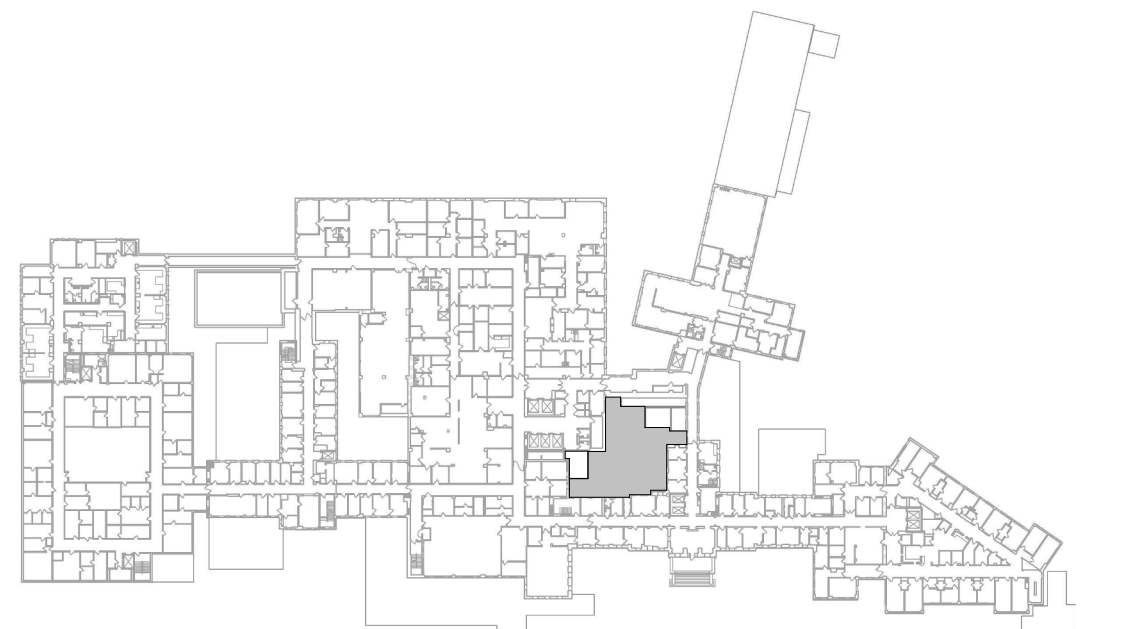
Project Number
581-22-700
Building Number
01S
Drawing Number
S-104
027 OF 693



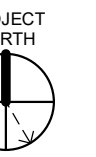
- PLAN NOTES:**
- REFER TO SHEETS S-001 THRU S-003 FOR GENERAL NOTES AND SHEET S-002 FOR TYPICAL DETAILS. THESE NOTES & DETAILS SHALL BE USED WHERE APPLICABLE, WHETHER SPECIFICALLY REFERENCED OR NOT.
 - REFER TO FOUNDATION AND SLAB-ON-GRADE NOTES ON SHEET S-001 & SPECIFICATIONS FOR GEOTECHNICAL RECOMMENDATIONS AND PAD PREPARATION.
 - REFER TO ARCHITECTURAL AND MECHANICAL DRAWINGS FOR ANY DIMENSIONS, OPENINGS, AND ELEVATIONS NOT SHOWN.
 - SEE DETAILS REFERENCED ON PLANS FOR TOP AND BOTTOM FOOTING ELEVATIONS.
 - ATTACH CFS STUDS TO HSS COLUMN IN WALL PER D9/S-526

- LEGEND:**
- W14x221) INDICATES BEAM AND BEAM SIZE (#OF STUDS)
- GRAVITY MOMENT FRAME CONNECTION PER D7 / S-511
- LATERAL MOMENT FRAME CONNECTION PER D7 / S-511
- CFH INDICATES COLD FORMED HEADER PER C3 / S-525
- INDICATES DIRECTION OF DECK SPAN
- INDICATES METAL STUD SHEAR WALL PER 3 / S-527
- X'X'X' INDICATES APPROXIMATE LENGTH OF SHEAR WALL
- INDICATES LOCATION OF SHEAR WALL SHEATHING
- INDICATES SIMPSON HOLDOWN TYPE AND SIZE PER
- D1 COMPOSITE DECK PER B1 / S-512
- DECK OPENING
- MICROPILE
- TB TIE BEAM PER D1 / S-503

KEY PLAN



SHOWN AREA



F1 FOUNDATION PLAN - DATA CENTER - PHASE 1B

1/4" = 1'-0"

Revisions:	Date:

CONSULTANT

ENGINEERING DISCIPLINE:



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33301 9th Avenue South
Suite 300
Federal Way, WA 98003-2600
TEL: (206) 431-2300
FAX: (206) 431-2250

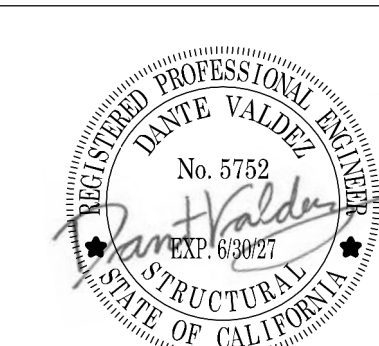
ARCHITECT/ENGINEER OF RECORD

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625 1ST AVE, STE 301
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(206) 590-2118
RAY SPEES



STAMP



Office of
Construction
and Facilities
Management



U.S. Department
of Veterans Affairs

Drawing Title

FOUNDATION PLAN - DATA
CENTER - PHASE 1

Approved:

Phase

100% CONSTRUCTION
DOCUMENTS

Project Title

EHRM INFRASTRUCTURE
UPGRADES - HUNTINGTON

Location

VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date

09/03/2025

Checked

CH

Drawn

JKM

Project Number

581-22-700

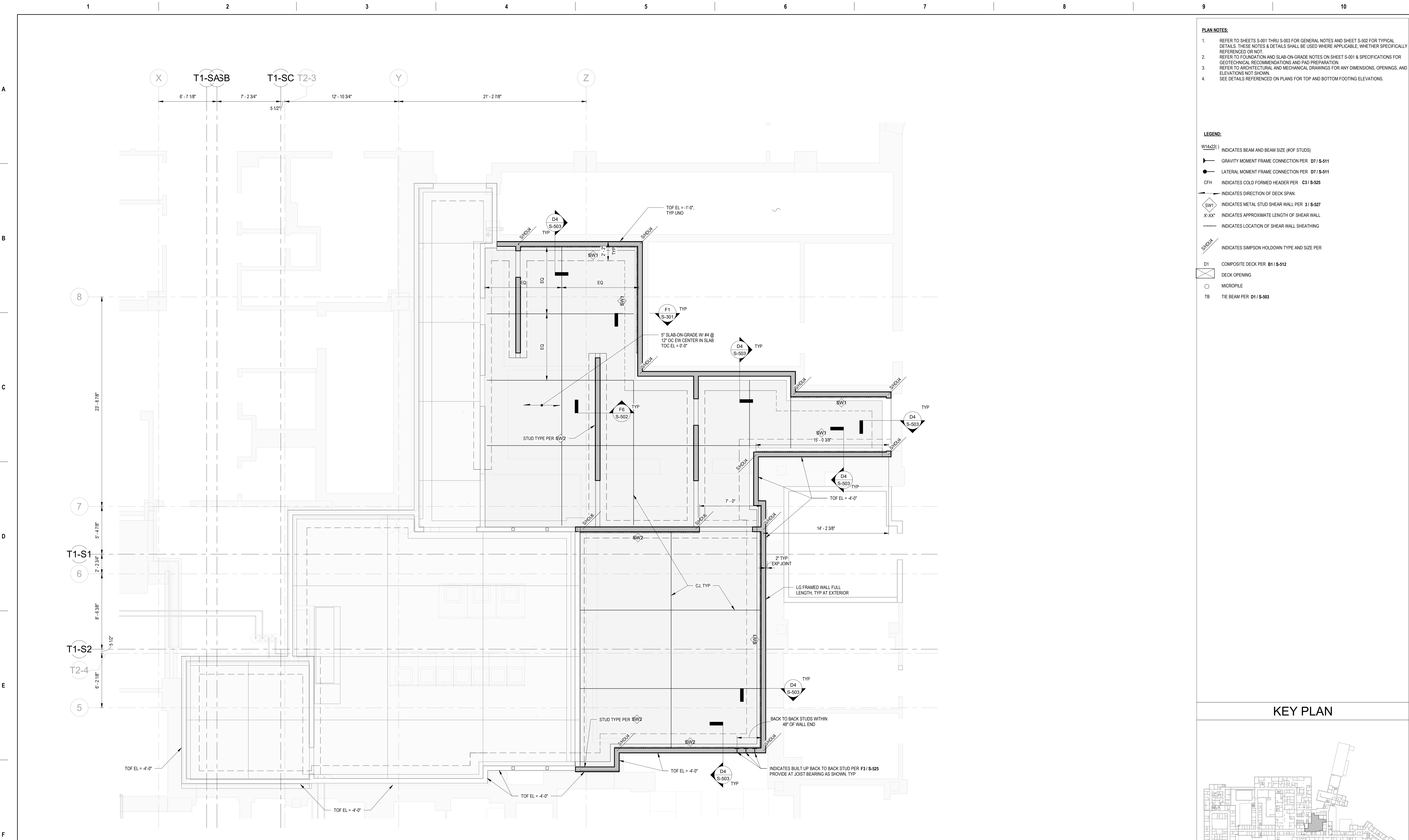
Building Number

GC115-1

Drawing Number

S-105-1B

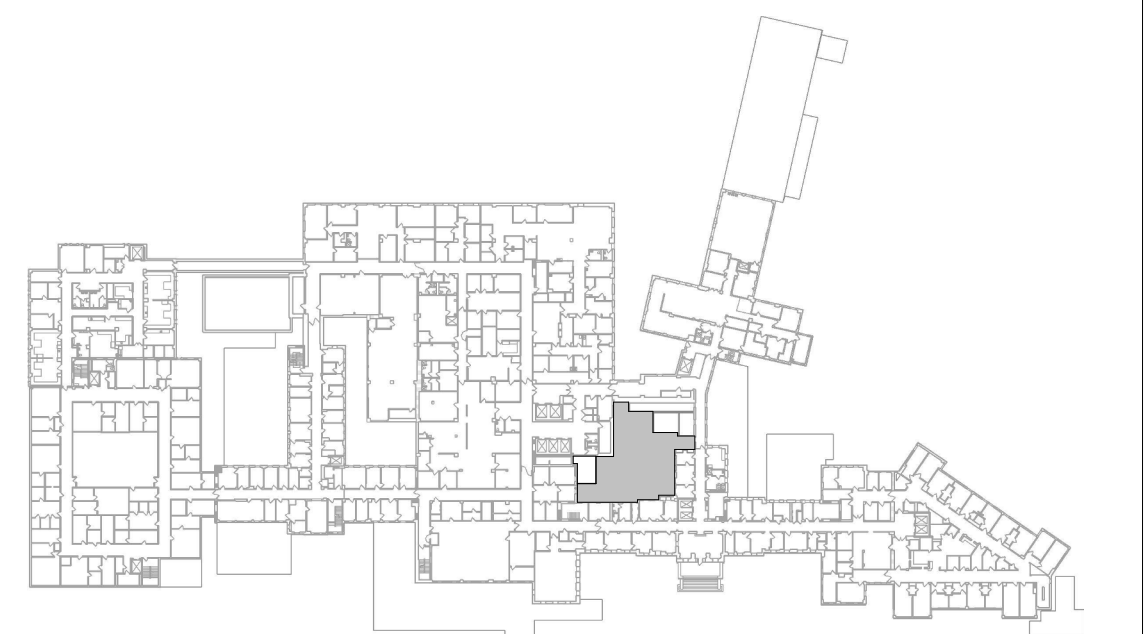
028 OF 693



- PLAN NOTES:**
- REFER TO SHEETS S-001 THRU S-003 FOR GENERAL NOTES AND SHEET S-502 FOR TYPICAL DETAILS. THESE NOTES & DETAILS SHALL BE USED WHERE APPLICABLE, WHETHER SPECIFICALLY REFERENCED OR NOT.
 - REFER TO FOUNDATION AND SLAB-ON-GRADE NOTES ON SHEET S-001 & SPECIFICATIONS FOR GEOTECHNICAL RECOMMENDATIONS AND PAD PREPARATION.
 - REFER TO ARCHITECTURAL AND MECHANICAL DRAWINGS FOR ANY DIMENSIONS, OPENINGS, AND ELEVATIONS NOT SHOWN.
 - SEE DETAILS REFERENCED ON PLANS FOR TOP AND BOTTOM FOOTING ELEVATIONS.

- LEGEND:**
- W14x221) INDICATES BEAM AND BEAM SIZE (#OF STUDS)
- GRAVITY MOMENT FRAME CONNECTION PER D7 / S-511
- LATERAL MOMENT FRAME CONNECTION PER D7 / S-511
- CFH INDICATES COLD FORMED HEADER PER C3 / S-525
- INDICATES DIRECTION OF DECK SPAN
- SW1) INDICATES METAL STUD SHEAR WALL PER 3 / S-527
- X'X'X' INDICATES APPROXIMATE LENGTH OF SHEAR WALL
- INDICATES LOCATION OF SHEAR WALL SHEATHING
- SHP04 INDICATES SIMPSON HOLDOWN TYPE AND SIZE PER
- D1 COMPOSITE DECK PER B1 / S-512
- DECK OPENING
- MICROPILE
- TB TIE BEAM PER D1 / S-503

KEY PLAN



SHOWN AREA



F1 FOUNDATION PLAN - DATA CENTER - PHASE 2B
1/4" = 1'-0"

		CONSULTANT ENGINEERING DISCIPLINE:  WSP USA Inc. 33301 9th Avenue South Suite 300 Federal Way, WA 98003-2600 TEL: (206) 431-2300 FAX: (206) 431-2250	ARCHITECT/ENGINEER OF RECORD A/E: SPEEDS DESIGN BUILD 625 1ST AVE, STE 301 SEATTLE, WA 98104 (206) 590-2118 RAY SPEEDS  SPEEDSDSIGNBUILD	STAMP 	Office of Construction and Facilities Management VA U.S. Department of Veterans Affairs	Drawing Title FOUNDATION PLAN - DATA CENTER - PHASE 2 Approved:	Phase 100% CONSTRUCTION DOCUMENTS	Project Title EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON	Project Number 581-22-700 Building Number GC115-1 Drawing Number S-106-2B 029 OF 693			
Revisions:										Location VA HUNTINGTON 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300	Issue Date 09/03/2025	Checked CH
Date:												

A

B

C

D

E

F

A

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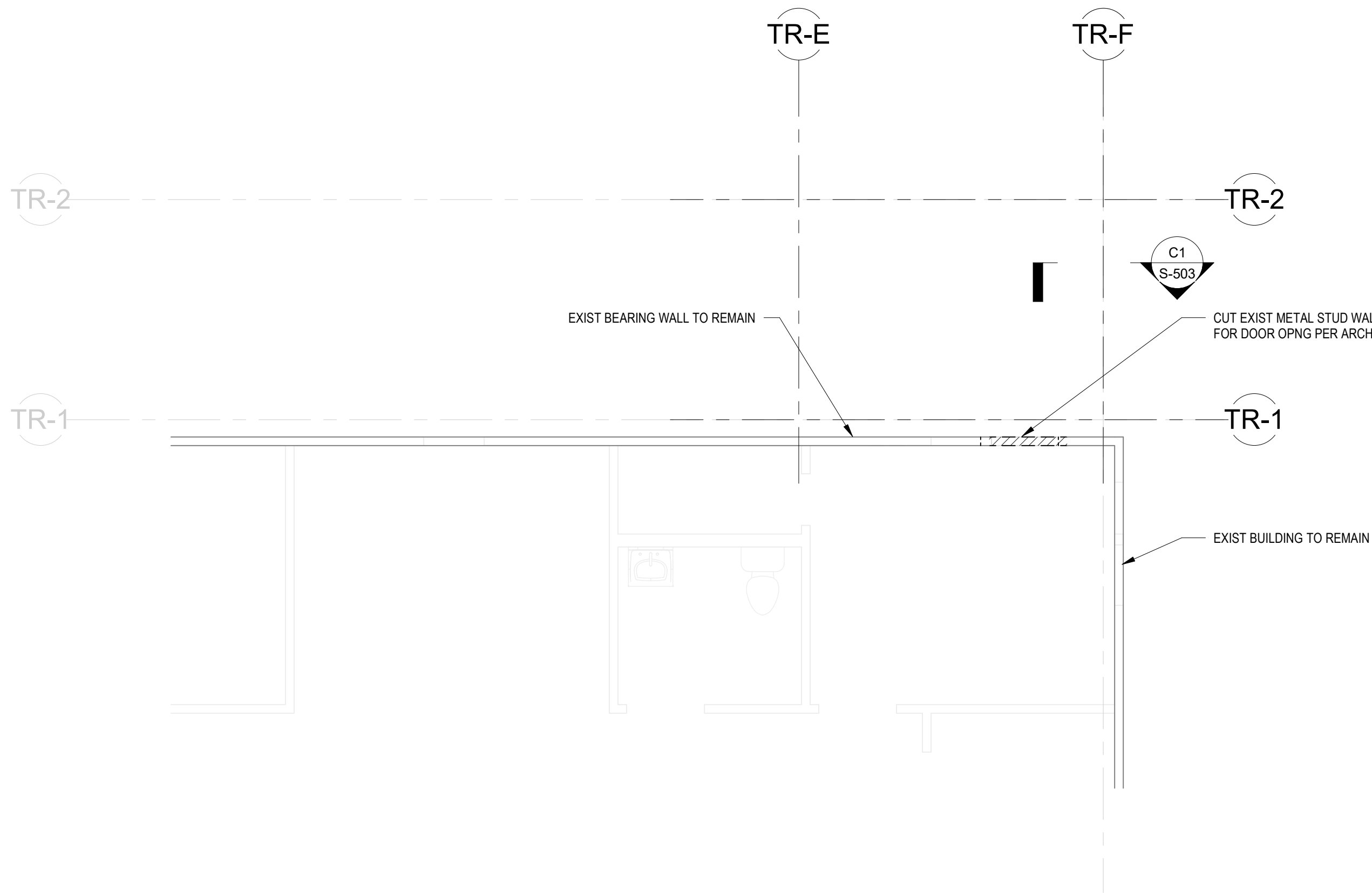
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NOTES:

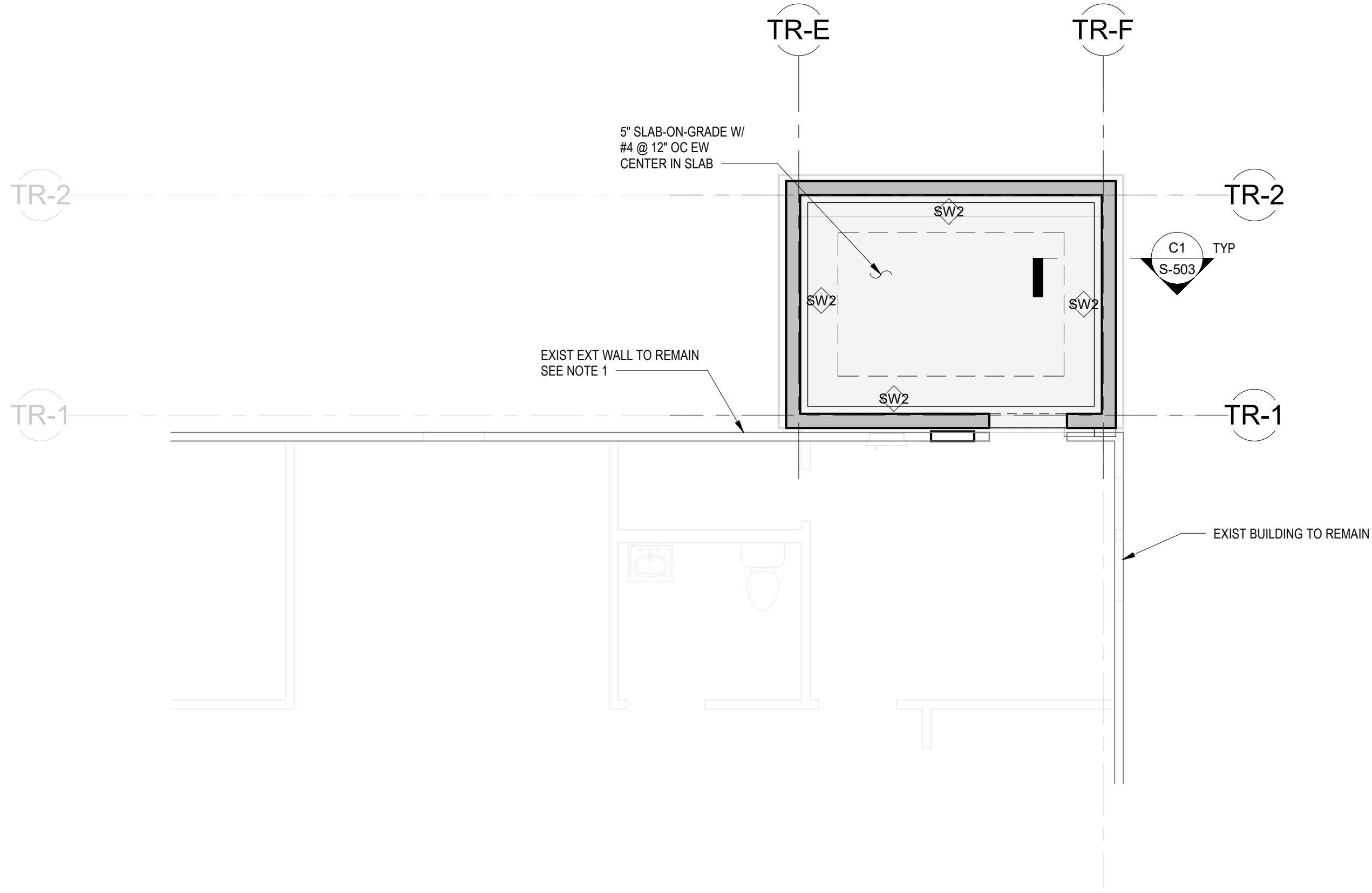
1. OPENING FRAMING:
A. IF EXISTING WALL IS WOOD FRAMED, REFER TO D6 / S-515 FOR FRAMING AT NEW OPENING
B. IF EXISTING WALL IS CFS FRAMED, REFER TO F3 / S-525 FOR FRAMING AT NEW OPENING

LEGEND:

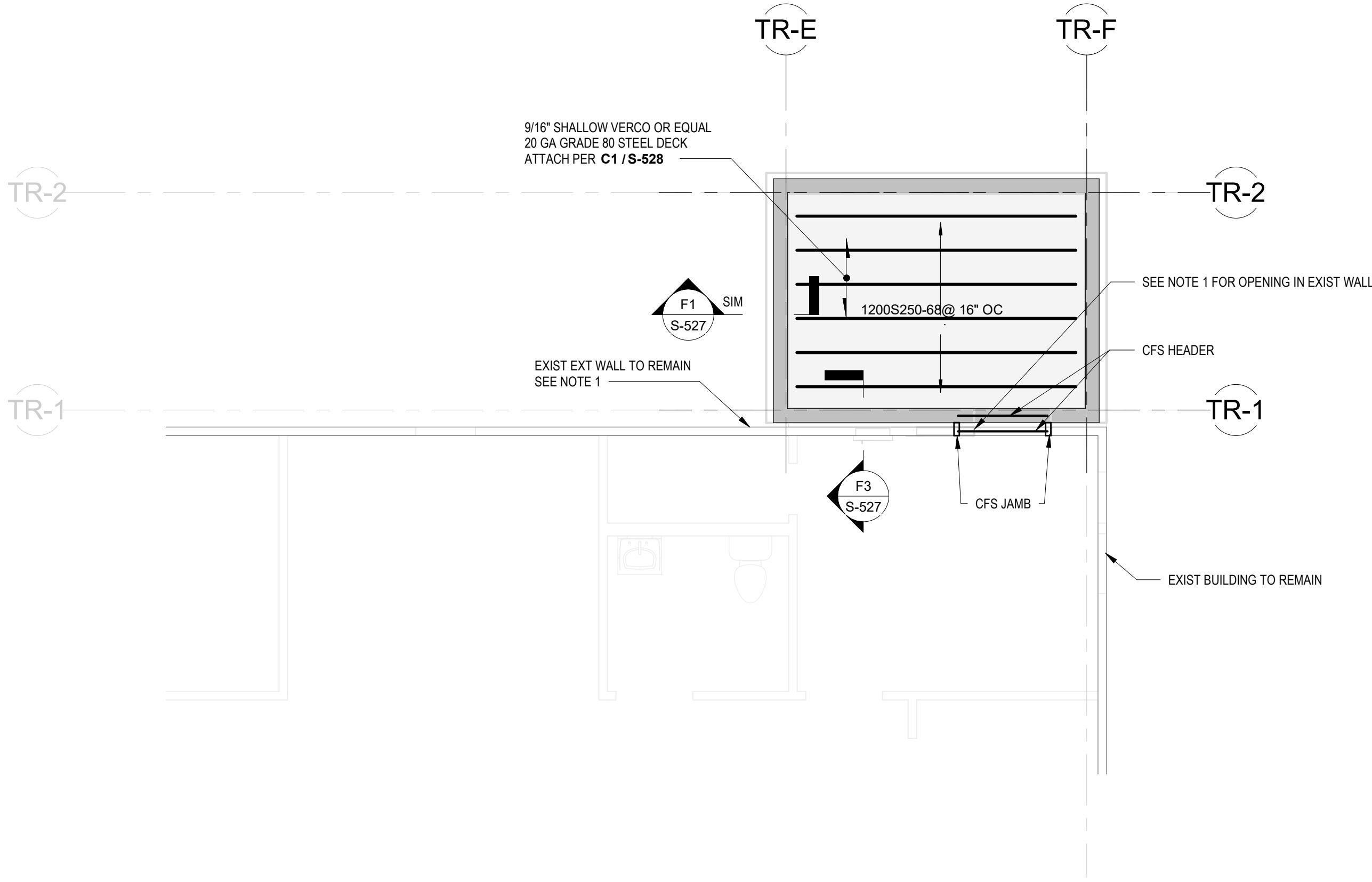
- EXISTING WALL TO BE DEMOLISHED
W14(22) INDICATES BEAM AND BEAM SIZE (#OF STUDS)
GRAVITY MOMENT FRAME CONNECTION PER D7 / S-511
LATERAL MOMENT FRAME CONNECTION PER D7 / S-511
CFH INDICATES COLD FORMED HEADER PER C3 / S-525
INDICATES DIRECTION OF DECK SPAN
INDICATES METAL STUD SHEAR WALL PER 3 / S-527
X'-XX" INDICATES APPROXIMATE LENGTH OF SHEAR WALL
INDICATES LOCATION OF SHEAR WALL SHEATHING
INDICATES SIMPSON HOLDDOWN TYPE AND SIZE PER
D1 COMPOSITE DECK PER B1 / S-512
DECK OPENING
MICROPILE
TIE BEAM PER D1 / S-503



C1 DEMOLITION PLAN TR1
1/4" = 1'-0"

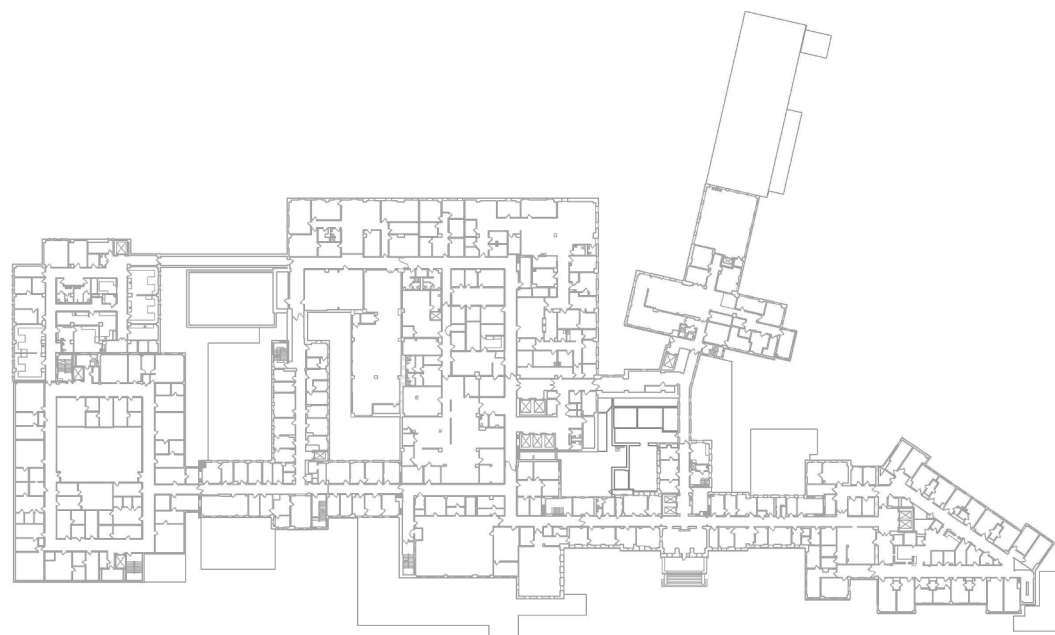


F1 FOUNDATION PLAN TR1
1/4" = 1'-0"



F6 ROOF FRAMING PLAN TR1
1/4" = 1'-0"

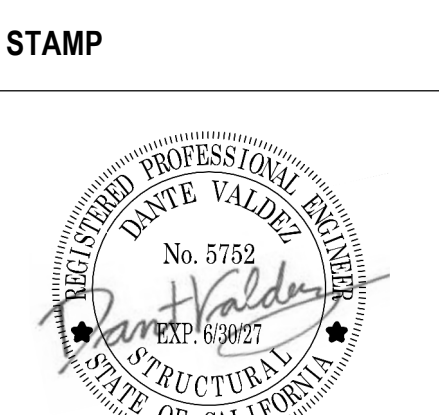
KEY PLAN



Revisions:	Date:

CONSULTANT
ENGINEERING DISCIPLINE:
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33301 9th Avenue South
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Federal Way, WA 98003-2600
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RAY SPEES
SDE SPEESDESIGNBUILD



Office of
Construction
and Facilities
Management
VA U.S. Department
of Veterans Affairs

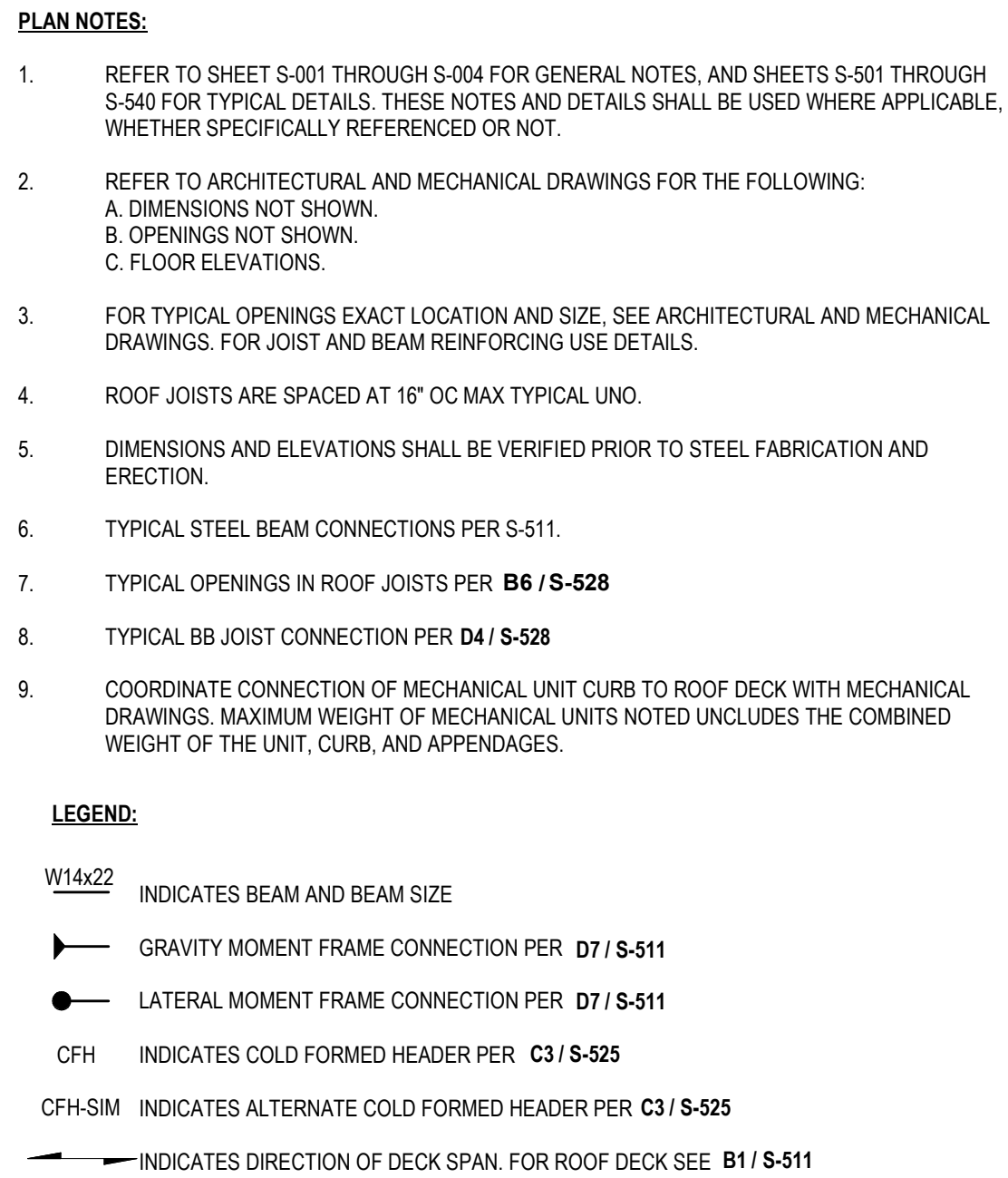
Drawing Title
STRUCTURAL PLANS - BUILDING
TR1
Approved:

Phase
100% CONSTRUCTION
DOCUMENTS

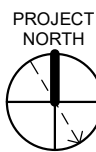
Project Title
EHRM INFRASTRUCTURE
UPGRADES - HUNTINGTON

Location
VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300
Issue Date
09/03/2025
Checked
CH
Drawn
Author

Project Number
581-22-700
Building Number
TRAILER 1
Drawing Number
S-107
030 OF 693



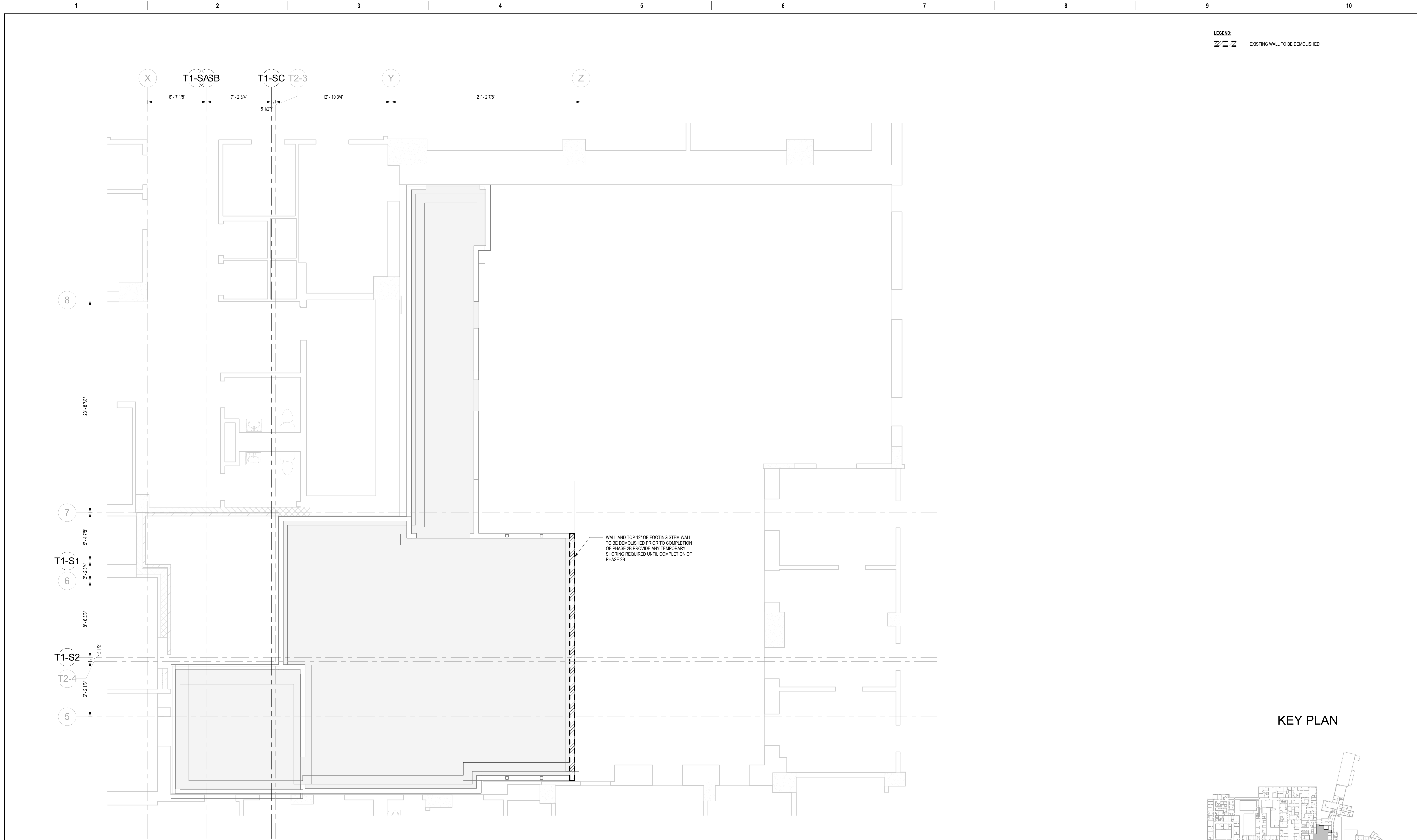
SHOWN AREA



031 OF 693

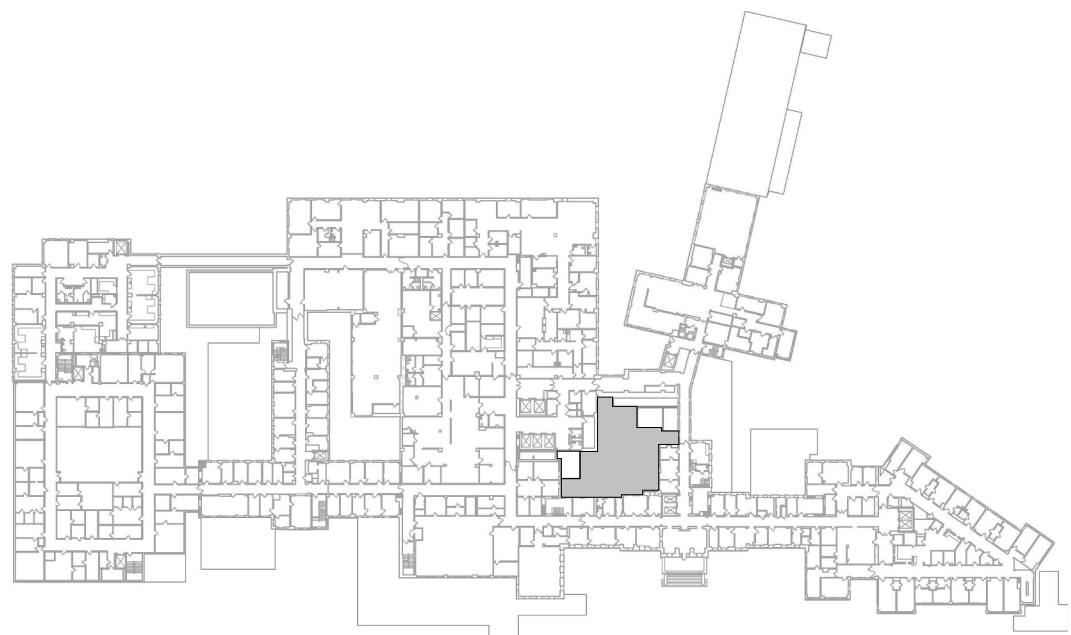
U.S. Department
of Veterans Affairs

Revisions:



LEGEND:
ZZZ EXISTING WALL TO BE DEMOLISHED

KEY PLAN



SHOWN AREA



F1 DEMOLITION PLAN - DATA CENTER - PHASE 2B
1/4" = 1'-0"

Revisions:	Date:

CONSULTANT
ENGINEERING DISCIPLINE:
wsp
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RAY SPEES
SDB SPEESDESIGNBUILD

STAMP
Professional Engineer Seal for RAY SPEES, No. 5732, State of California.

Office of Construction and Facilities Management
VA U.S. Department of Veterans Affairs

Drawing Title
DEMOLITION PLAN - DATA CENTER

Approved:

Phase
100% CONSTRUCTION DOCUMENTS

Project Title
EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

Location
VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

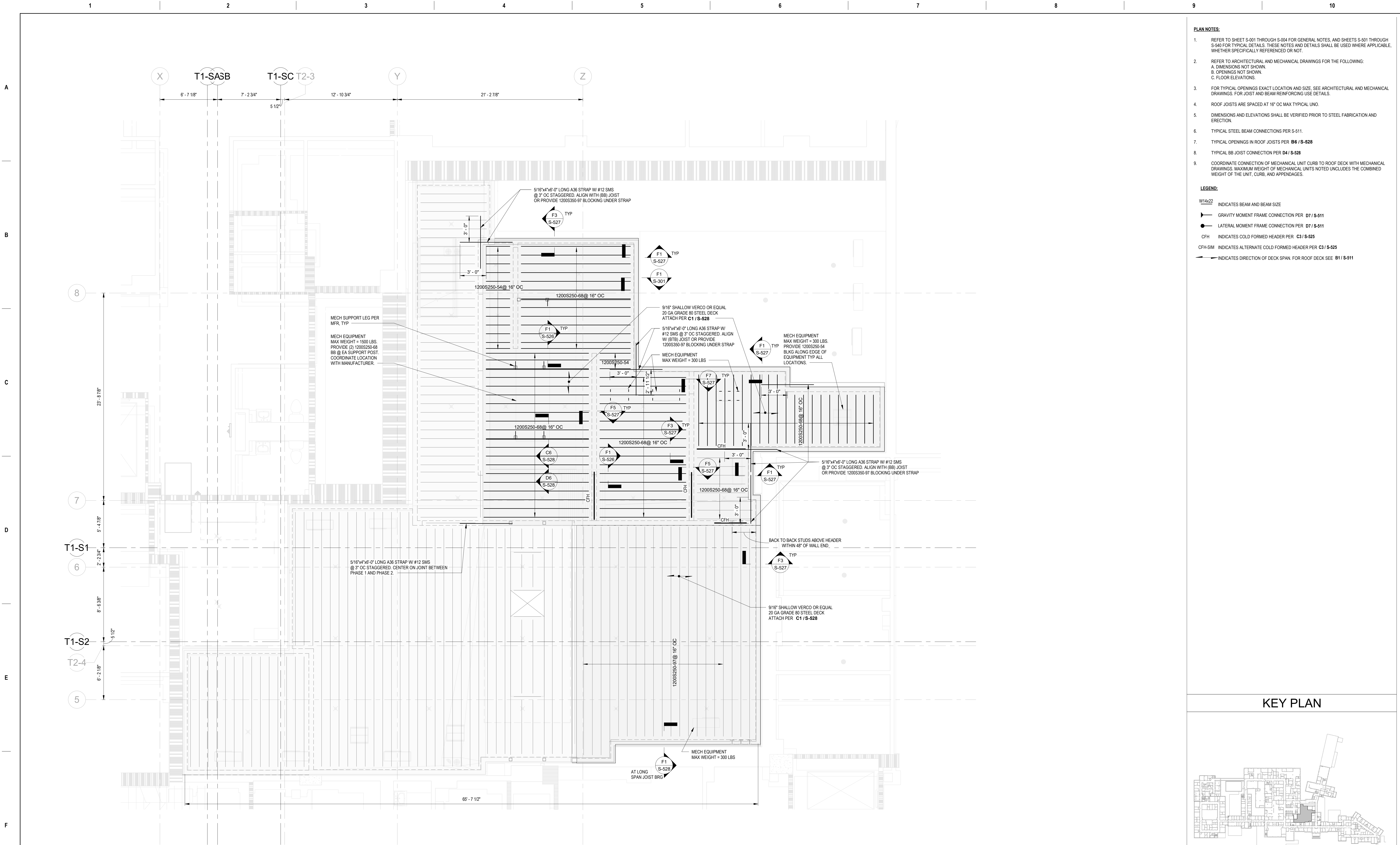
Issue Date
09/03/2025

Checked
CH

Drawn
Author

Project Number
581-22-700
Building Number
GC115-1

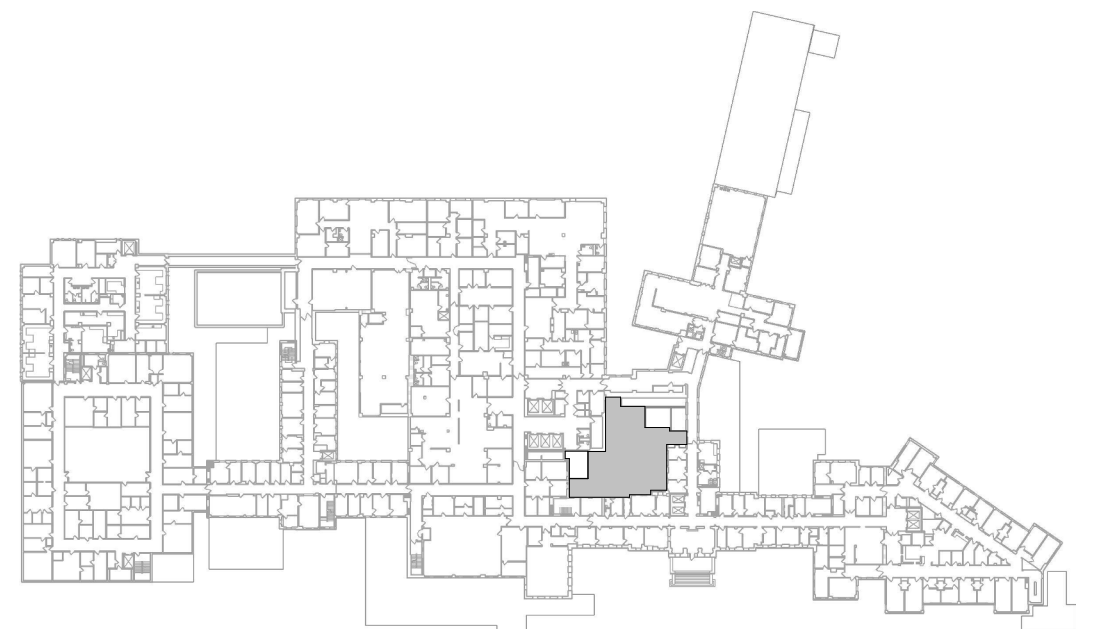
Drawing Number
SD105
032 OF 693



- PLAN NOTES:**
- REFER TO SHEET S-001 THROUGH S-004 FOR GENERAL NOTES, AND SHEETS S-501 THROUGH S-540 FOR TYPICAL DETAILS. THESE NOTES AND DETAILS SHALL BE USED WHERE APPLICABLE, WHETHER SPECIFICALLY REFERENCED OR NOT.
 - REFER TO ARCHITECTURAL AND MECHANICAL DRAWINGS FOR THE FOLLOWING:
A. DIMENSIONS NOT SHOWN.
B. OPENINGS NOT SHOWN.
C. FLOOR ELEVATIONS.
 - FOR TYPICAL OPENINGS EXACT LOCATION AND SIZE. SEE ARCHITECTURAL AND MECHANICAL DRAWINGS FOR JOIST AND BEAM REINFORCING USE DETAILS.
 - ROOF JOISTS ARE SPACED AT 16" OC MAX TYPICAL UNO.
 - DIMENSIONS AND ELEVATIONS SHALL BE VERIFIED PRIOR TO STEEL FABRICATION AND ERECTION.
 - TYPICAL STEEL BEAM CONNECTIONS PER S-511.
 - TYPICAL OPENINGS IN ROOF JOISTS PER **B6 / S-528**.
 - TYPICAL BB JOIST CONNECTION PER **D4 / S-528**.
 - COORDINATE CONNECTION OF MECHANICAL UNIT CURB TO ROOF DECK WITH MECHANICAL DRAWINGS. MAXIMUM WEIGHT OF MECHANICAL UNITS NOTED UNCLUSTERS THE COMBINED WEIGHT OF THE UNIT, CURB, AND APPENDAGES.

- LEGEND:**
- W14x22 INDICATES BEAM AND BEAM SIZE
- GRAVITY MOMENT FRAME CONNECTION PER **D7 / S-511**
- LATERAL MOMENT FRAME CONNECTION PER **D7 / S-511**
- CFH INDICATES COLD FORMED HEADER PER **C3 / S-525**
- CFH-SIM INDICATES ALTERNATE COLD FORMED HEADER PER **C3 / S-525**
- INDICATES DIRECTION OF DECK SPAN. FOR ROOF DECK SEE **B1 / S-511**

KEY PLAN



SHOWN AREA

F1 ROOF FRAMING PLAN - DATA CENTER - PHASE 2B

1/4" = 1'-0"

Revisions:	Date:

CONSULTANT

ENGINEERING DISCIPLINE:



WSP USA Inc.
33301 9th Avenue South
Suite 300
Federal Way, WA 98003-2600
TEL: (206) 431-2300
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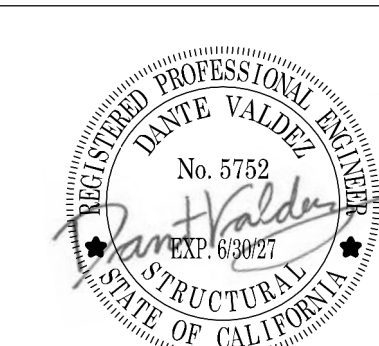
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(206) 590-2118
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STAMP



Office of
Construction
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Management



U.S. Department
of Veterans Affairs

Drawing Title

ROOF FRAMING PLAN - DATA
CENTER - PHASE 2

Approved:

Phase

100% CONSTRUCTION
DOCUMENTS

Project Title

EHRM INFRASTRUCTURE
UPGRADES - HUNTINGTON

Location

VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date

09/03/2025

Checked

CH

Drawn

Author

Project Number

581-22-700

Building Number

GC115-1

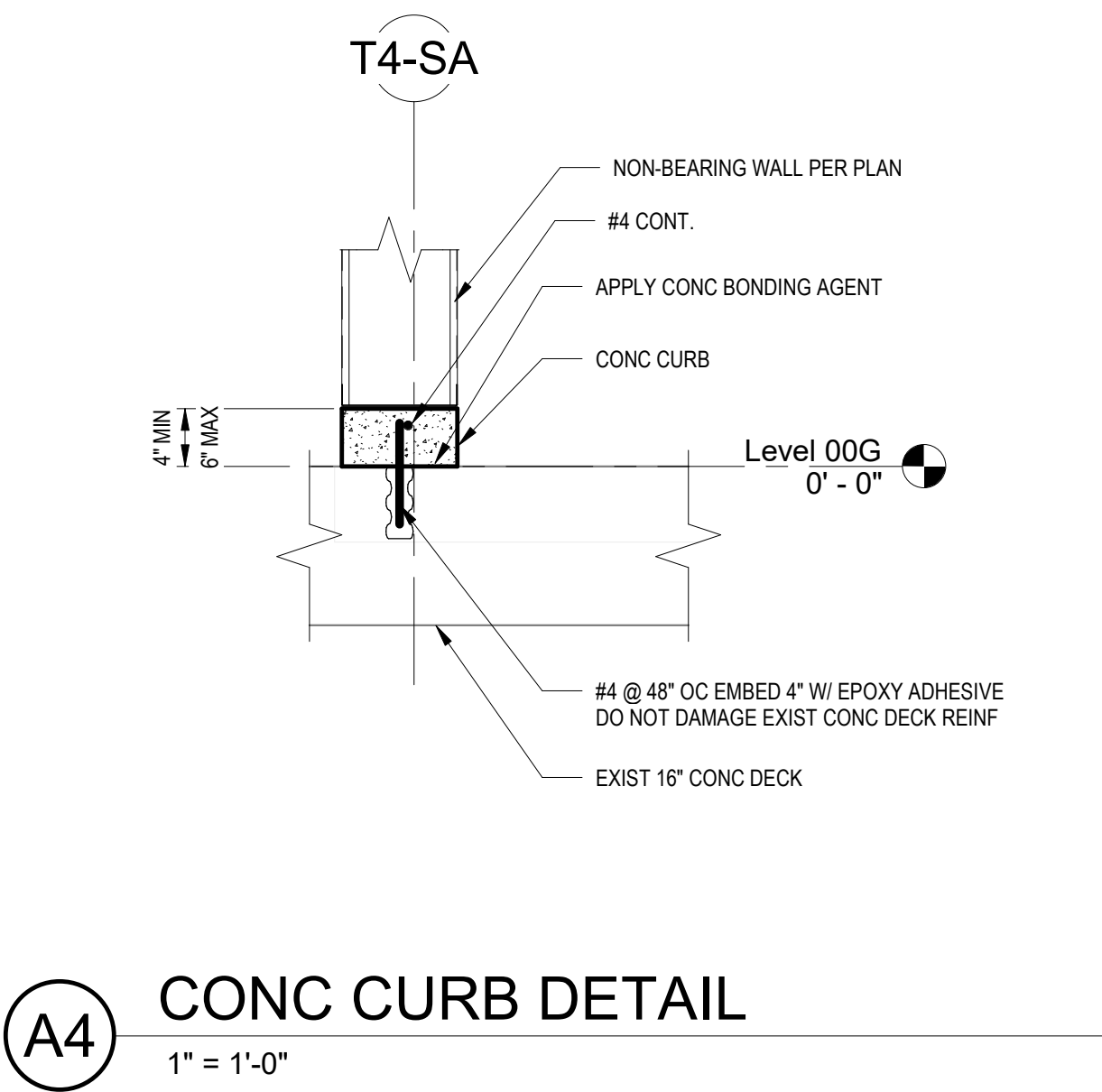
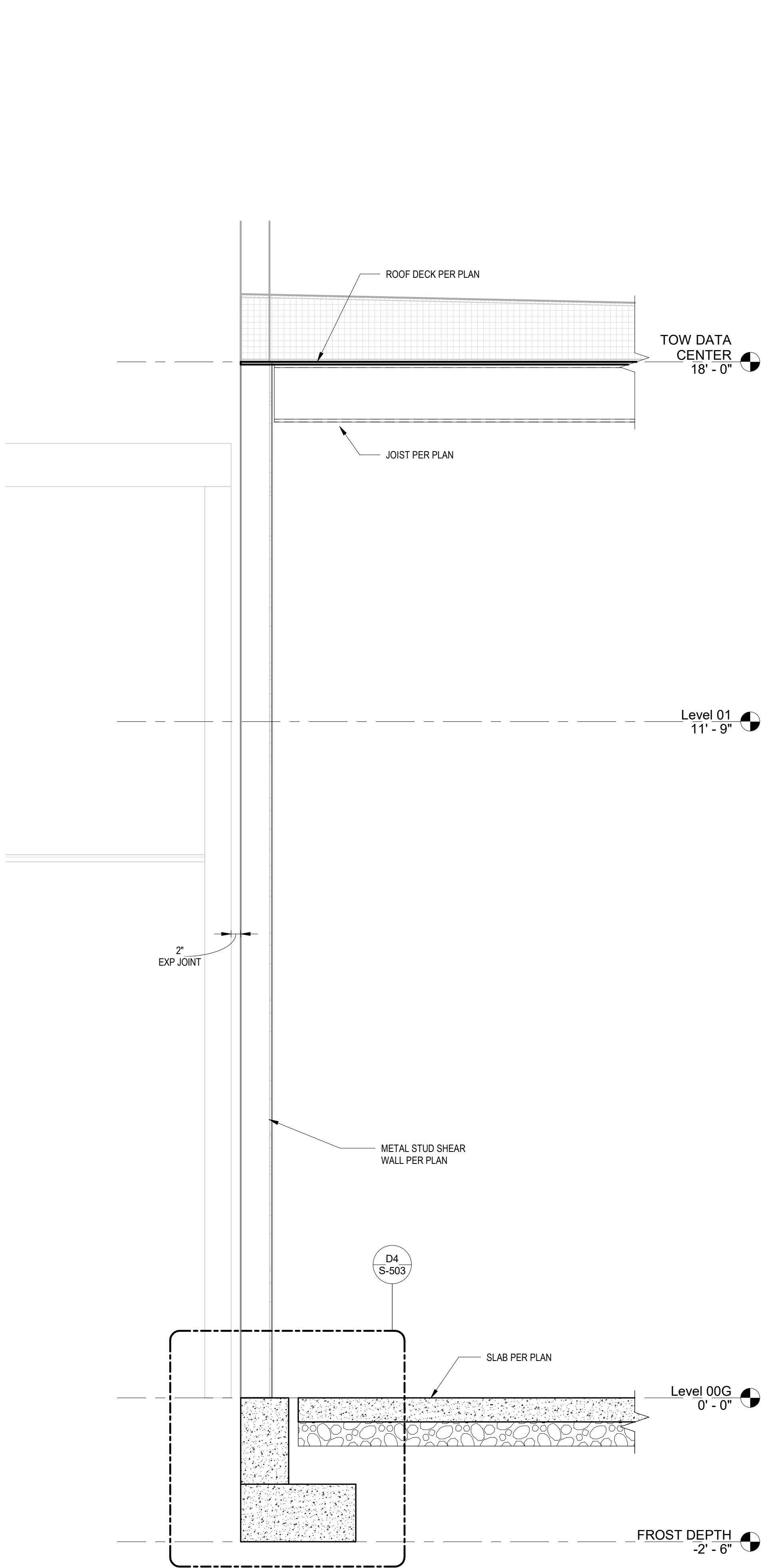
Drawing Number

S-116-2B

033 OF 693

A
B
C
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E
F

A
B
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E
F



F1 TYPICAL SECTION
3/4" = 1'-0"

Revisions:	Date:

CONSULTANT

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STAMP



Office of
Construction
and Facilities
Management

 U.S. Department
of Veterans Affairs

Drawing Title

BUILDING SECTIONS - DATA
CENTER

Approved:

Phase

100% CONSTRUCTION
DOCUMENTS

Project Title

EHRM INFRASTRUCTURE
UPGRADES - HUNTINGTON

Location

VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date

09/03/2025

Checked

CH

Drawn

JKM

Project Number

581-22-700

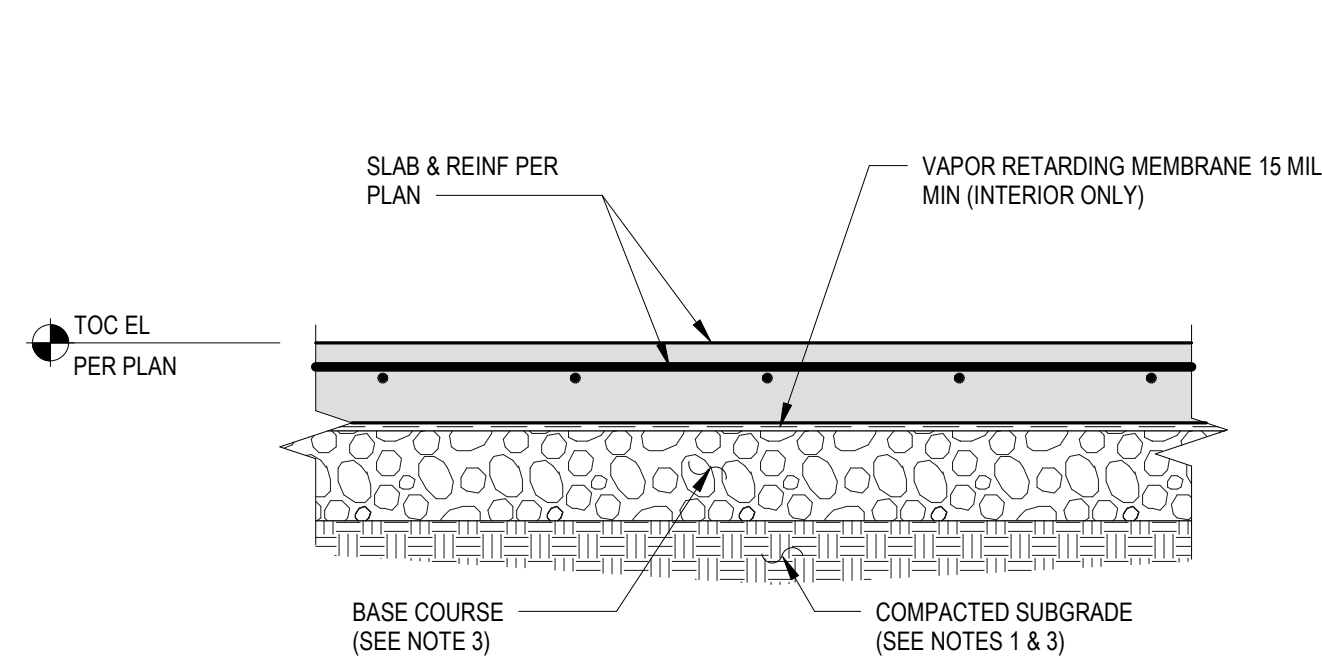
Building Number

GC115-1

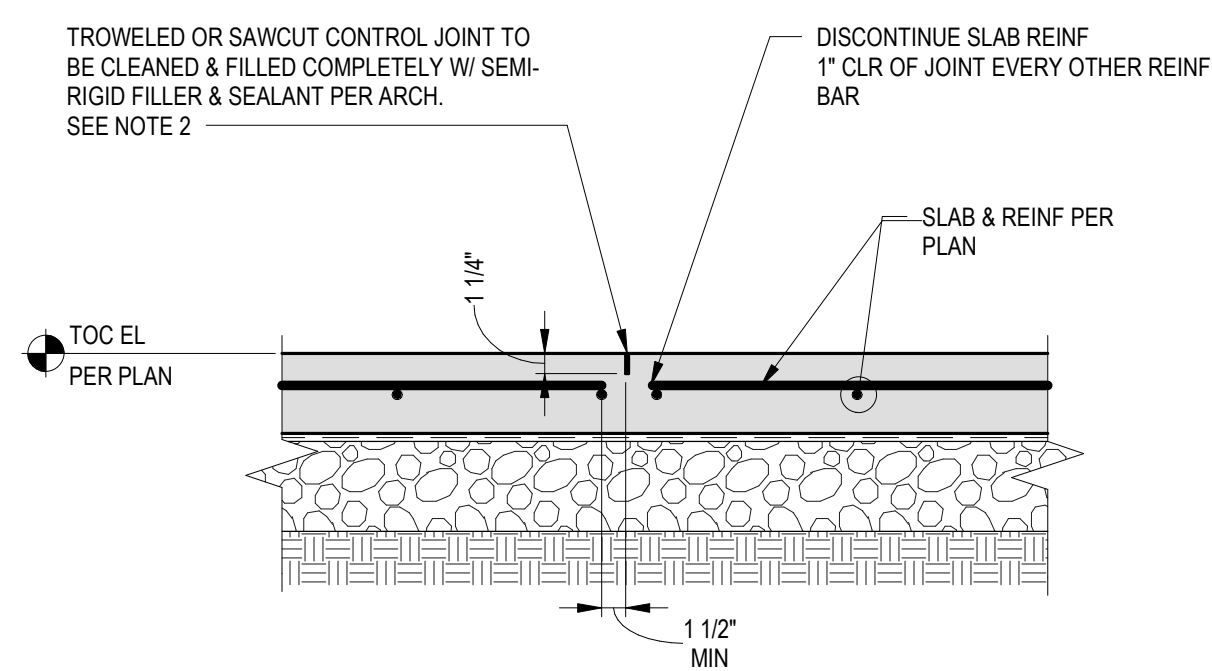
Drawing Number

S-301

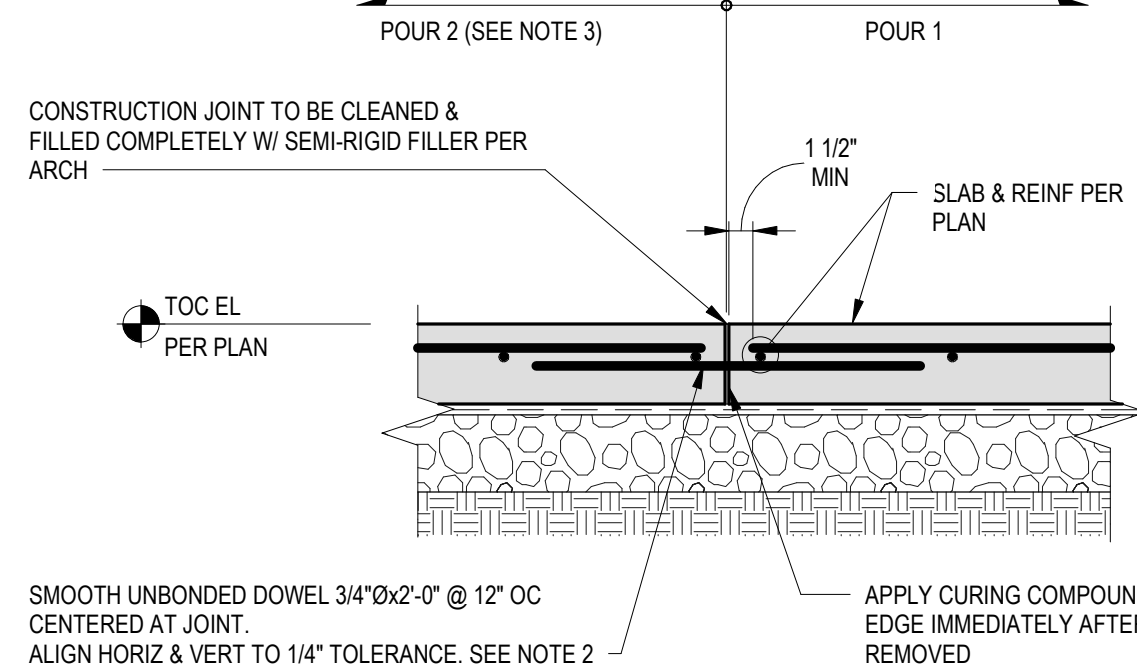
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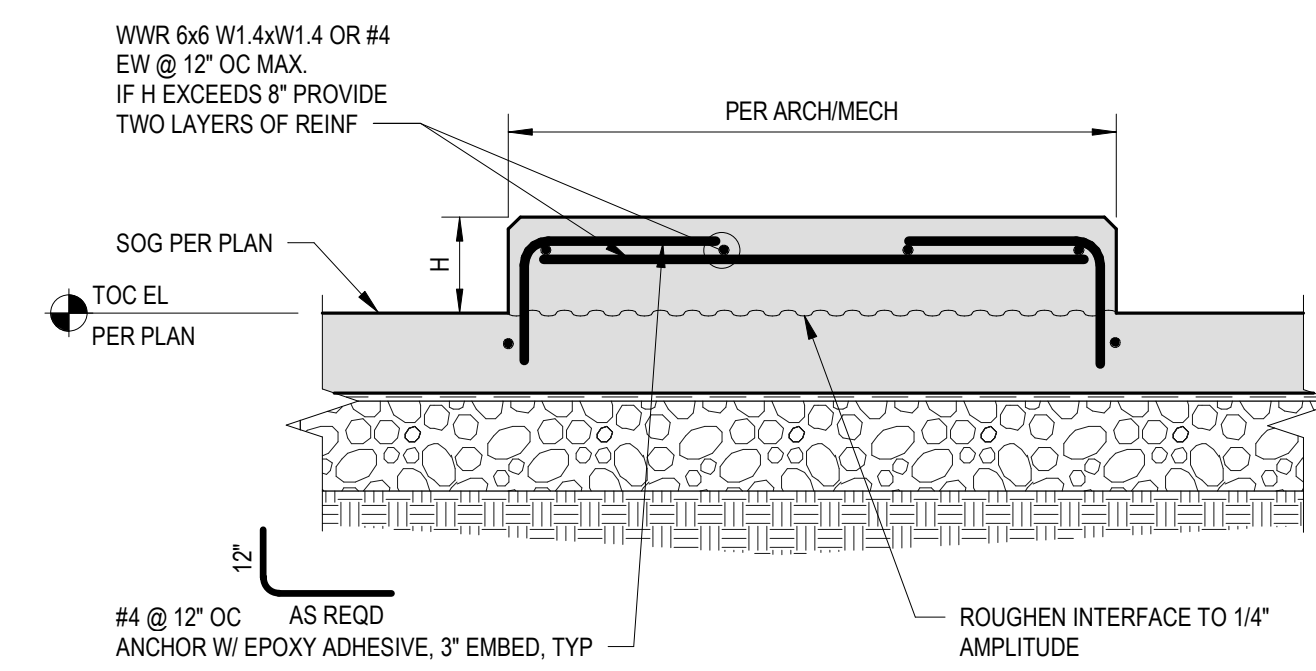
- NOTES:
- SEE GEOTECHNICAL REPORT FOR ADDITIONAL REQUIREMENTS.
 - SEE PLANS FOR LOCATION OF COLUMN BLOCKOUTS, JOINTS AND SLAB DEPRESSIONS.
 - 4" MIN SAND AND GRAVEL BASE COURSE UNO IN GEOTECHNICAL REPORT. COMPACT SUBGRADE PER GEOTECHNICAL REPORT.



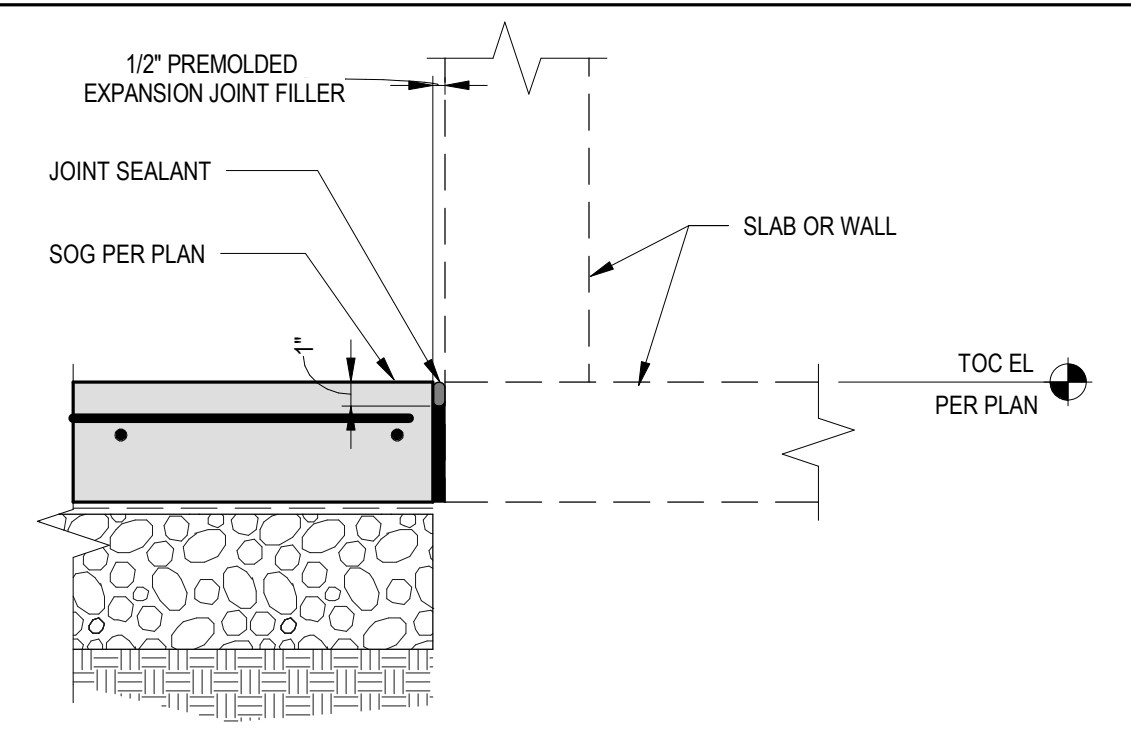
- NOTES:
- LOCATE CONTROL JOINTS PER PLAN.
 - SAWCUT JOINTS AFTER PEAK HEAT OF HYDRATION BUT NO LATER THAN 12 HOURS AFTER CONCRETE HAS BEEN MIXED.
 - FOR SLAB SUBGRADE SEE TYPICAL SLAB-ON-GRADE DETAIL.



- NOTES:
- LOCATE CONSTRUCTION JOINTS PER PLAN.
 - OPTION: PLASTIC SLEEVES MAY BE CAST IN POUR 1 (SPEED DOWEL OR SIMILAR) SO DOWELS MAY BE INSTALLED AFTER EDGE FORM REMOVAL.
 - DELAY POUR 2 MINIMUM 24 HOURS AFTER POUR 1.
 - FOR SLAB SUBGRADE SEE TYPICAL SLAB-ON-GRADE DETAIL.

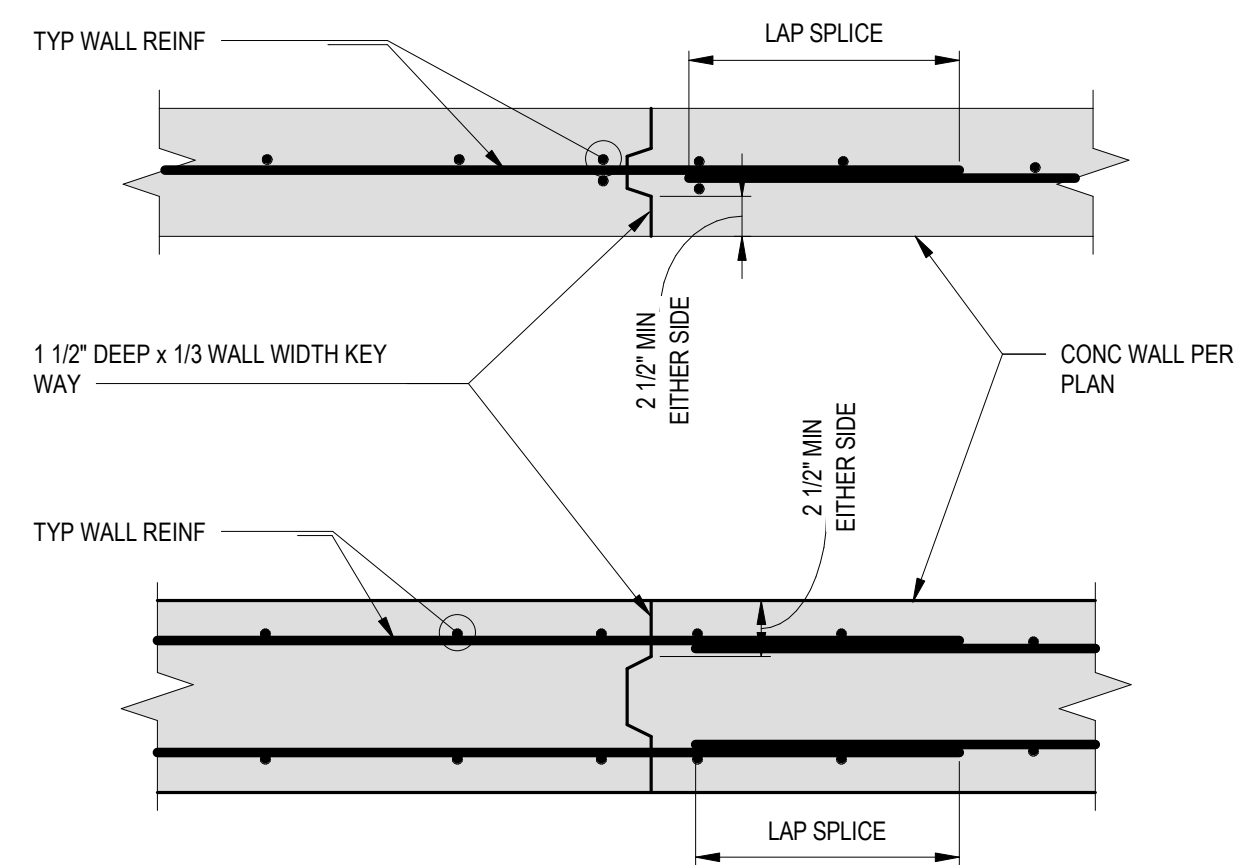


- NOTES:
- SEE ARCHITECTURAL/MECHANICAL DRAWINGS FOR PAD SIZE AND LOCATION.
 - PROVIDE INSERT AS REQUIRED BY ARCHITECTURAL AND MECHANICAL DRAWINGS.
 - 4" SHS12"



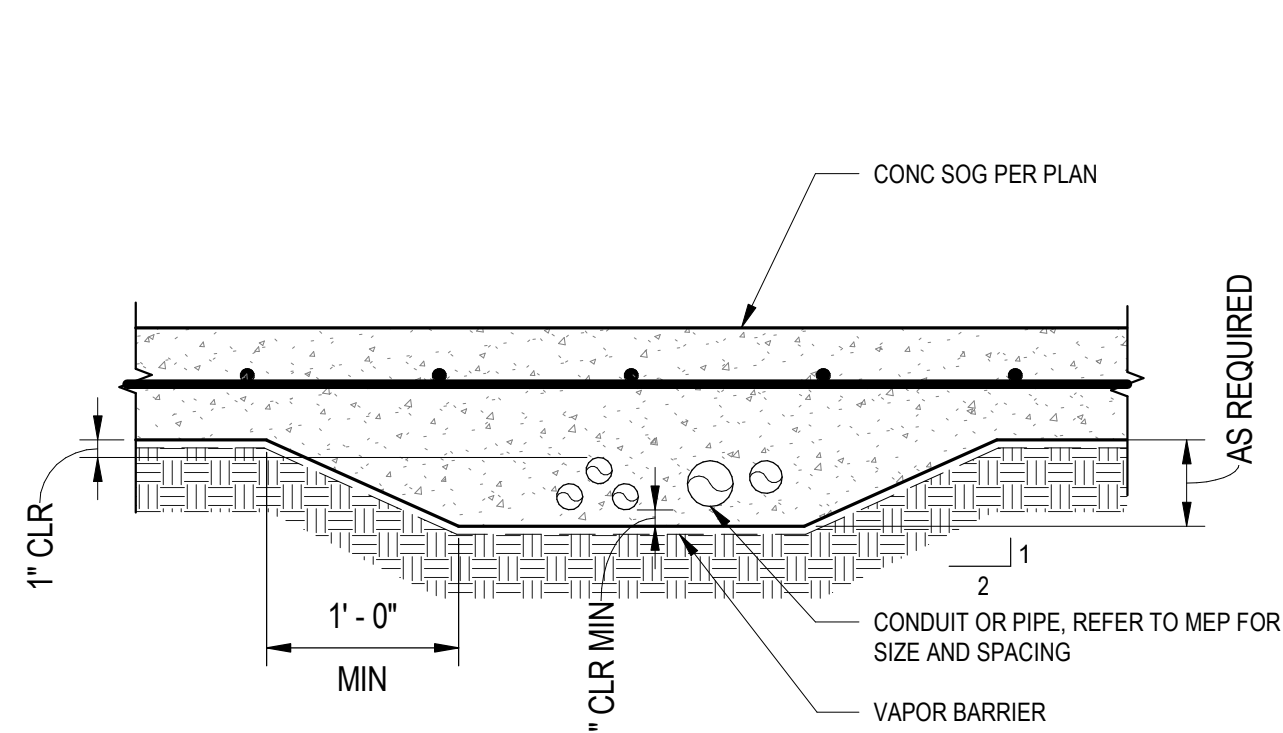
B

B1 TYP SLAB-ON-GRADE DETAIL
1" = 1'-0"

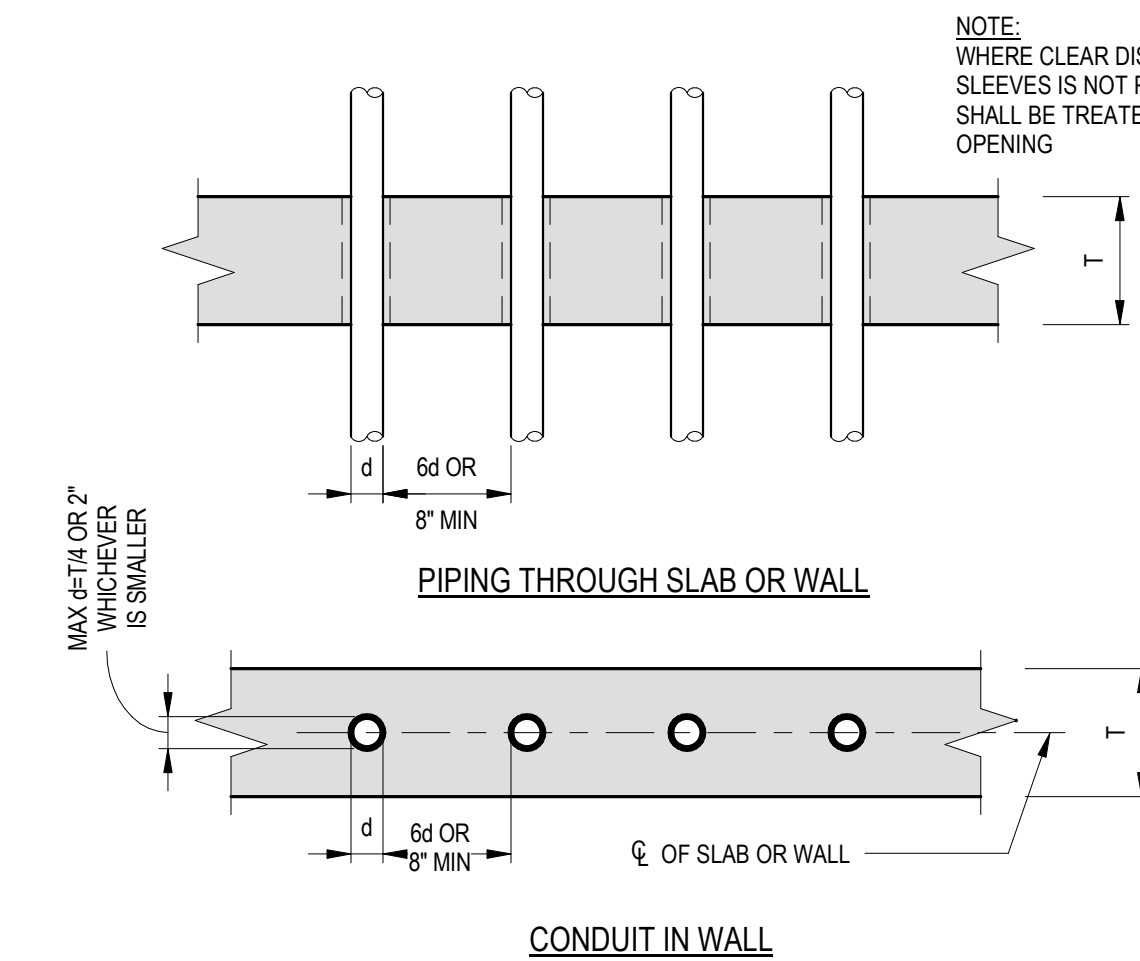


NOTE:
OBTAIN APPROVAL OF ENGINEER FOR LOCATION OF CONSTRUCTION JOINTS.

B3 SLAB CONTROL JOINT
1" = 1'-0"

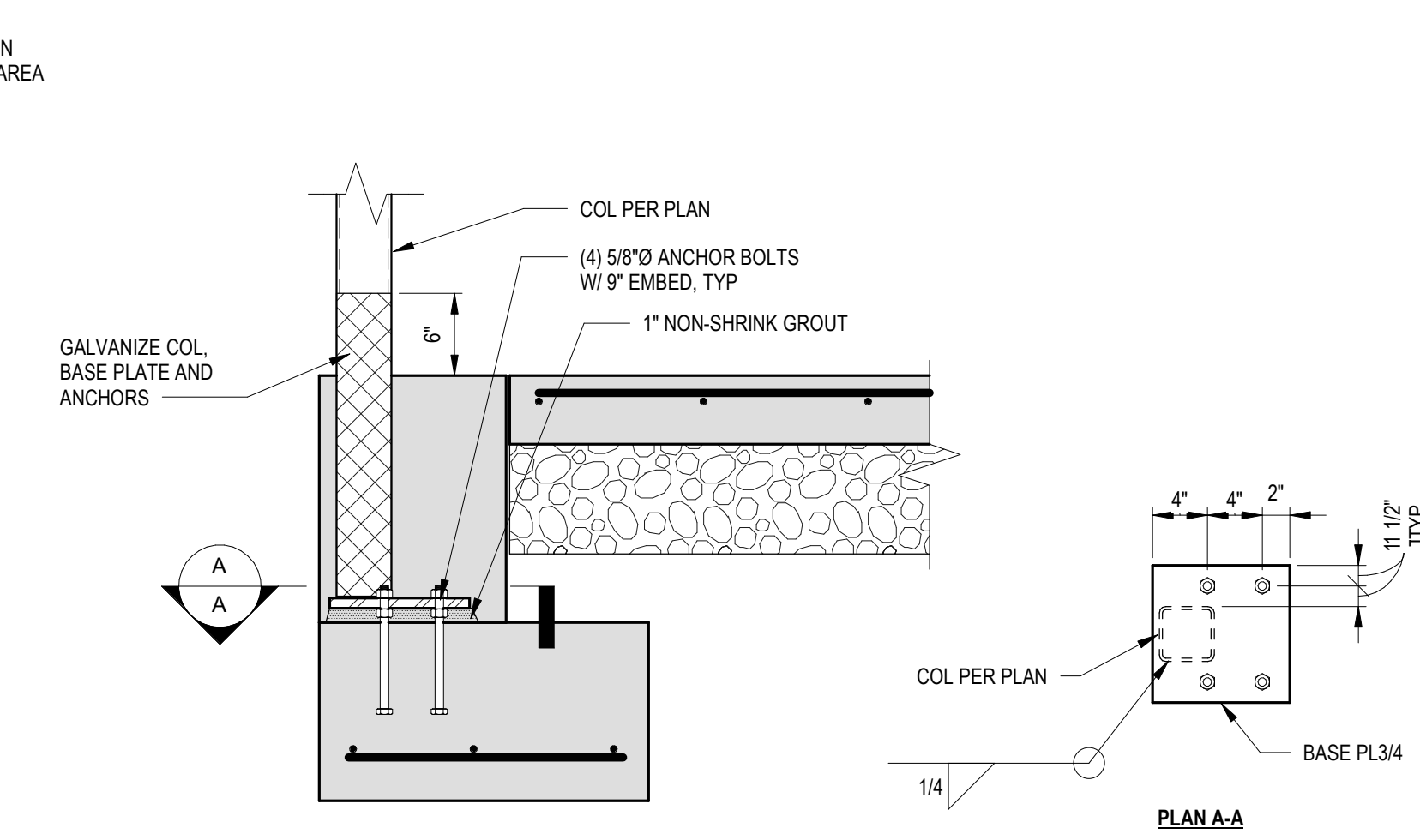


B5 SLAB CONSTRUCTION JOINT
1" = 1'-0"



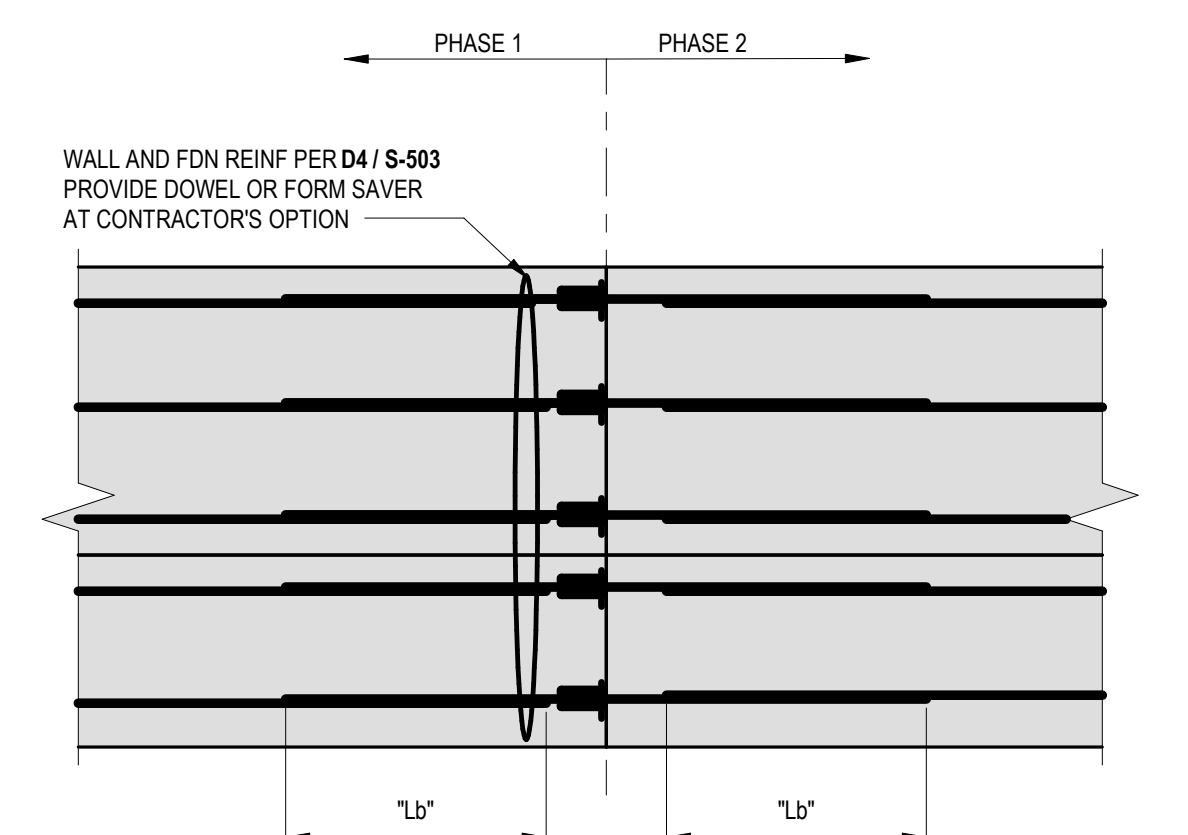
NOTE:
WHERE CLEAR DISTANCE BETWEEN SLEEVES IS NOT PROVIDED, THIS AREA SHALL BE TREATED AS A SLAB OPENING.

B7 HOUSEKEEPING PAD DETAIL
1" = 1'-0"



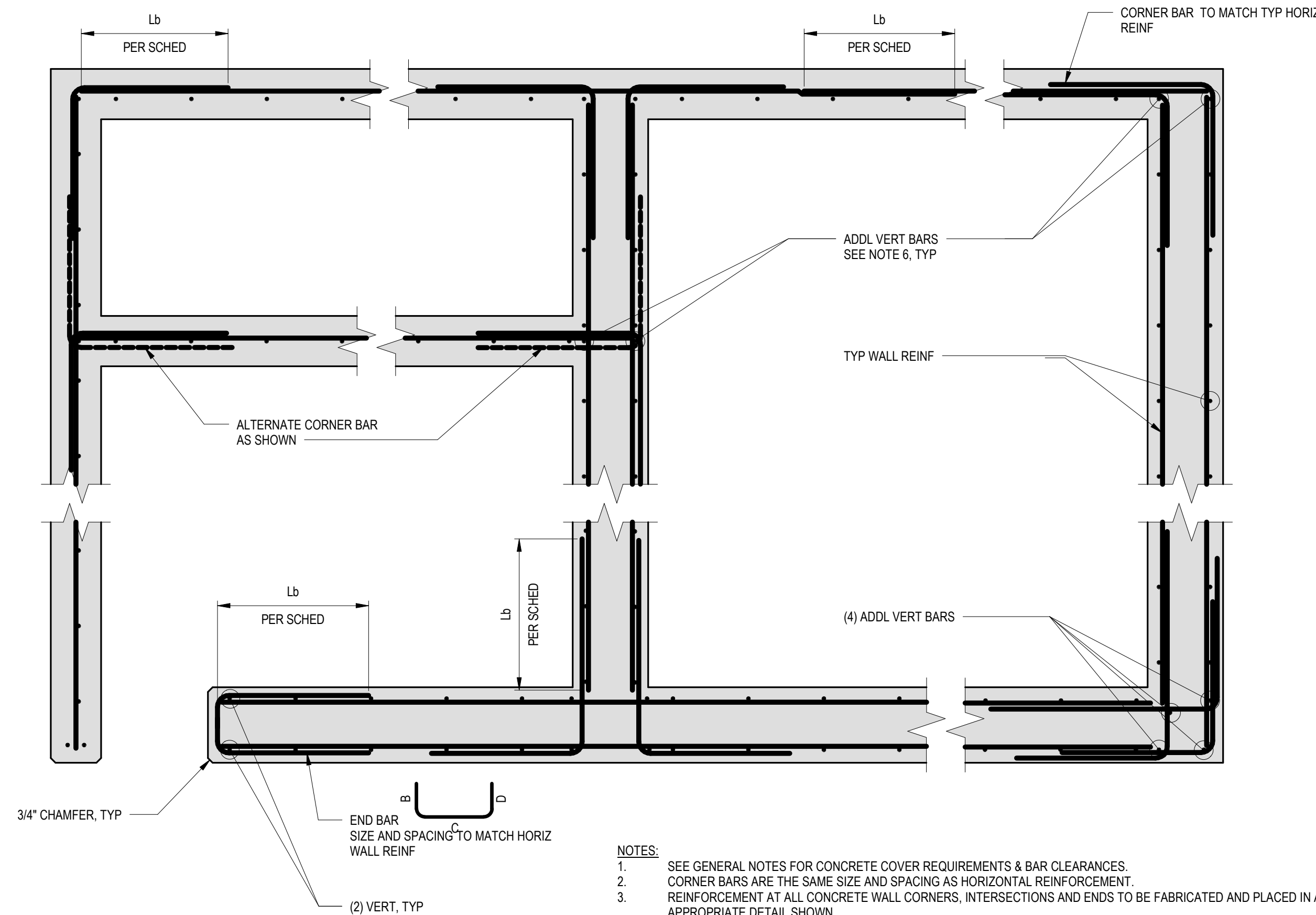
NOTE:
FOR INFORMATION NOT SHOWN REFER TO D4 / S-503 .

B9 TYP SLAB ISOLATION JOINT
1 1/2" = 1'-0"



C

C1 WALL CONSTRUCTION JOINTS
1" = 1'-0"



- NOTES:
- SEE GENERAL NOTES FOR CONCRETE COVER REQUIREMENTS & BAR CLEARANCES.
 - CORNER BARS ARE THE SAME SIZE AND SPACING AS HORIZONTAL REINFORCEMENT.
 - REINFORCEMENT AT ALL CONCRETE WALL CORNERS, INTERSECTIONS AND ENDS TO BE FABRICATED AND PLACED IN ACCORDANCE WITH APPROPRIATE DETAIL SHOWN.
 - DETAIL ALSO APPLIES TO ALL CONTINUOUS WALL FOOTINGS.
 - SINGLE CURTAIN REINFORCEMENT SHALL BE PLACED AT WALL CENTERLINE UNLESS NOTED OTHERWISE.
 - VERTICAL REINFORCEMENT SHOWN IS ADDITIONAL IF TYPICAL VERTICAL REINFORCEMENT IS NOT IN PROPER LOCATION.
 - 90° STANDARD HOOK MAY BE SUBSTITUTED FOR CORNER BARS. USE 90° STANDARD HOOK FOR EMBEDMENT LESS THAN 24" PAST FACE OF WALL.

C3 TYP SOG SLAB AT CONDUIT
1" = 1'-0"

C5 PIPING & CONDUIT AT SLAB OR WALL
1" = 1'-0"

$f_c = 4,000 \text{ psi}$
 $f_y = 60,000 \text{ psi}$

SIZE	Ld	Ldt	Lb	Lbt	Ldh
#4	19	25	25	32	9
#5	24	31	31	40	12
#6	28 (43)	37 (55)	37 (55)	48 (72)	14
#7	42 (62)	54 (81)	54 (81)	70 (105)	17
#8	47 (71)	62 (92)	62 (92)	80 (120)	19
#9	54 (80)	70 (104)	70 (104)	90 (136)	21
#10	60 (90)	78 (117)	78 (117)	102 (153)	24
#11	67 (100)	87 (130)	87 (130)	113 (170)	27

$f_c = 5,000 \text{ psi}$
 $f_y = 75,000 \text{ psi}$

SIZE	Ld	Ldt	Lb	Lbt	Ldh
#4	21	28	28	36	11
#5	27	34	34	45	13
#6	32 (48)	41 (62)	41 (62)	54 (81)	16
#7	46 (70)	60 (90)	60 (90)	78 (118)	19
#8	53 (80)	69 (103)	69 (103)	90 (134)	21
#9	60 (90)	78 (117)	78 (117)	101 (152)	24
#10	67 (101)	88 (131)	88 (131)	114 (171)	27
#11	75 (112)	97 (146)	97 (146)	126 (190)	30

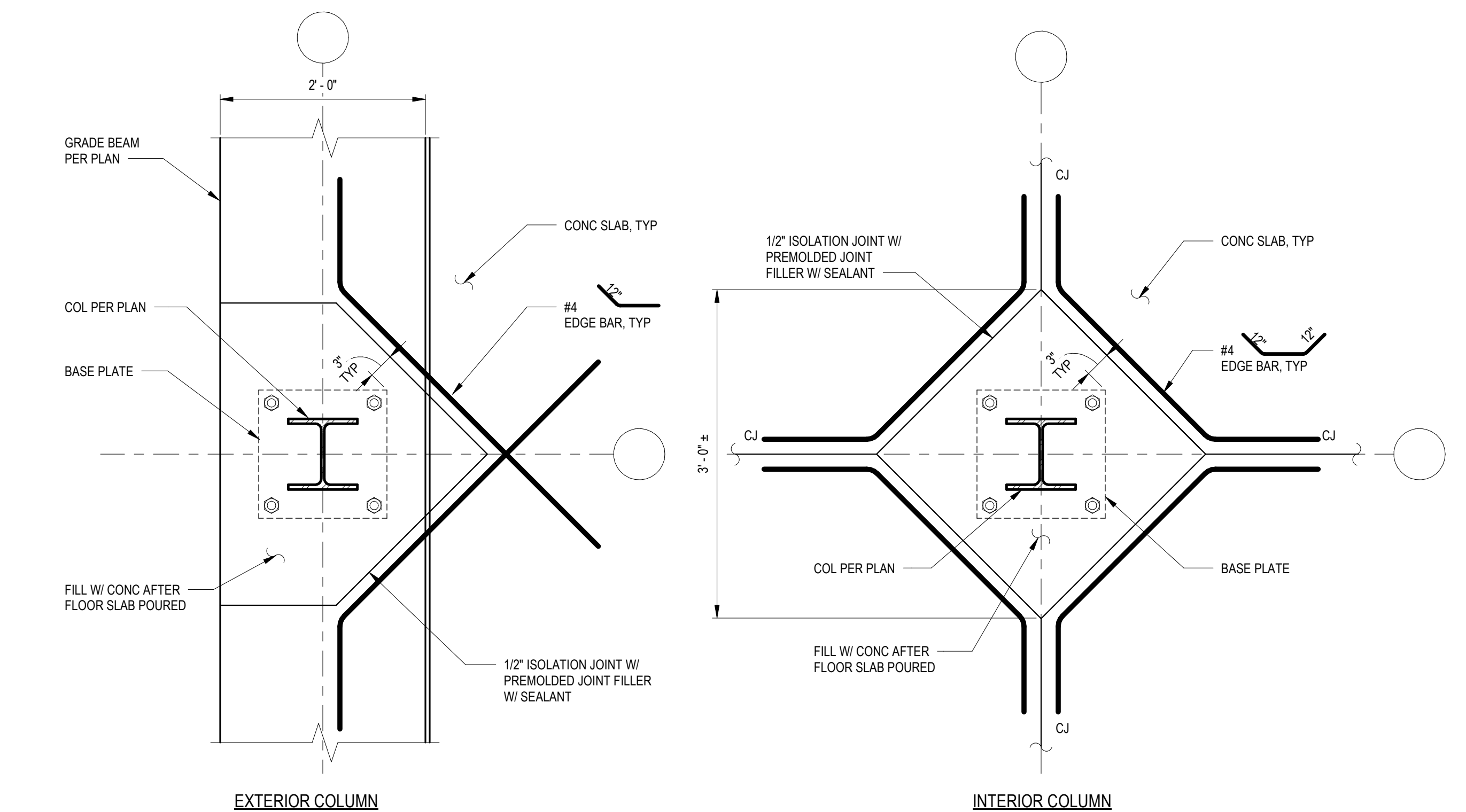
- NOTES:
- USE THE LENGTH IN THESE SCHEDULES, UNLESS NOTED OTHERWISE.
 - USE LENGTH IN ("") WHEN BAR COVER IS d_b OR LESS OR BAR CLEAR SPACING IS $2d_b$ OR LESS.
 - TOP BAR IS HORIZONTAL BAR WITH MORE THAN 12" OF FRESH CONCRETE CAST BELOW IT.

ABBREVIATIONS

d_b = BAR DIAMETER
 L_d = TENSION DEVELOPMENT LENGTH
 L_{dt} = TENSION DEVELOPMENT LENGTH FOR A TOP BAR
 L_b = CLASS B LAP SPlice LENGTH, 1.3 L_d
 L_{bt} = CLASS B LAP SPlice LENGTH FOR A TOP BAR, 1.3 L_{dt}
 L_{dh} = TENSION DEVELOPMENT LENGTH FOR A STANDARD HOOK
 f_c = CONCRETE COMPRESSIBLE STRENGTH AT 28 DAYS
 f_y = BAR YIELD STRENGTH

F5 DEVELOPMENT & LAP SPLICE LENGTH SCHED - CONCRETE
1/2" = 1'-0"

C7 HSS COLUMN BASE
1" = 1'-0"



C9 FDTN JOINT NEW TO EXISTING
1" = 1'-0"

D

E

F

F1 TYP HORIZ REINF PLACING DETAIL FOR CONCRETE WALLS AND STEM WALLS
3/4" = 1'-0"

Revisions:	Date:

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PROFESSIONAL ENGINEER
DATE MADE: 05/20/2025
No. 5732
RAY SPEES
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STATE OF CALIFORNIA

Office of
Construction
and Facilities
Management

VA U.S. Department
of Veterans Affairs

F7 TYP ISOLATION JOINT AT COLUMN
1" = 1'-0"

Drawing Title
TYPICAL CONCRETE DETAILS

Approved:

Phase
100% CONSTRUCTION DOCUMENTS

Project Title
EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

Location
VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date
09/03/2025

Checked
CH

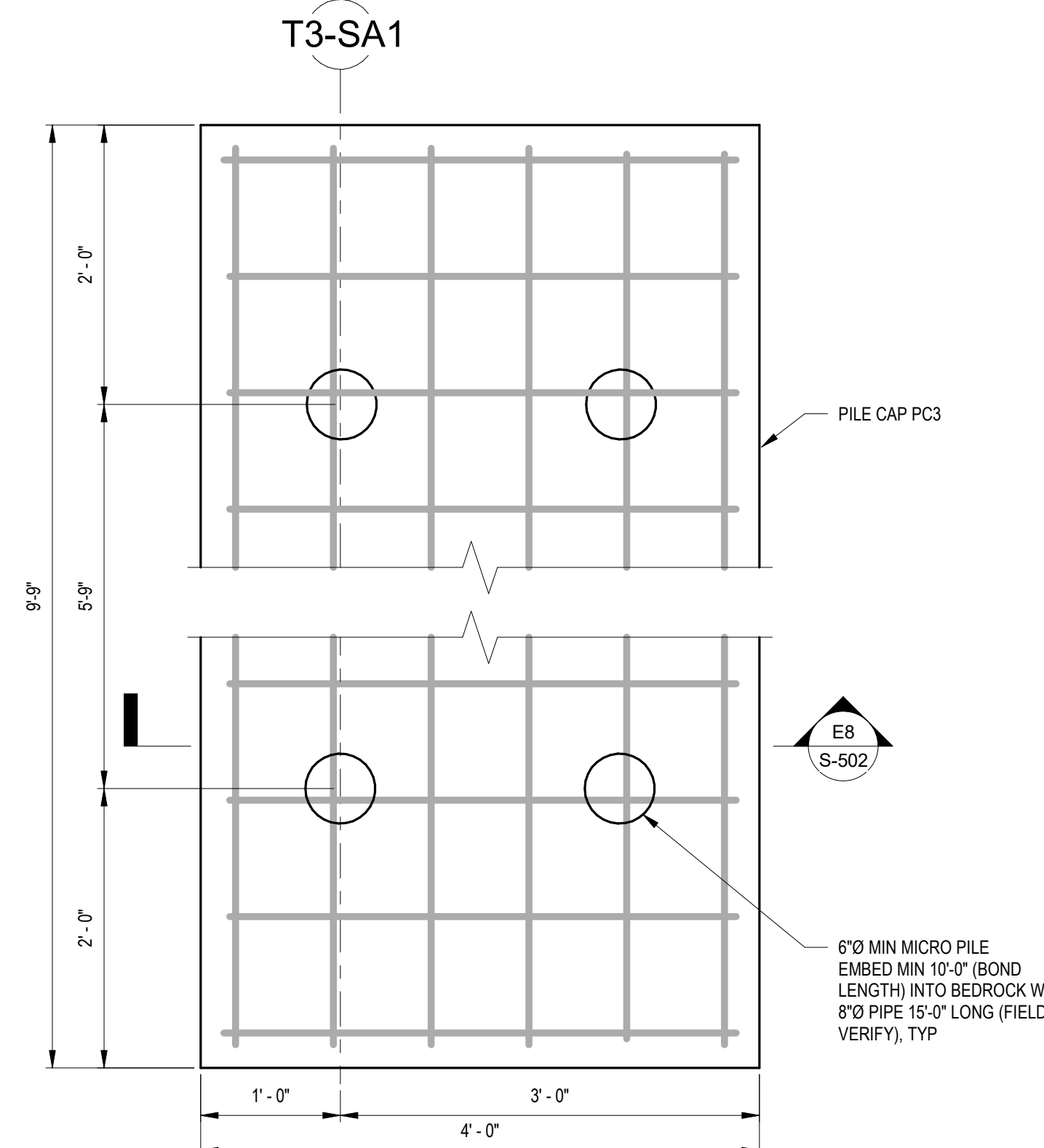
Drawn
JKM

Project Number
581-22-700

Building Number

Drawing Number
S-501

035 OF 693



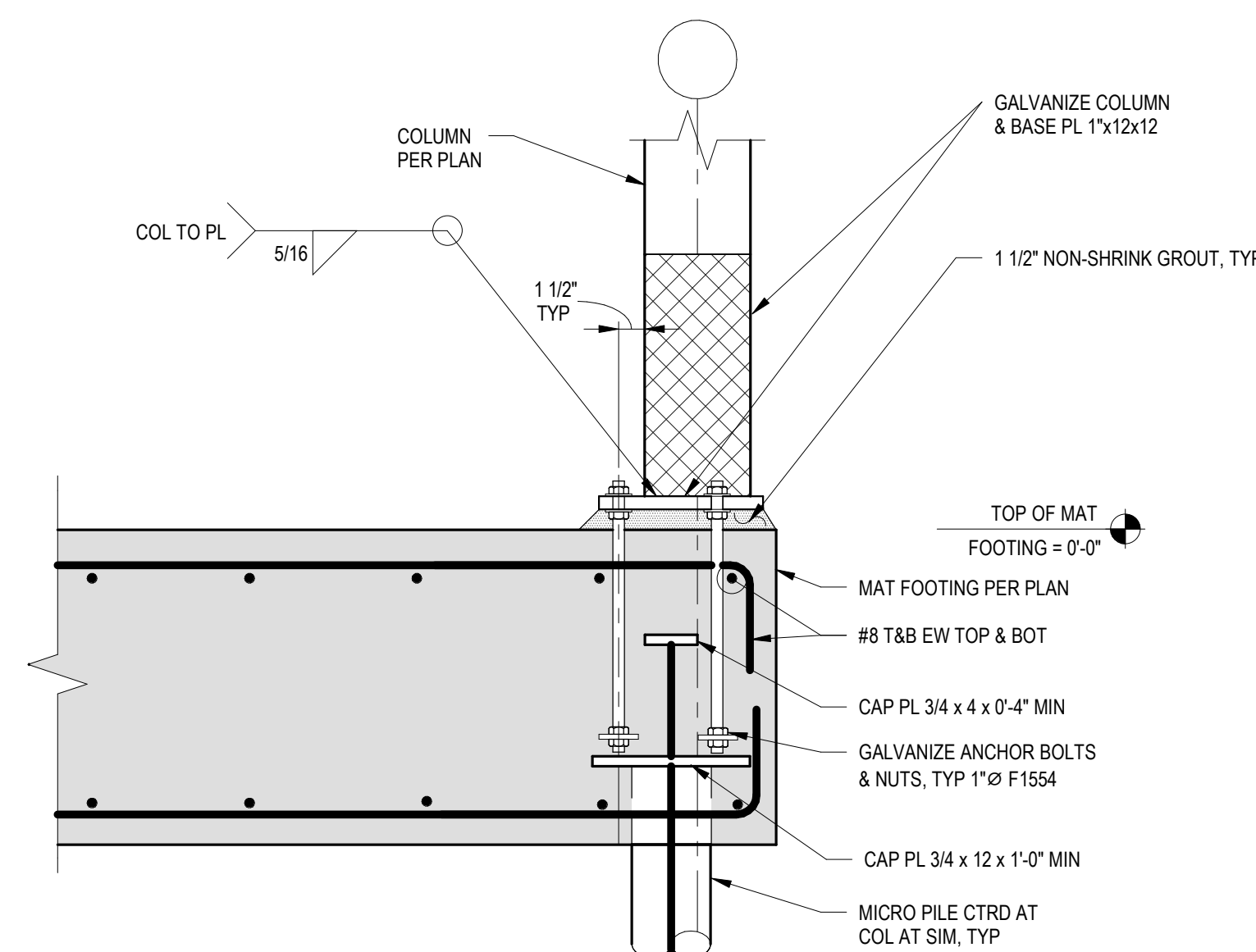
NOTES:

1. EACH MICRO PILE TO BE DESIGNED FOR MAXIMUM REACTION OF 70 IPS (LRFD)
2. SEE DETAIL C1/S-502 FOR PROOF LOAD TESTING CRITERIA.

NOTES:

1. EACH MICRO PILE TO BE DESIGNED FOR MAXIMUM REACTION OF 50 KIPS (LRFD)
2. SEE DETAIL C1/S-502 FOR PROBE LOAD TESTING CRITERIA

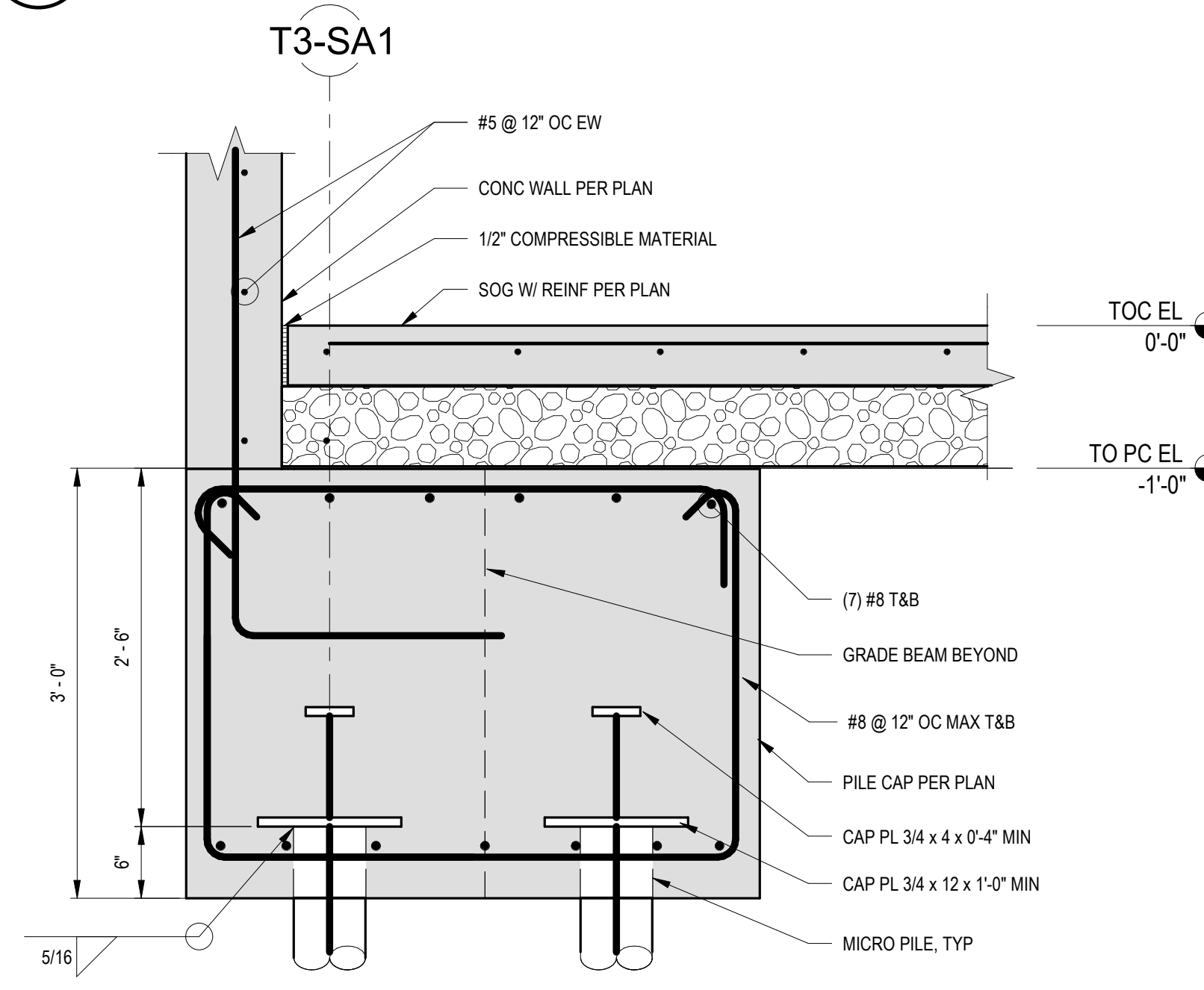
C8 PILE CAP PC3
1" = 1'-0"



NOTES:

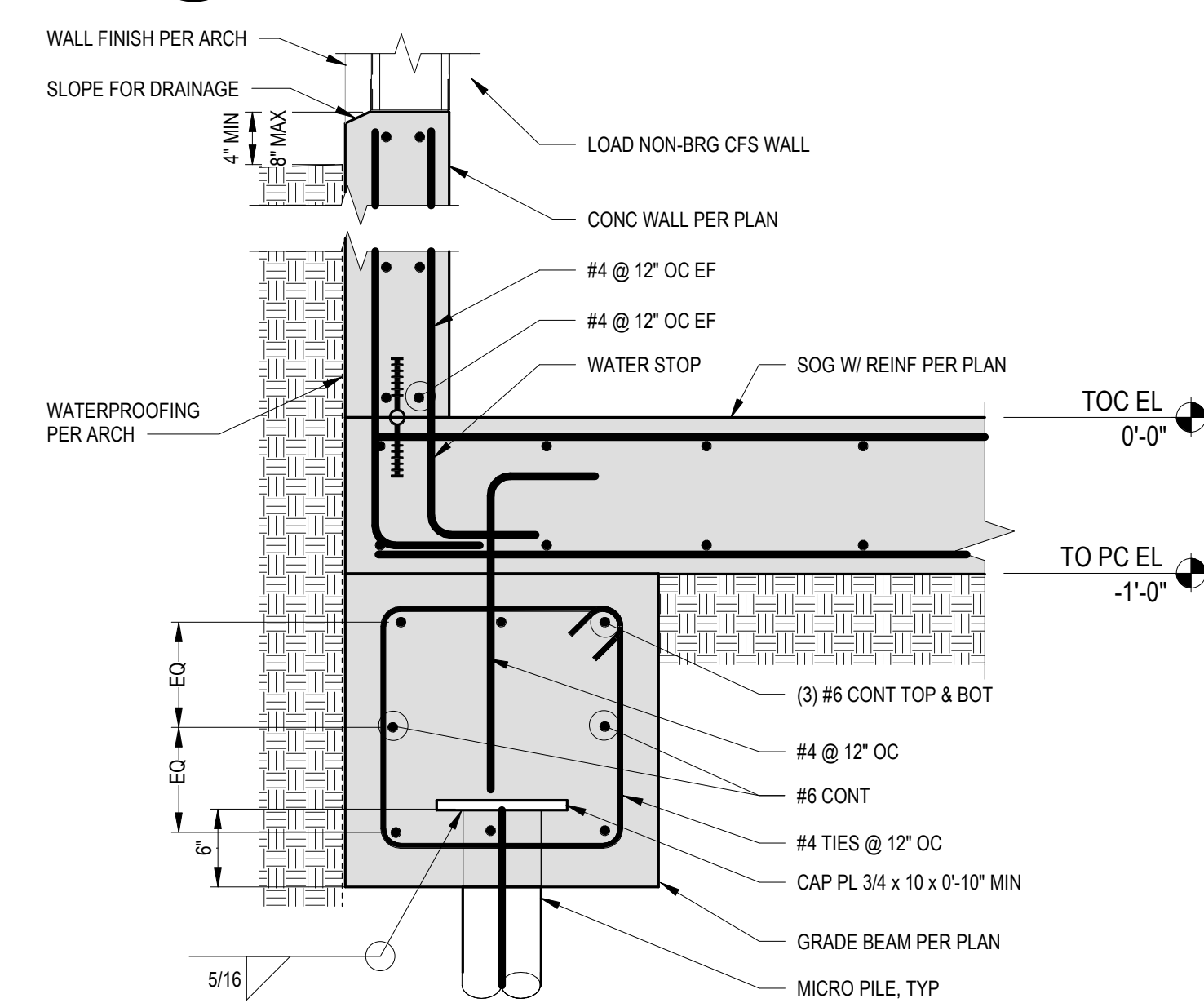
1 ADDITIONAL INFORMATION NOT SHOWN PER C1 / S-502

E8 PILE CAP SECTION-PC3
1" = 1'-0"



COLUMN CENTERED ON EXIST WALL

BASE PL5/8 W/ (4) 3/4"Ø EPOXY ANCHORS.
REMOVE TOPPING AROUND BASE PL



F9 CONC PEDESTAL
1" = 1'-0"

Project Title EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON			Project Number 581-22-700	
			Building Number	
Location VA HUNTINGTON 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300			Drawing Number S-502	
Issue Date 08/03/2025	Checked CH	Drawn JKM	036 OF 693	

A

B

C

D

E

F

A

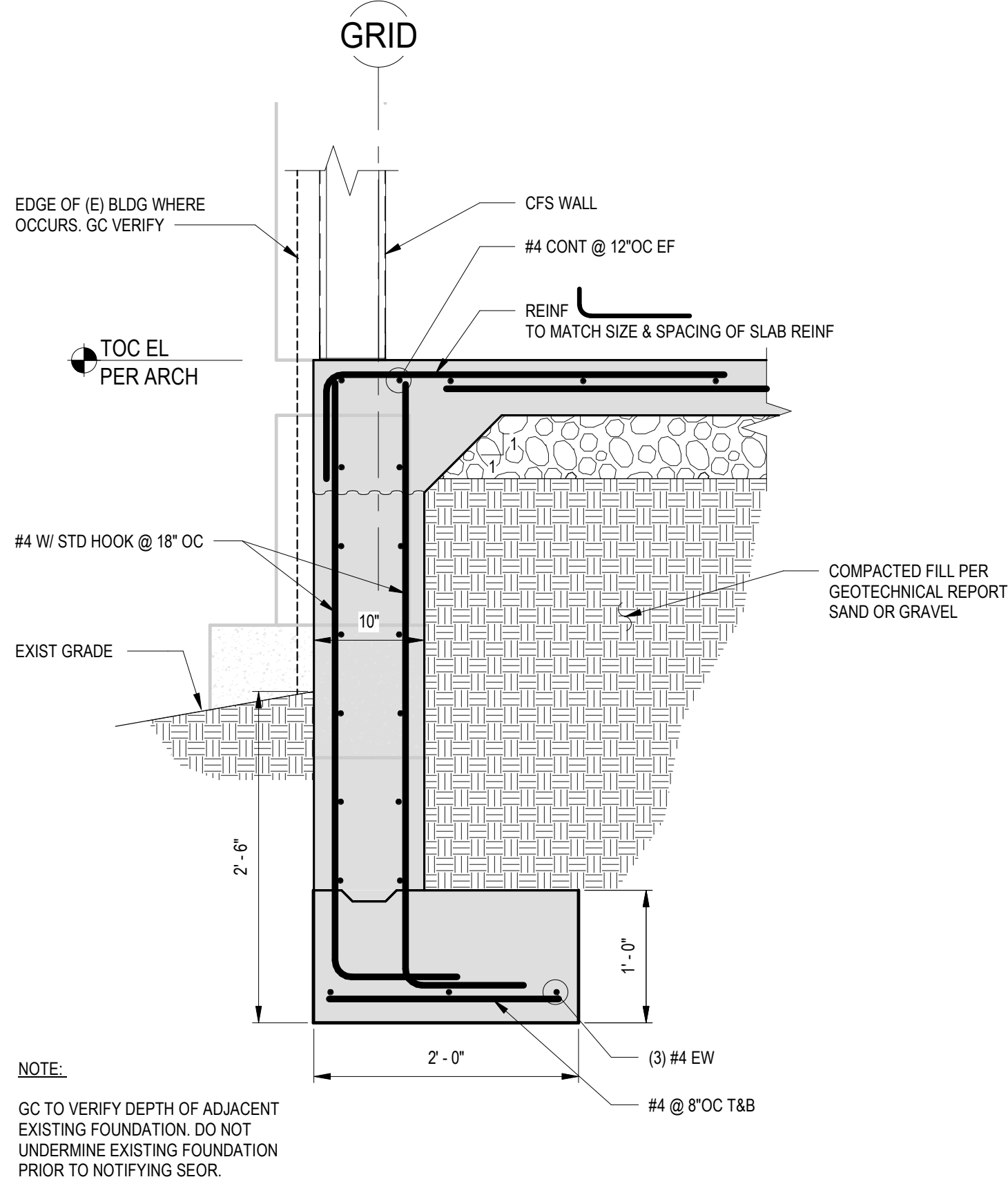
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C

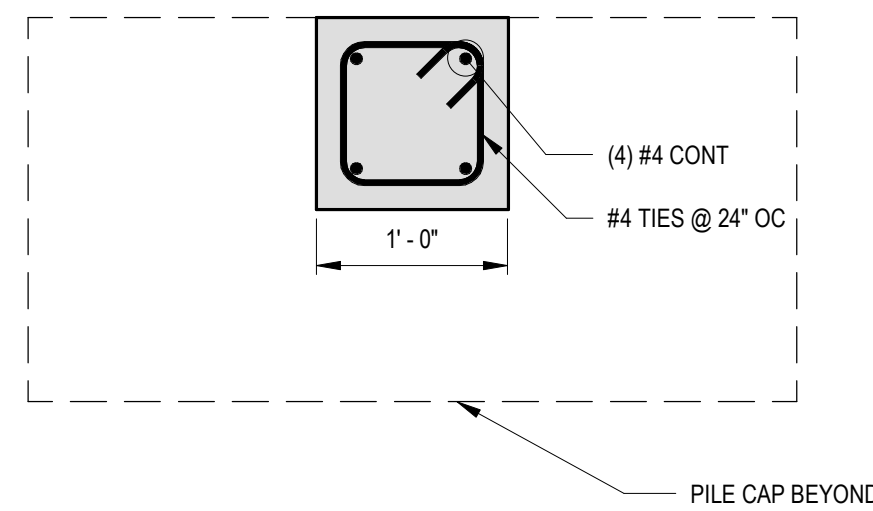
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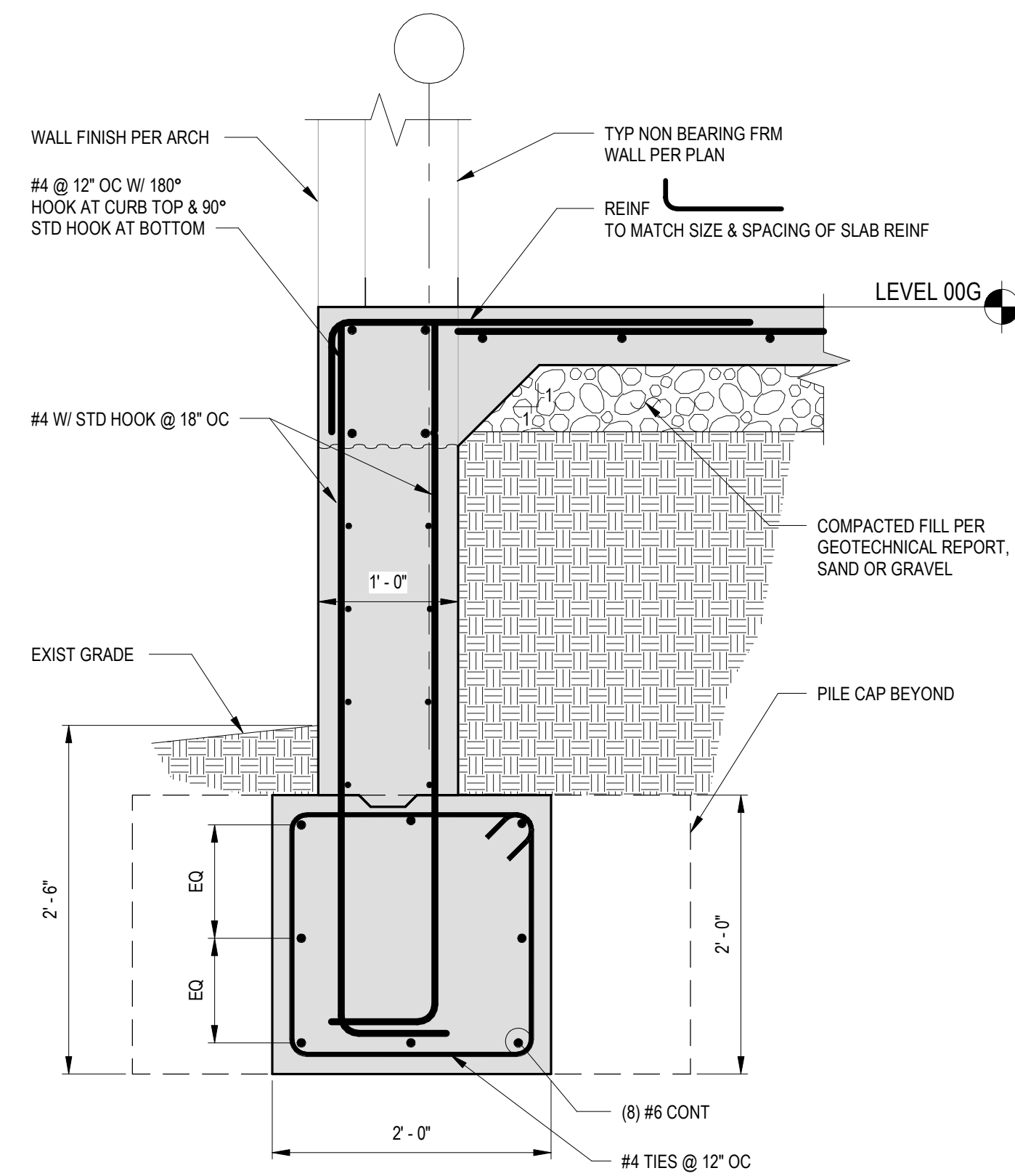
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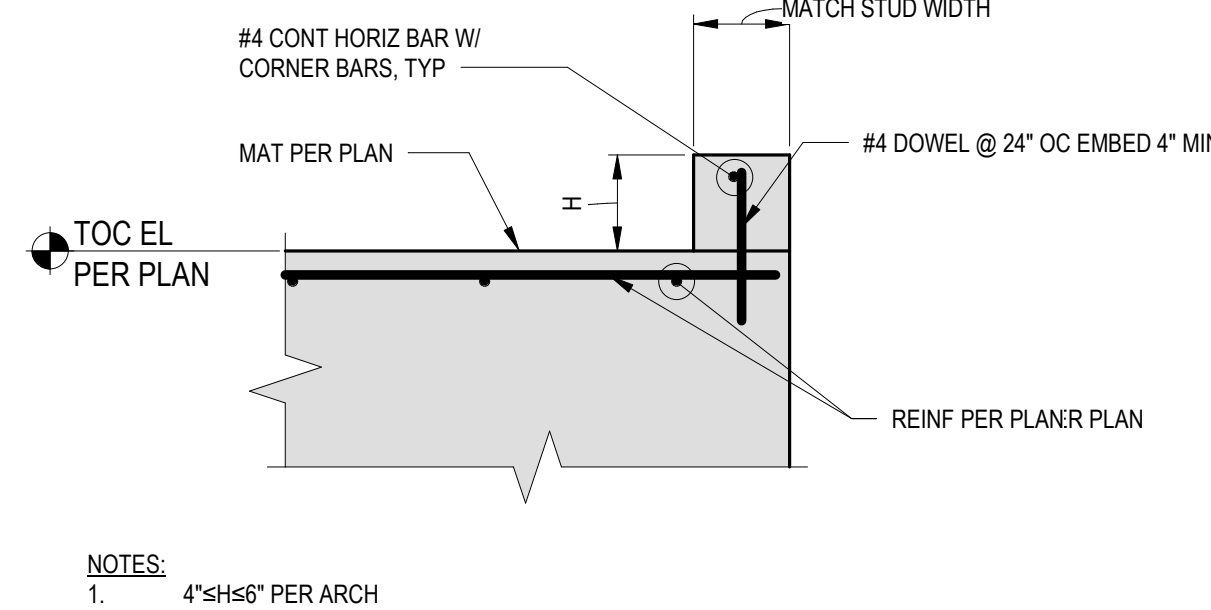
C1 WALL FOUNDATION DETAIL
1" = 1'-0"



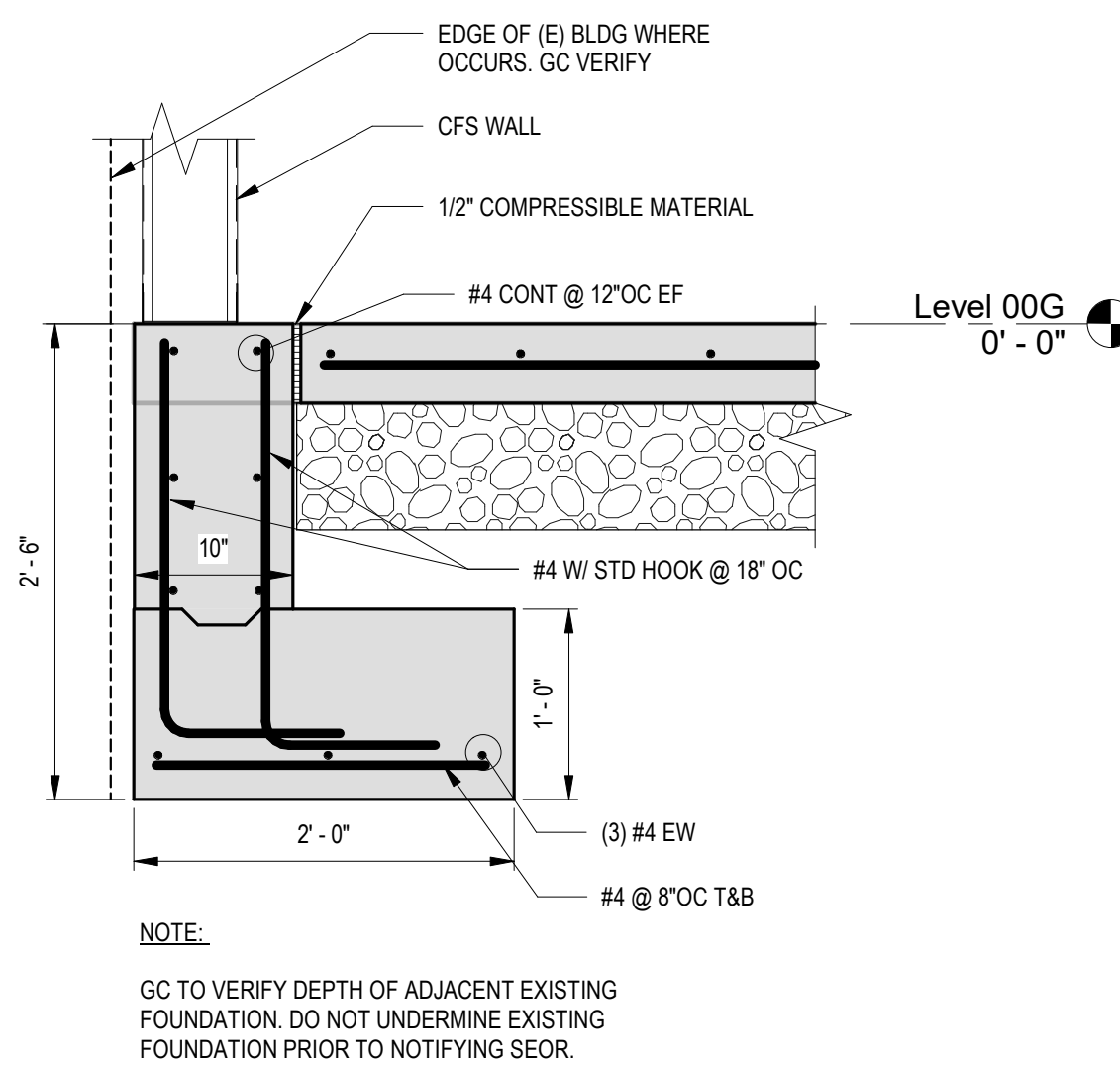
D1 TIE BEAM
1" = 1'-0"



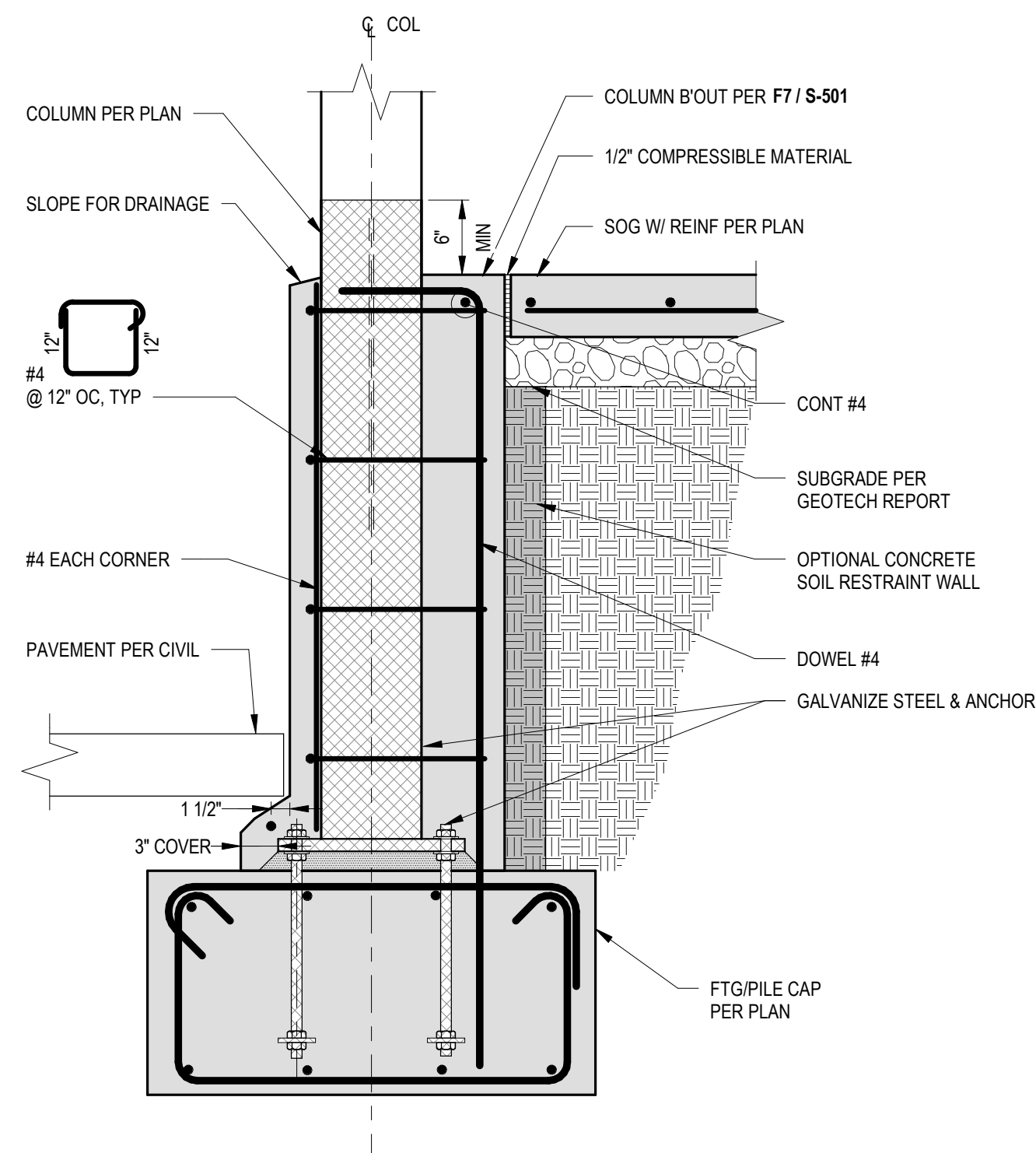
F1 SECTION - GRADE BEAM GB1
1" = 1'-0"



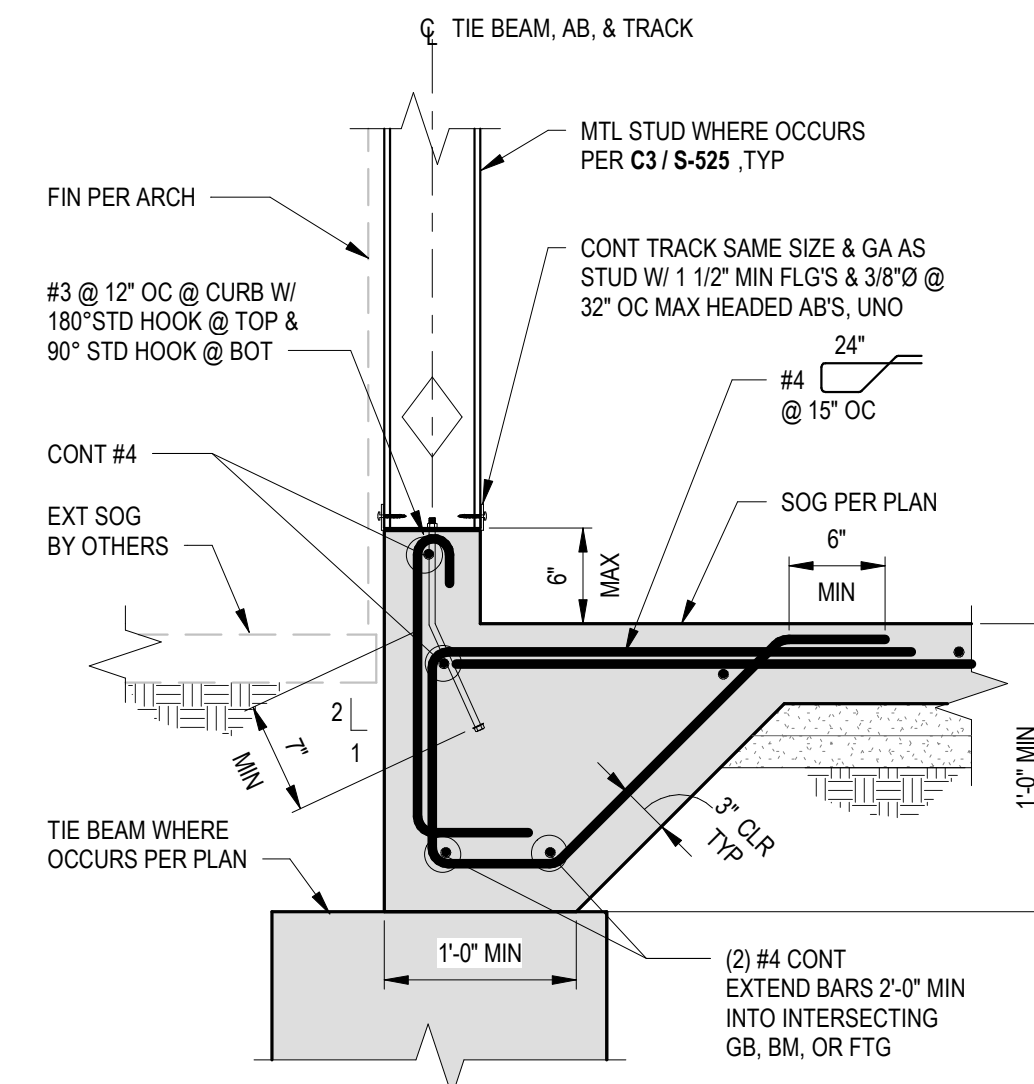
B4 CURB DETAIL
1" = 1'-0"



D4 WALL FOUNDATION DETAIL
1" = 1'-0"



F4 WALL FOUNDATION DETAIL
1" = 1'-0"



B6 TYPICAL EXTERIOR FRAMED WALL CONNECTION
1" = 1'-0"

Revisions:	Date:

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RAY SPEES
REGISTERED PROFESSIONAL ENGINEER
CIVIL
STATE OF CALIFORNIA

Office of
Construction
and Facilities
Management
VA U.S. Department
of Veterans Affairs

Drawing Title
FOUNDATION DETAILS
Approved:

Phase
100% CONSTRUCTION
DOCUMENTS

Project Title
EHRM INFRASTRUCTURE
UPGRADES - HUNTINGTON
Location
VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300
Issue Date
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Checked
CH
Drawn
Author

Project Number
581-22-700
Building Number
Drawing Number
S-503
037 OF 693

A

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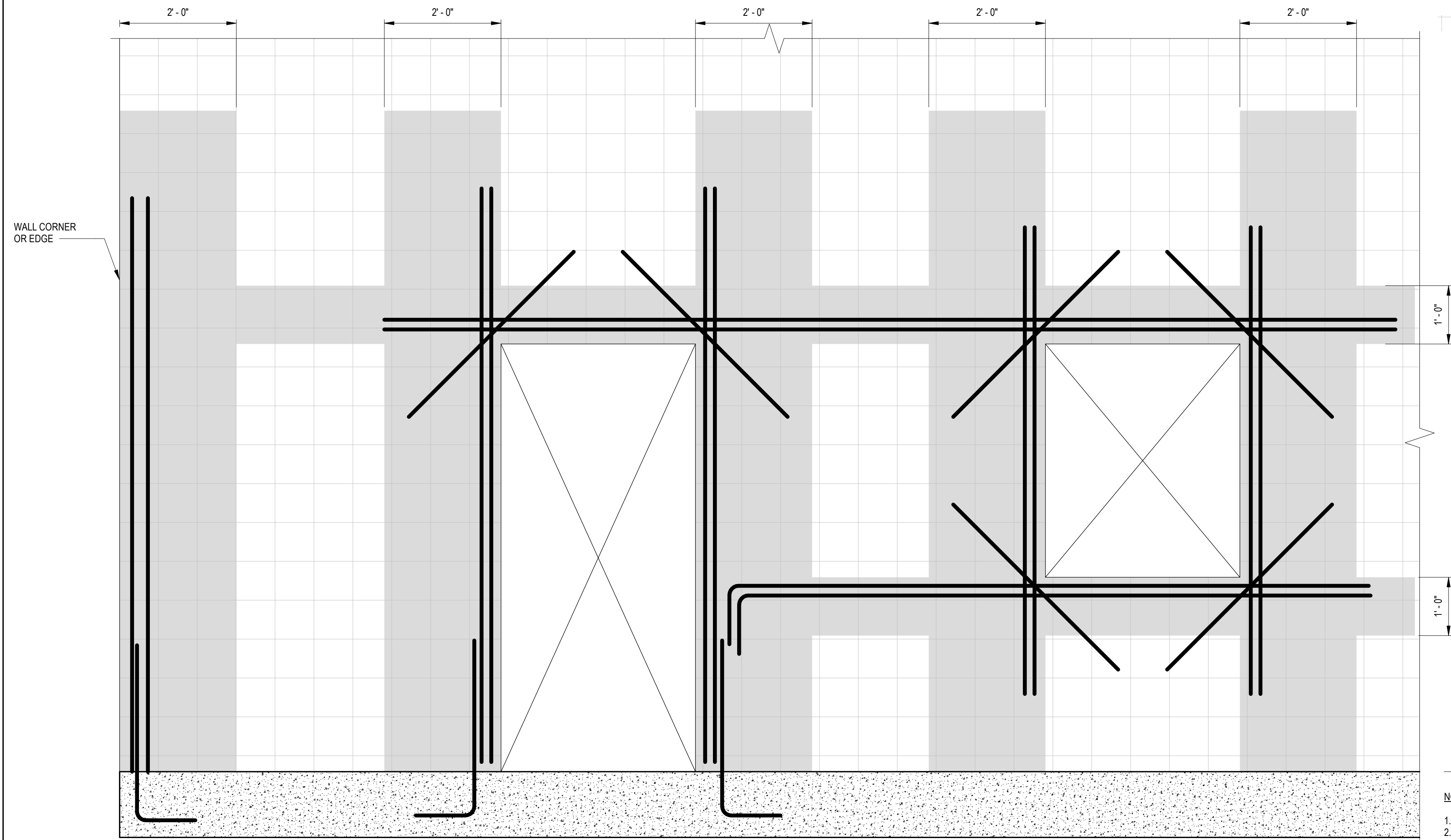
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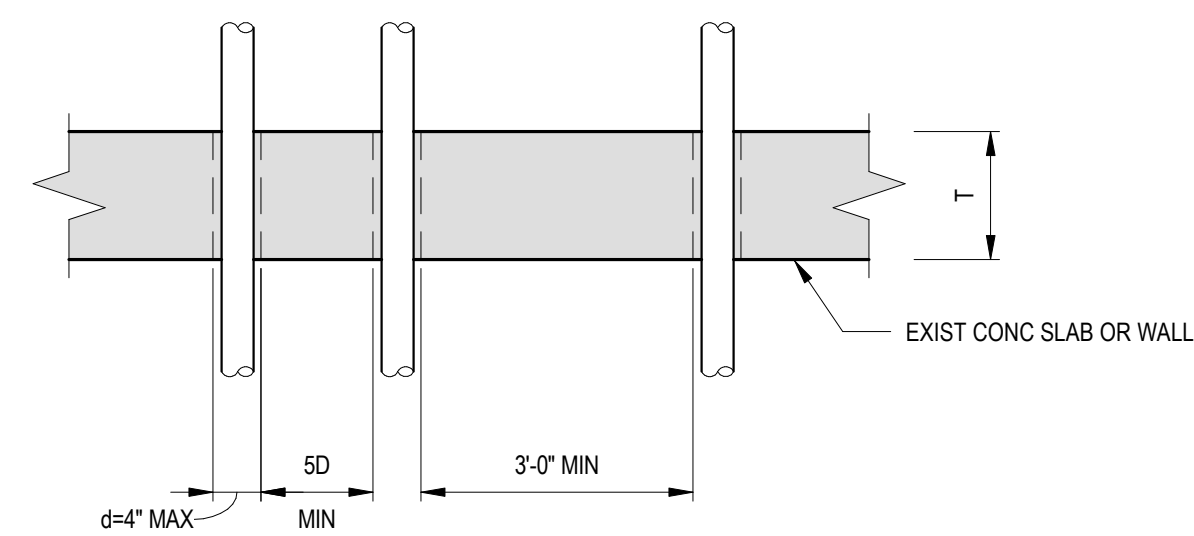
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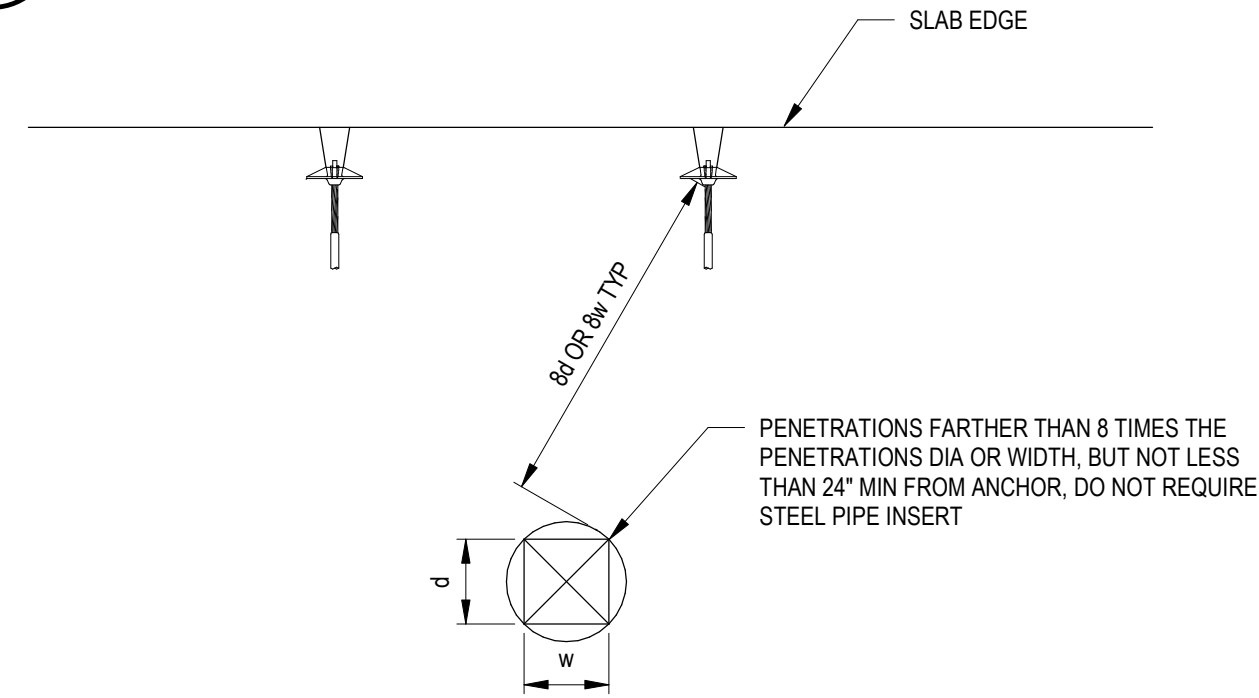
TOC EL
PER PLAN

- NOTES:
- DO NOT CORE IN THE SHADED AREAS.
 - SEE B6 / S-504 FOR MORE INFORMATION.



- NOTES:
- LOCATE EXISTING REINFORCEMENT PRIOR TO DRILLING AND DO NOT CUT OR DAMAGE EXISTING REINFORCEMENT.
 - DO NOT DRILL MORE THAN 2 HOLES WITHIN 3'-0" UNLESS APPROVED BY COR AND SUBMITTED TO SEOR.
 - DO NOT DRILL THROUGH BEAMS.

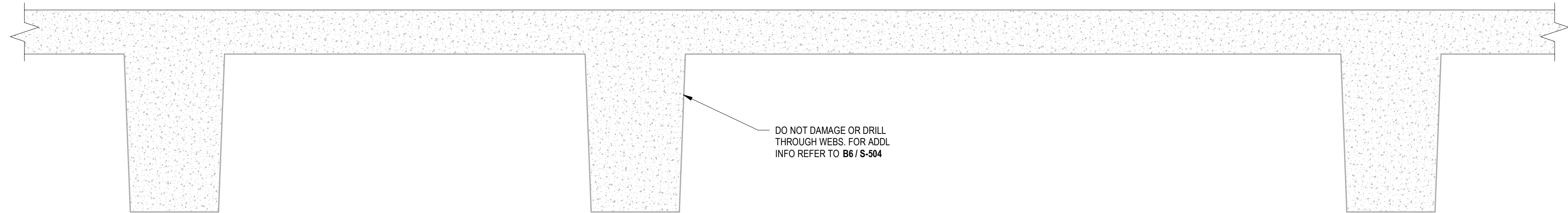
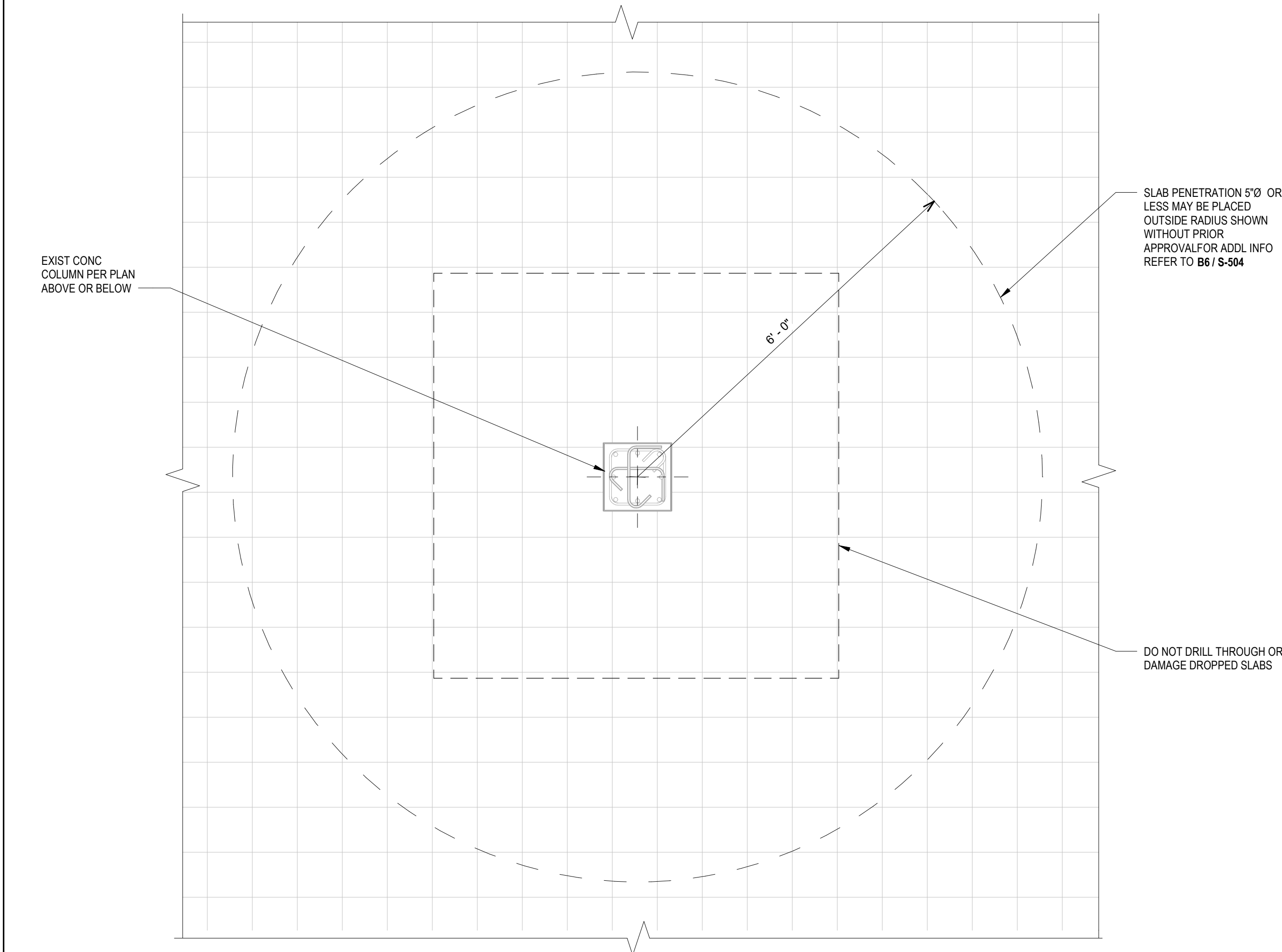
B6 PIPING & CONDUIT AT EXISTING SLAB OR WALL
1" = 1'-0"



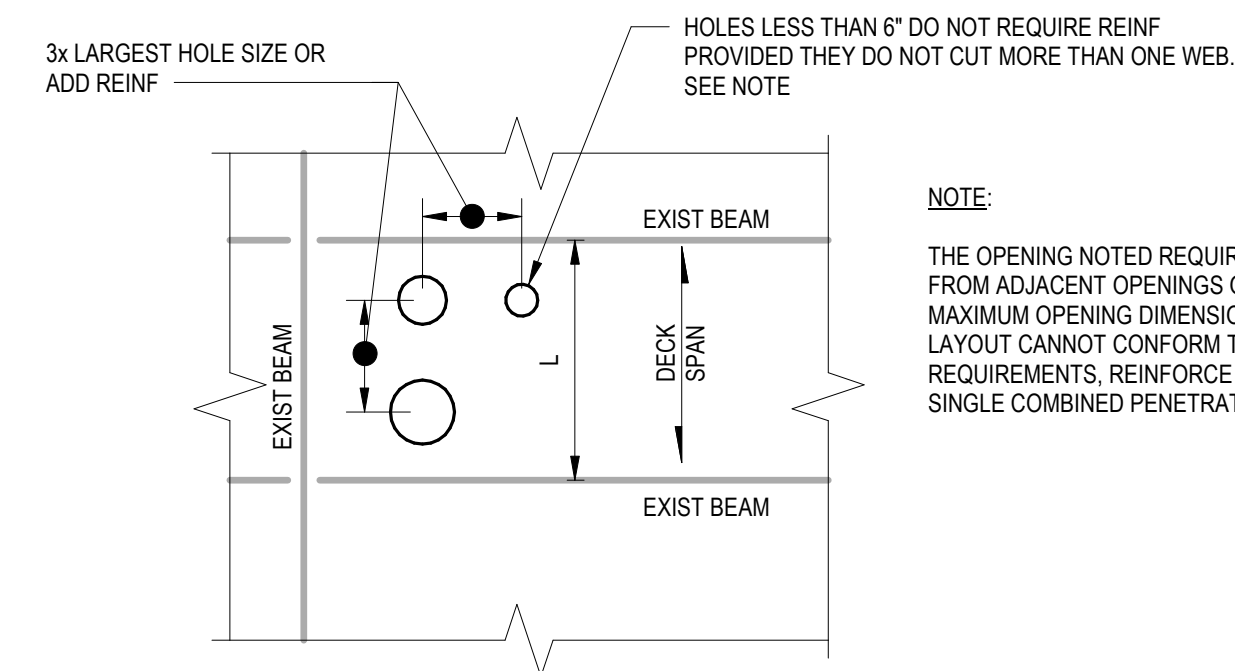
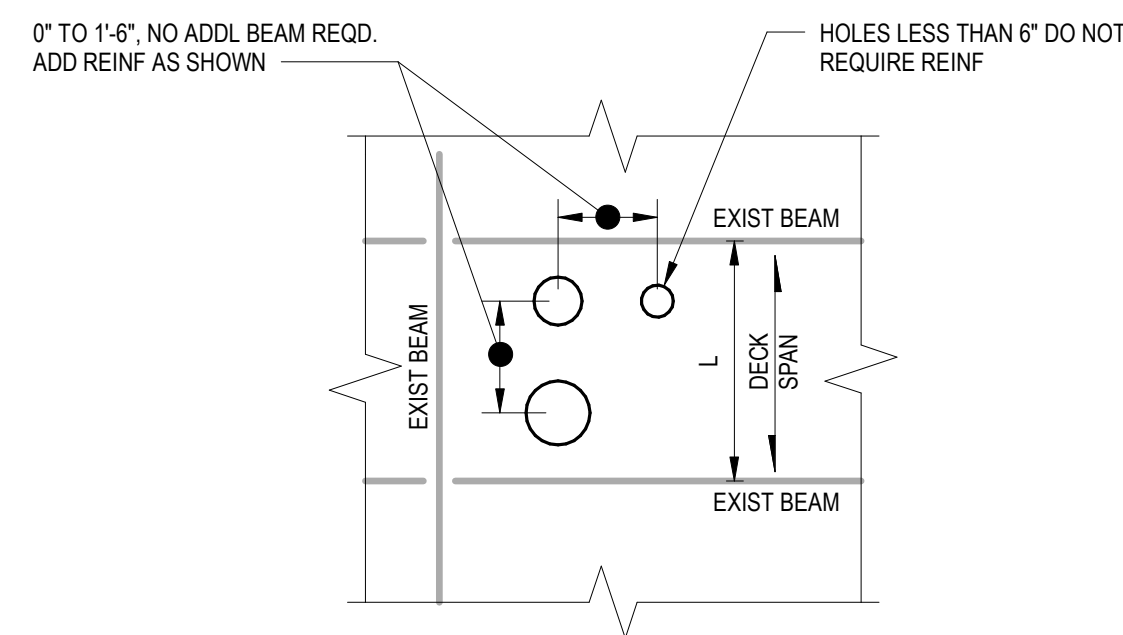
- NOTES:
- LOCATE ALL EXISTING REINFORCEMENT, TENDONS, AND PT ANCHOR HEADS PRIOR TO DRILLING SUCH THAT CORING AND CUTTING DO NOT DAMAGE ANY EXISTING ANCHOR HEADS, PT TENDONS OR REINFORCEMENT.
 - NEW PENETRATIONS @ 8'-0" OC MIN.
 - NEW ROUND OPENINGS TO BE 8" MAX. NEW SQUARE OR RECTANGULAR OPENING LENGTH/WIDTH TO BE 1'-0" MAX.
 - BEFORE ANY WORK STARTS, LOCATE AND DOCUMENT EXISTING REINFORCING THAT HAS THE POTENTIAL TO INTERFERE W/ CONSTRUCTION. IDENTIFY AND DOCUMENT ALL THIS REINFORCEMENT IN A SHOP DRAWING SUBMITTAL AND SUBMIT TO ARCHITECT FOR STANDARD SHOP DRAWING REVIEW AND APPROVAL TIME DURATION.

C6 PENETRATIONS AT EXIST POST TENSION SLABS
1" = 1'-0"

C1 PENETRATIONS IN EXISTING CONCRETE OR CMU WALL
3/4" = 1'-0"



E5 OPENINGS IN EXIST RIBBED SLABS
1" = 1'-0"



NOTE:
THE OPENING NOTED REQUIRES A CLEAR SPACING FROM ADJACENT OPENINGS OF THREE TIMES THE MAXIMUM OPENING DIMENSION. IF THE REQUIRED LAYOUT CANNOT CONFORM TO THESE REQUIREMENTS, REINFORCE THE GROUP AS A SINGLE COMBINED PENETRATION.

F1 PENETRATIONS IN EXISTING CONCRETE SLABS WITH COLUMNS AND NO BEAMS
3/4" = 1'-0"

Revisions:	Date:

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Construction
and Facilities
Management

U.S. Department
of Veterans Affairs

F7 ROOF DECK OPNG LESS THAN 5'-6"
1" = 1'-0"

Drawing Title

TYPICAL PENETRATIONS IN EXISTING STRUCTURE

Approved:

Phase

100% CONSTRUCTION DOCUMENTS

Project Title

EHRM INFRASTRUCTURE UPGRADES - HUNTINGTON

Project Number

581-22-700

Building Number

Location

VA HUNTINGTON
1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300

Issue Date

09/03/2025

Checked

CH

Drawn

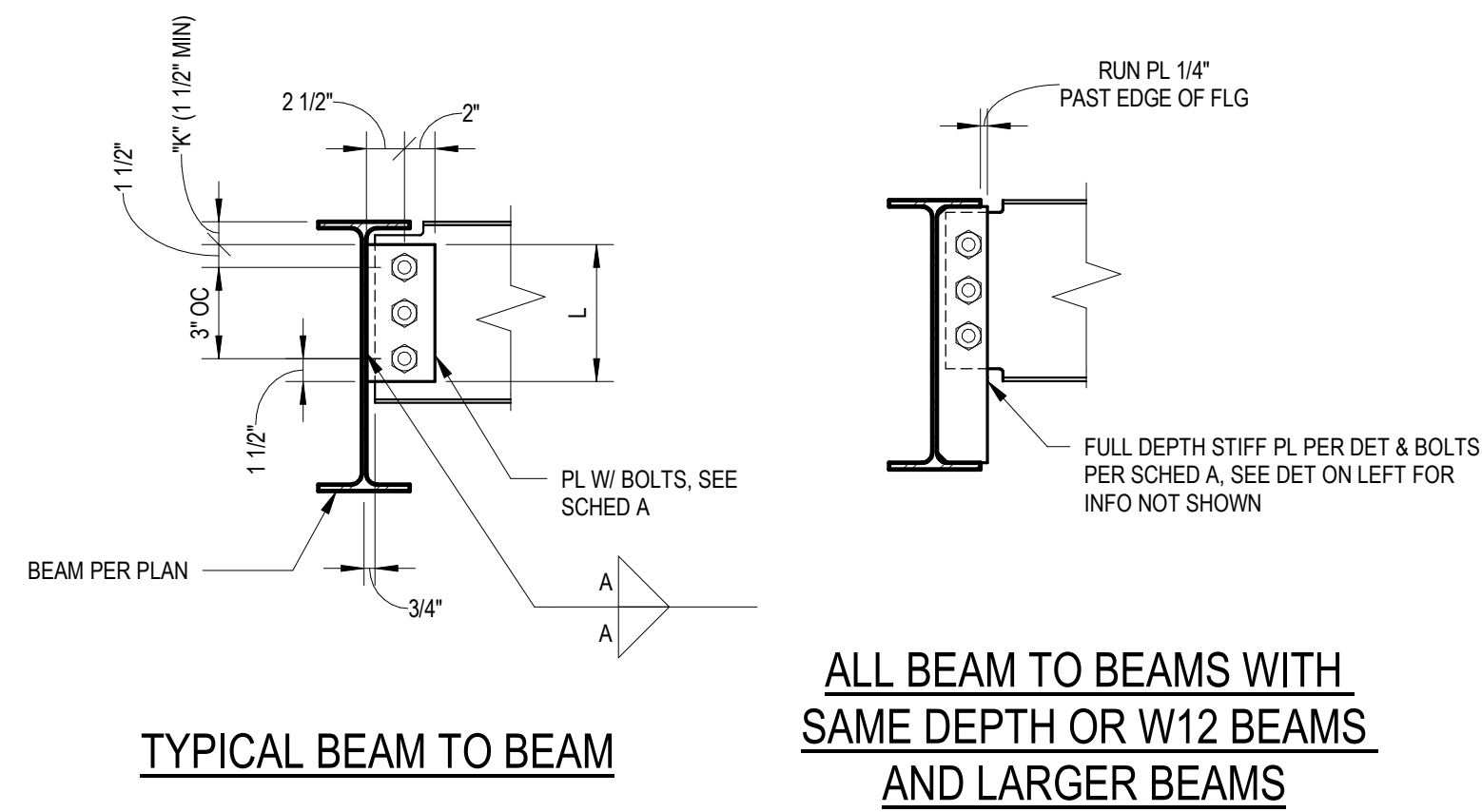
Author

Drawing Number

S-504

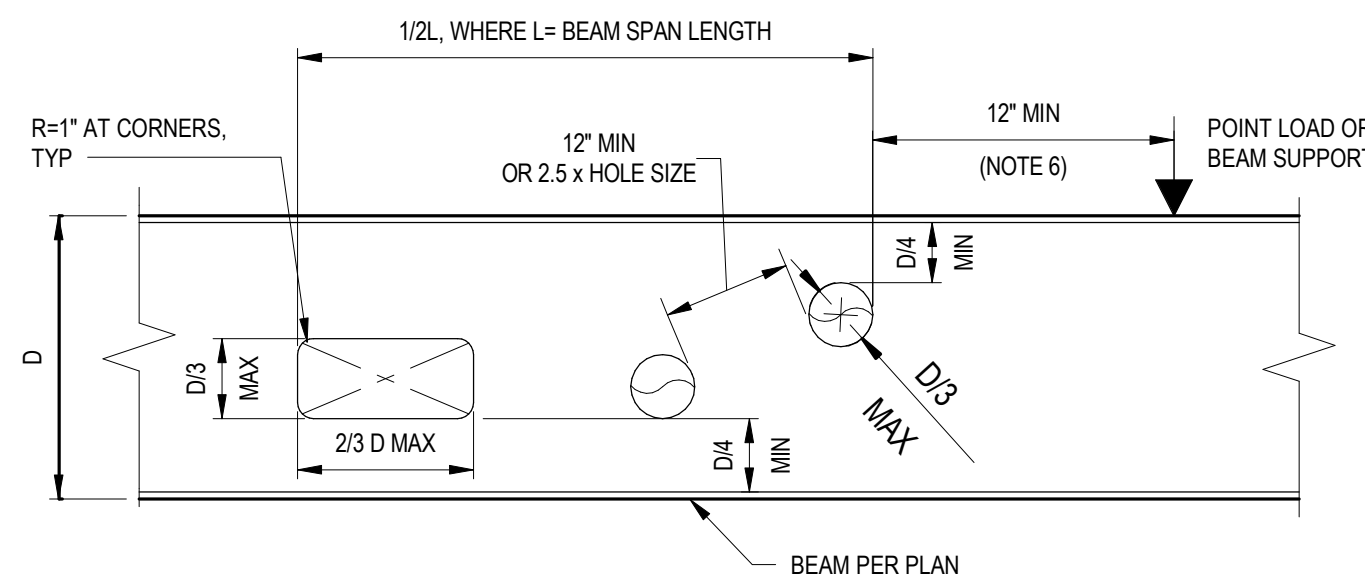
038 OF 693

A



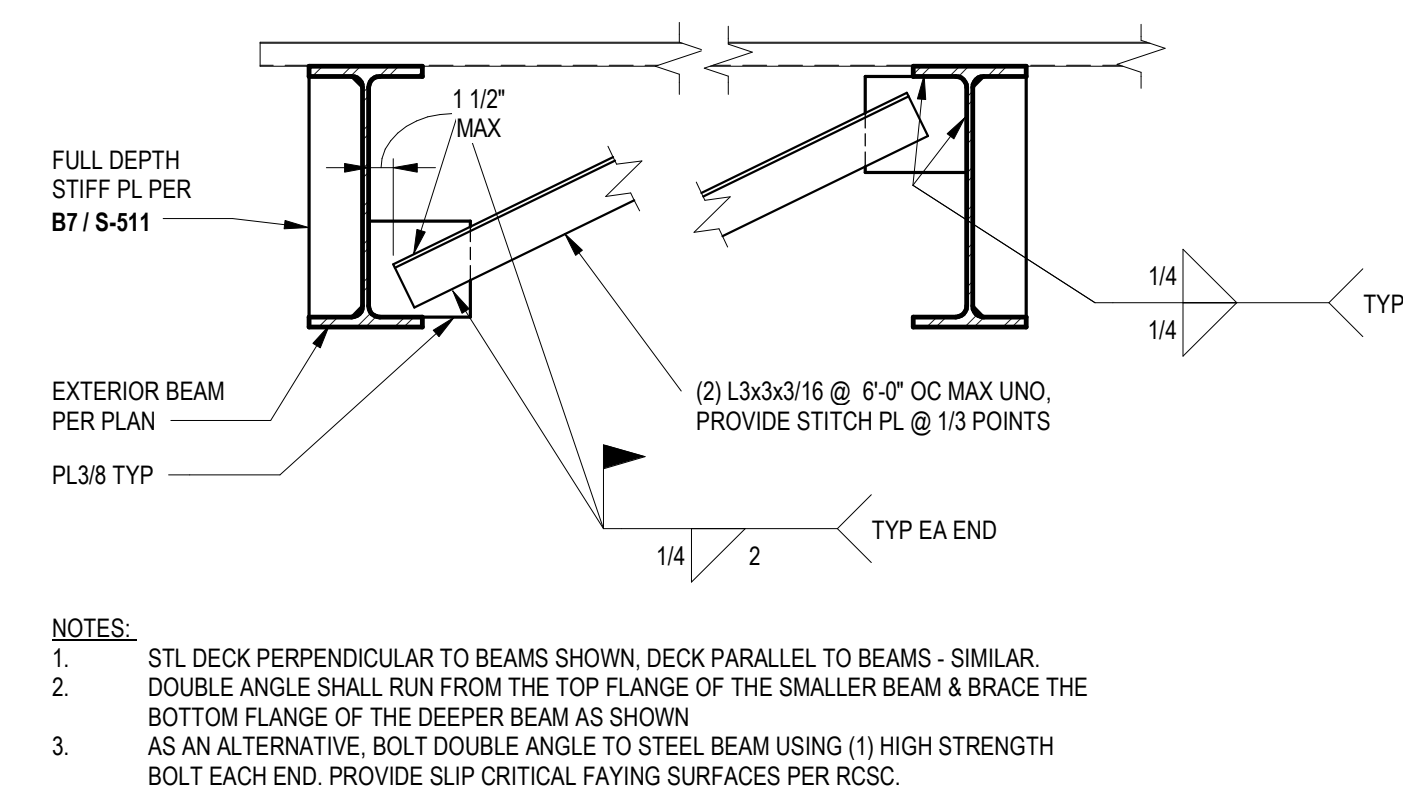
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B1 TYPICAL SHEAR TAB CONNECTION
1" = 1'-0"



C

D1 BLOCKOUTS IN BEAM WEB
1" = 1'-0"



E

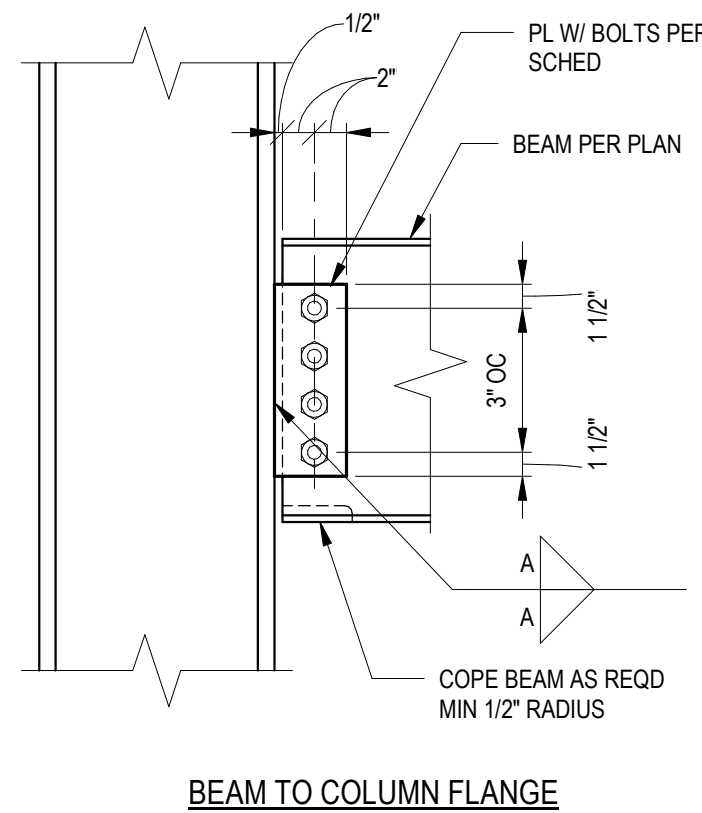
E1 TYPICAL BEAM BOTTOM FLANGE BRACING
1" = 1'-0"

F

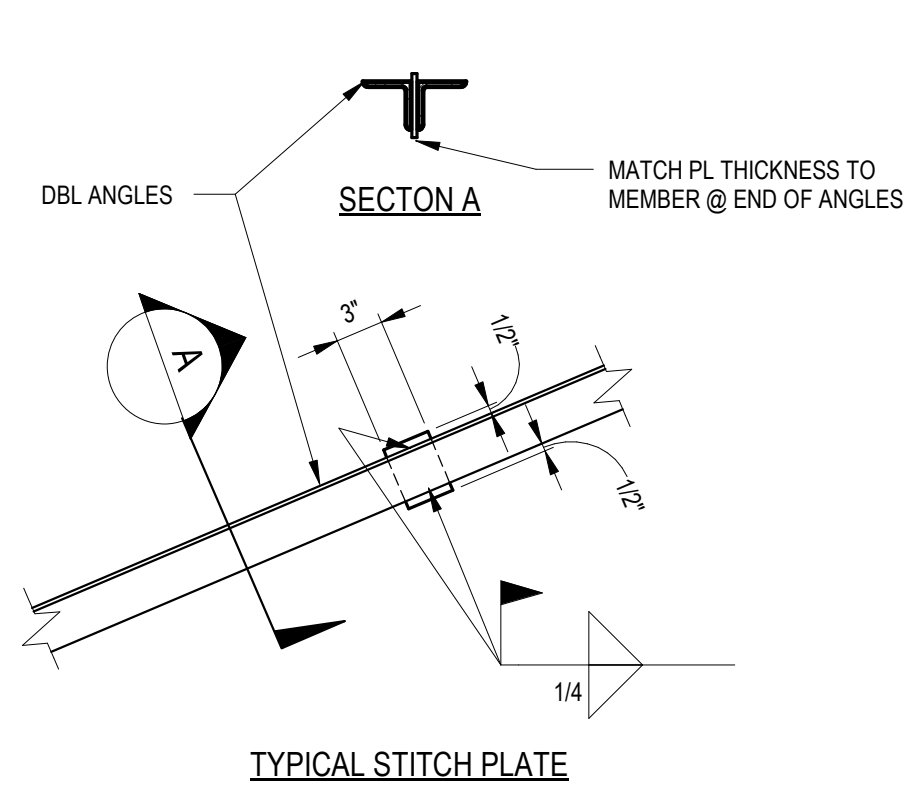
TABLE A										
BEAM SECTION	MAXIMUM REACTION				TOP COPE ONLY				TOP & BOTTOM COPE	
	MIN NUMBER OF BOLTS REQUIRED 7/8"Ø A325/F1852 UNLESS NOTED OTHERWISE	MAX ASD REACTION RvD (KIPS)	MAX LRFD REACTION RvD (KIPS)	A36 PLATE THICKNESS (INCH)	WELD SIZE A (INCH)	Fy (BEAM) = 50 KSI		Fy (BEAM) = 50 KSI		
						MIN WEB THICKNESS (INCH)	MAX COPE LENGTH (INCH)	MIN WEB THICKNESS (INCH)	MAX COPE LENGTH (INCH)	
C8,MC8,W8,W10	2	22	33	3/8	1/4	0.25	4	0.25	-	
C12,MC12,W12,W14	3	39	58			0.25	4	0.25	2 1/2	
W16	4	52	78			0.25	5	0.26	3	
W18	5	65	97			0.25	9	0.27	5	
W21	6	78	117			0.25	11	0.30	6	
W24	7	91	137			0.25	14	0.29	11	
W27,W30	8	104	156			0.25	16	0.28	14	
W33,W36	9	116	174			0.25	16	0.27	14	

NOTES:
1. BEAMS SHALL HAVE STANDARD ROUND HOLES (STD) AND SHEAR TAB PLATES MAY HAVE HORIZONTAL SHORT SLOTTED HOLES (HSSH) IF NOT SLIP CRITICAL.
2. PROVIDE THE MINIMUM QUANTITY OF BOLTS SHOWN IN TABLE A.
3. MAXIMUM COPE DEPTH SHALL BE 2" FOR DEPTHS UP TO W18 AND 3" FOR BEAMS W21 OR GREATER. CHANNELS (C AND MC) BE COPE WITHOUT PRIOR APPROVAL FROM THE STRUCTURAL ENGINEER.
4. COPED BEAMS SHALL BE CHECKED FOR MINIMUM WEB THICKNESS AND MAXIMUM COPE LENGTH PER TABLE A.

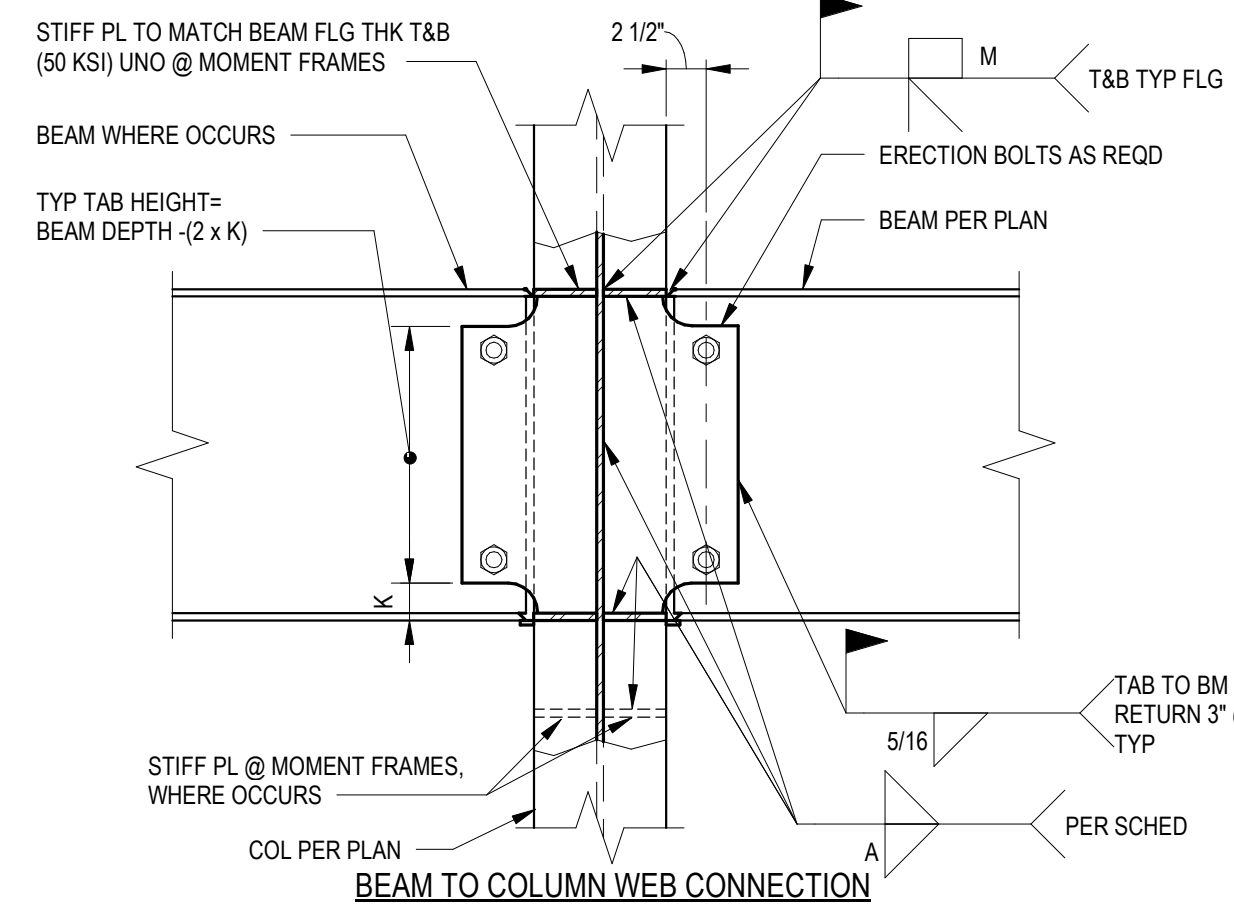
B3 TYPICAL BOLT SCHEDULE
1" = 1'-0"



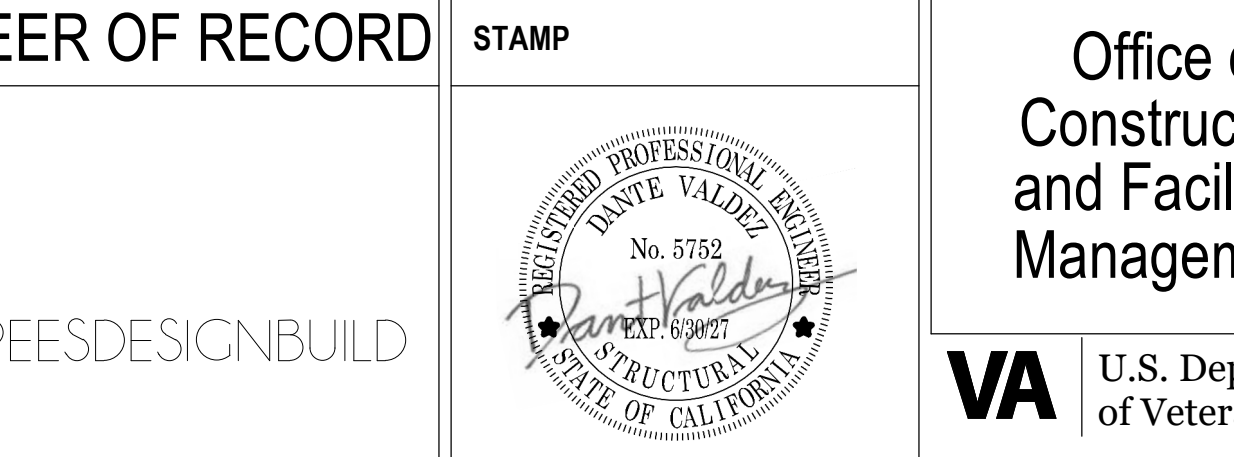
D3 TYPICAL SHEAR TAB CONNECTION
1" = 1'-0"



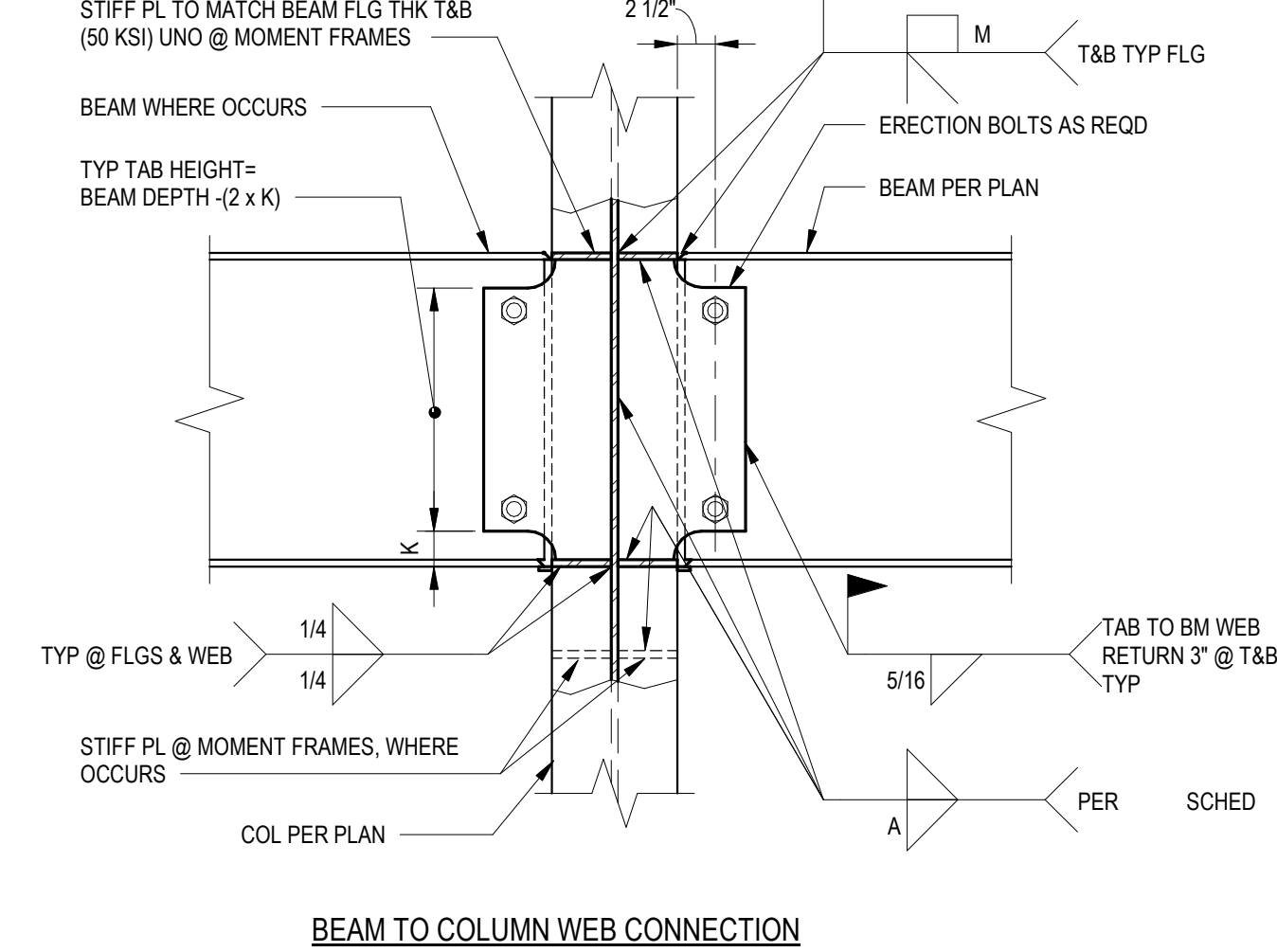
E5 DECK TO EXIST STRUCURE CONN
1" = 1'-0"



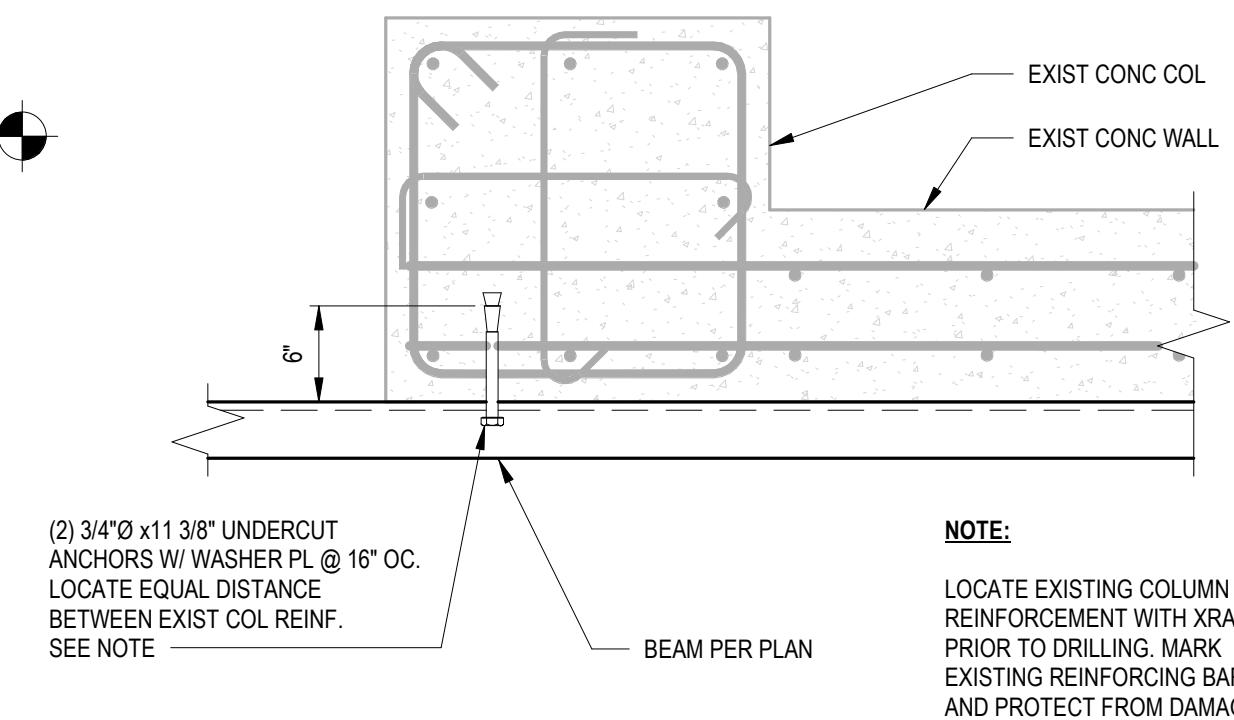
F5 LATERAL BEAM TO COLUMN MOMENT CONNECTIONS
1" = 1'-0"



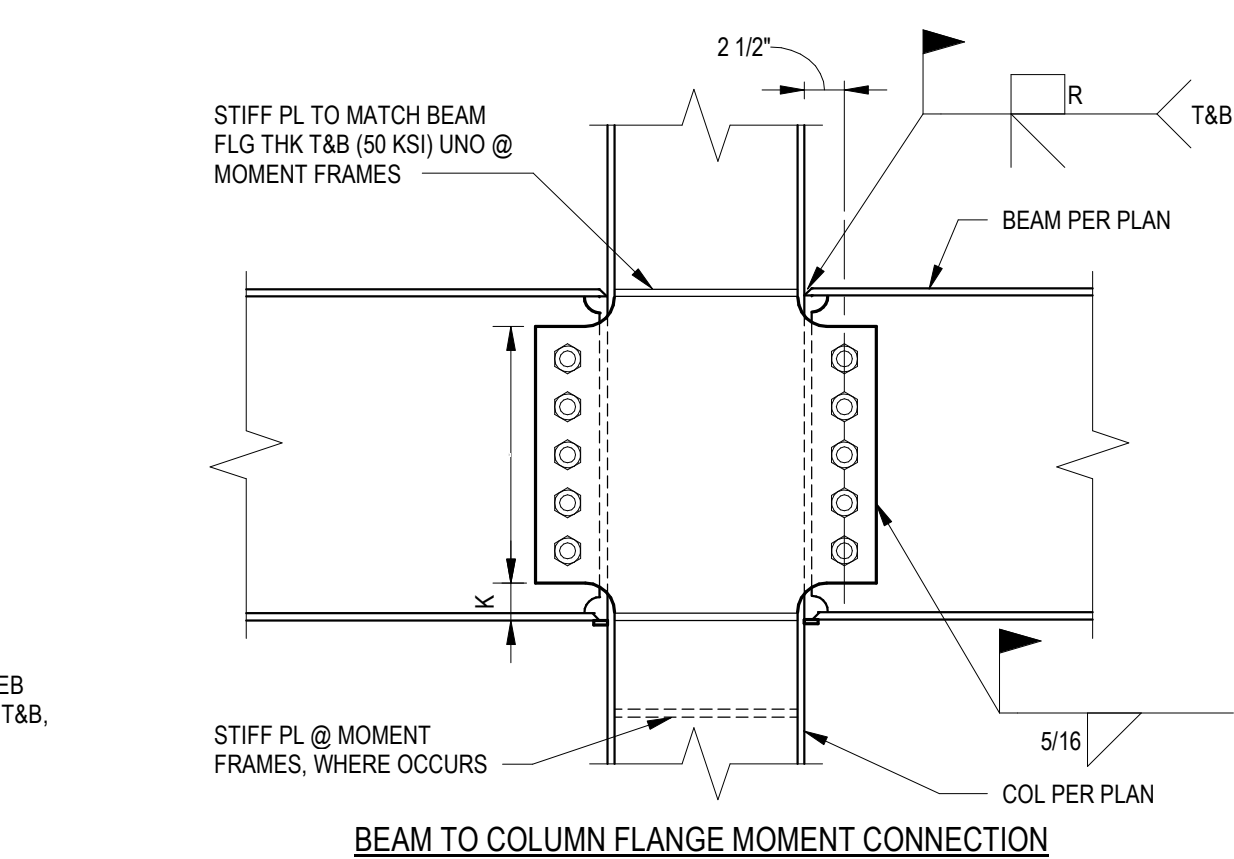
B7 TYPICAL STIFFENER PLATES
1" = 1'-0"



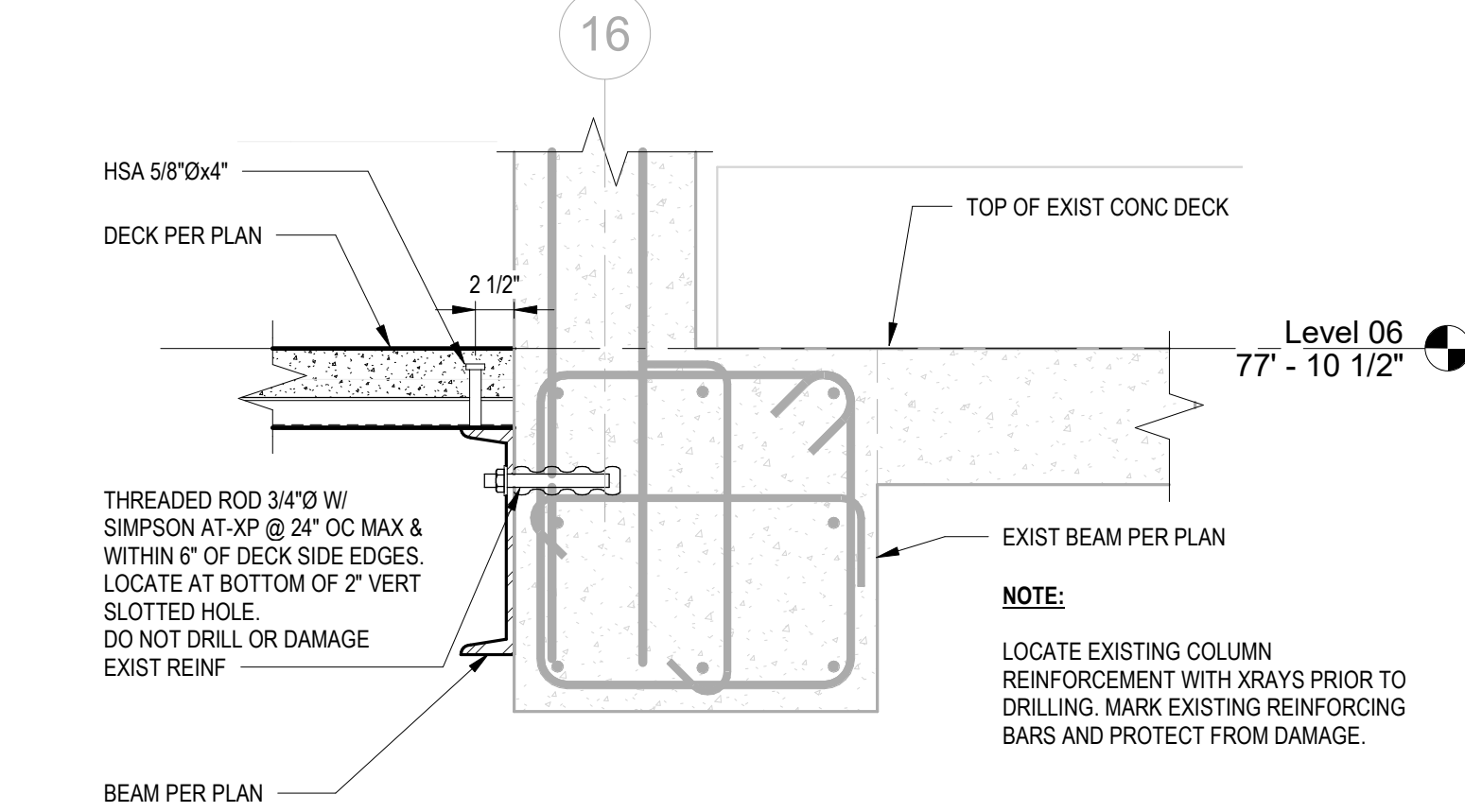
D7 GRAVITY BEAM TO COLUMN MOMENT CONNECTION
1" = 1'-0"



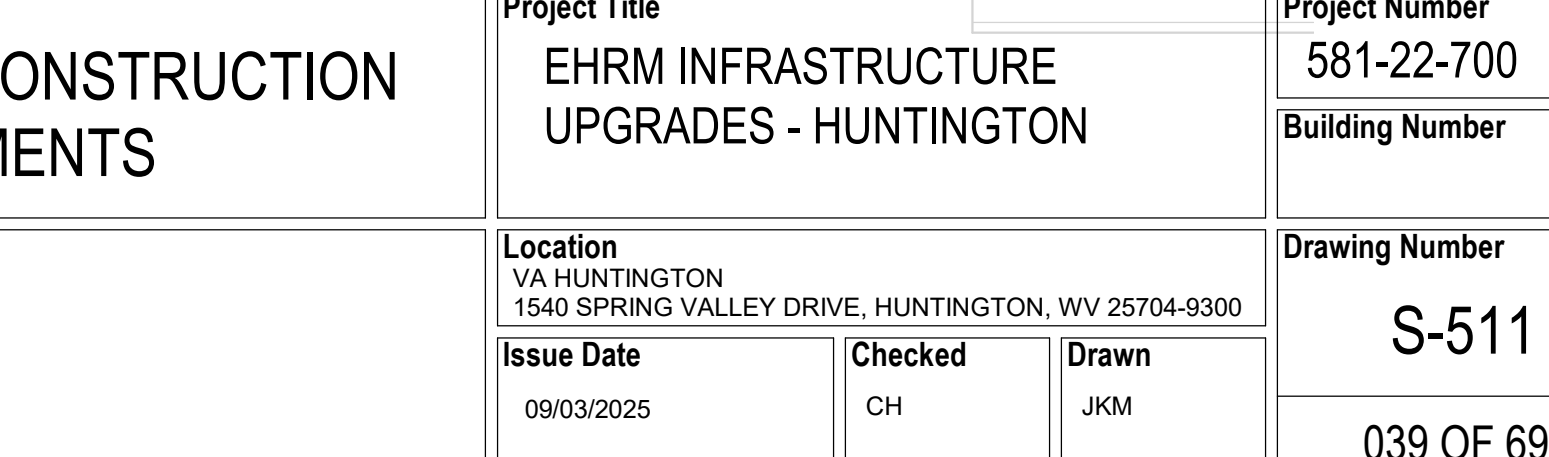
E7 CONN TO EXIST SHEAR WALL
1" = 1'-0"



E9 BEAM CONN TO EXIST SHEAR WALL
1" = 1'-0"



F9 BEAM CONN TO EXIST BEAM
1" = 1'-0"



Revisions:	Date:	CONSULTANT ENGINEERING DISCIPLINE: WSP WSP USA Inc. 33301 9th Avenue South Suite 300 Federal Way, WA 98003-2600 TEL: (206) 431-2300 FAX: (206) 431-2250	ARCHITECT/ENGINEER OF RECORD A/E: SPEES DESIGN BUILD 625 1ST AVE, STE 301 SEATTLE, WA 98104 (206) 590-2118 RAY SPEES SPEESDESIGNBUILD	STAMP Professional Engineer No. 5732 RAY SPEES STATE OF CALIFORNIA	Office of Construction and Facilities Management U.S. Department of Veterans Affairs	Drawing Title TYPICAL STRUCTURAL STEEL DETAILS Approved:	Phase 100% CONSTRUCTION DOCUMENTS	Project Title EHM INFRASTRUCTURE UPGRADES - HUNTINGTON	Project Number 581-22-700 Building Number
								Location VA HUNTINGTON 1540 SPRING VALLEY DRIVE, HUNTINGTON, WV 25704-9300	Drawing Number S-511 039 OF 693
								Issue Date 09/03/2025	Checked CH
								Drawn JKM	